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(12) **United States Patent**
Madisetti et al.

(10) **Patent No.:** **US 12,405,977 B1**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **METHOD AND SYSTEM FOR OPTIMIZING USE OF RETRIEVAL AUGMENTED GENERATION PIPELINES IN GENERATIVE ARTIFICIAL INTELLIGENCE APPLICATIONS**

(58) **Field of Classification Search**
CPC G06F 16/3329; G06F 40/284
See application file for complete search history.

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Widerman Malek PL

(57) **ABSTRACT**

A system and method of improving performance of LLMs including receiving context files, generating refined context files from the context files, sending the refined context files to h-LLMs, receiving a user prompt, generating a plurality of derived prompts from the user prompt, transmitting the plurality of derived prompts to the h-LLMs, receiving a plurality of h-LLM results, processing the plurality of h-LLM results to generate a responsive result, and transmitting the responsive result to a user interface.

21 Claims, 54 Drawing Sheets

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Arshdeep Bahga, Chandigarh (IN)

(73) Assignee: **Vijay Madisetti**, Alpharetta, GA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/812,707**

(22) Filed: **Aug. 22, 2024**

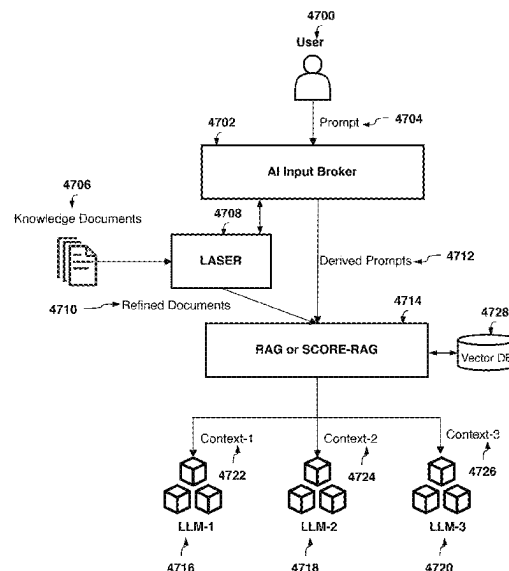
Related U.S. Application Data

(63) Continuation-in-part of application No. 18/744,199, filed on Jun. 14, 2024, now Pat. No. 12,306,859, which is a continuation-in-part of application No. 18/406,906, filed on Jan. 8, 2024, now Pat. No. 12,158,904, which is a continuation-in-part of application No. 18/470,487, filed on Sep. 20, 2023, now Pat. No. 12,147,461.

(60) Provisional application No. 63/551,548, filed on Feb. 9, 2024, provisional application No. 63/604,909, filed on Dec. 1, 2023, provisional application No. 63/604,910, filed on Dec. 1, 2023, provisional application No. 63/602,675, filed on Nov. 27, 2023.

(51) **Int. Cl.**
G06F 16/3329 (2025.01)
G06F 40/284 (2020.01)

(52) **U.S. Cl.**
CPC **G06F 16/3329** (2019.01); **G06F 40/284** (2020.01)



(56)

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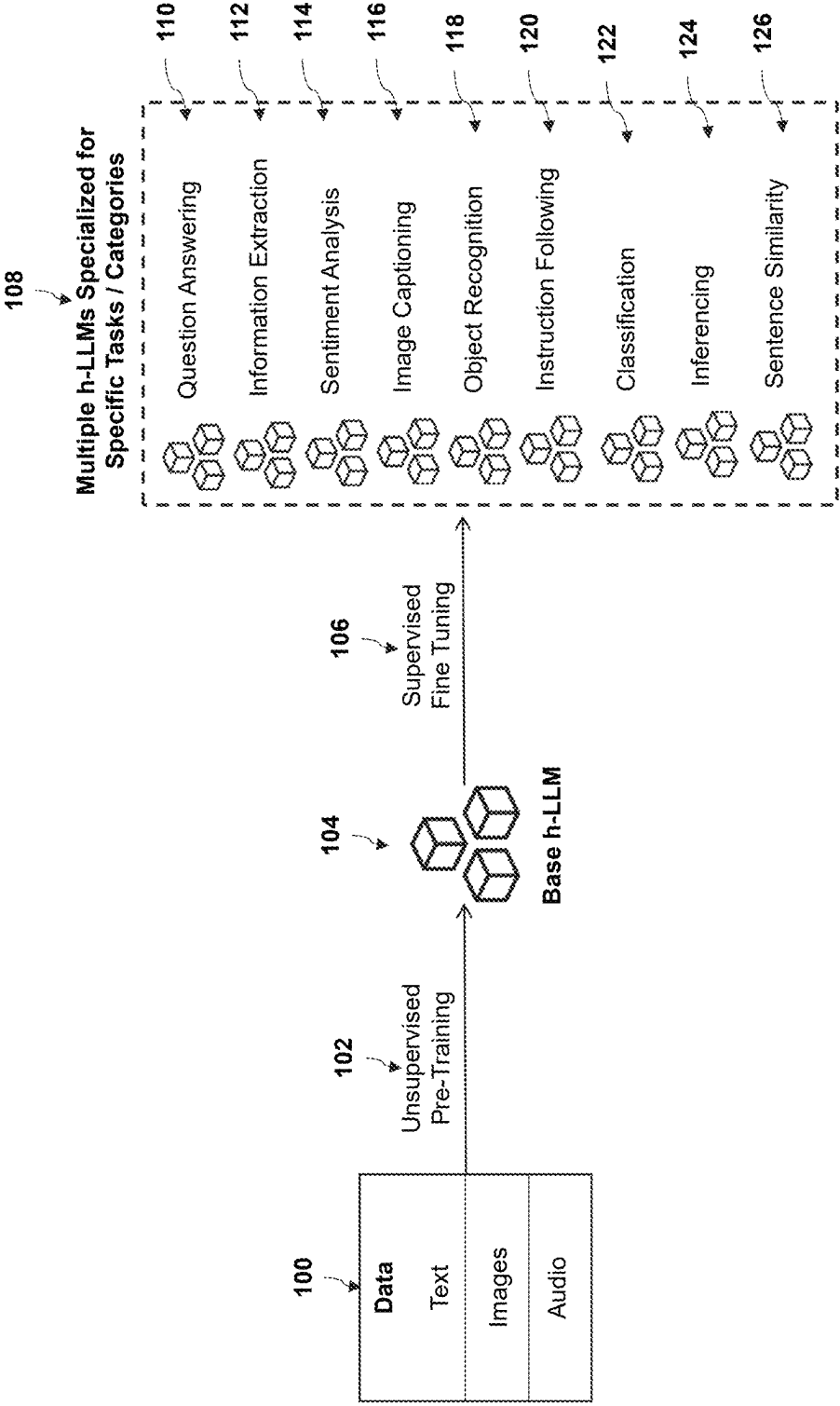


FIG. 1

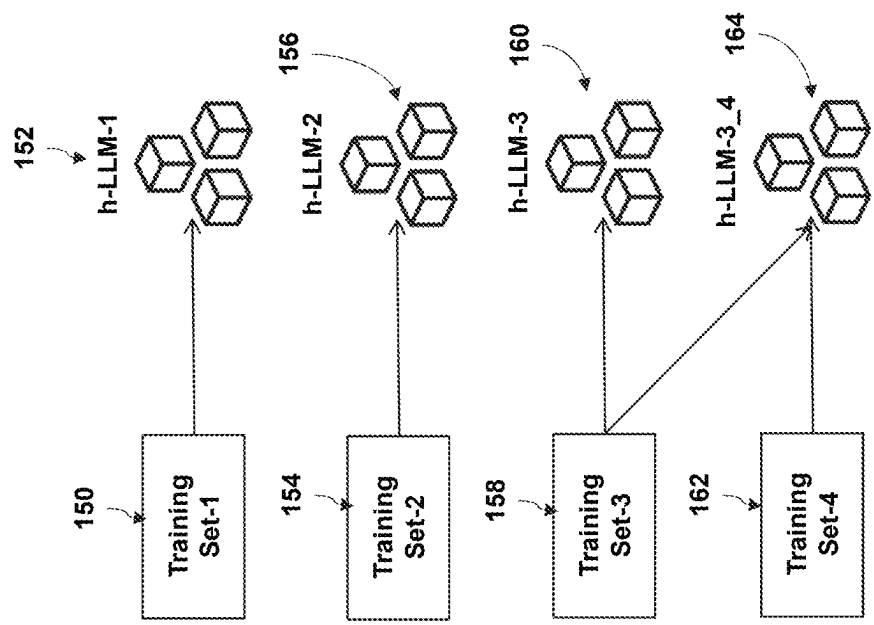


FIG. 2

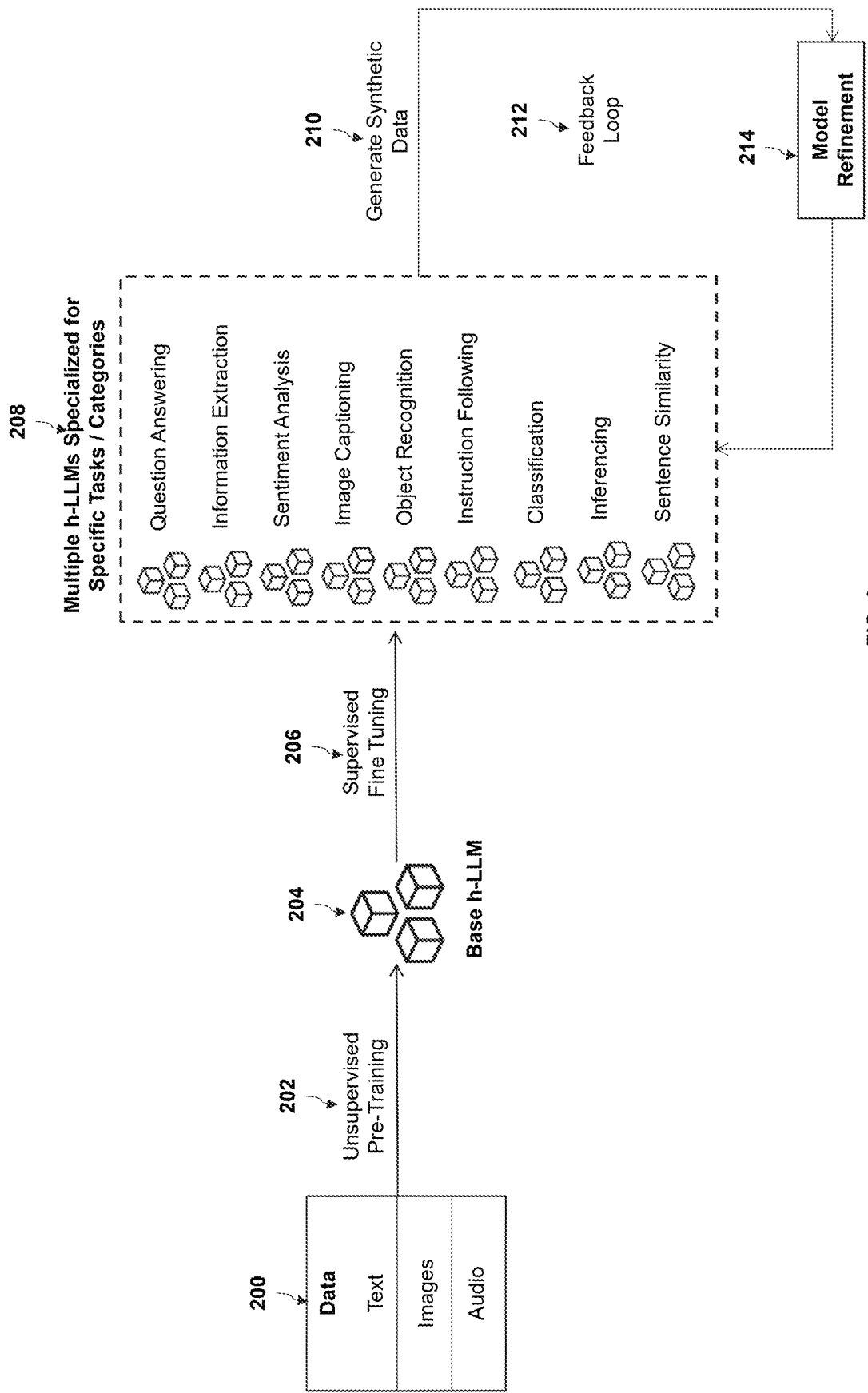


FIG. 3

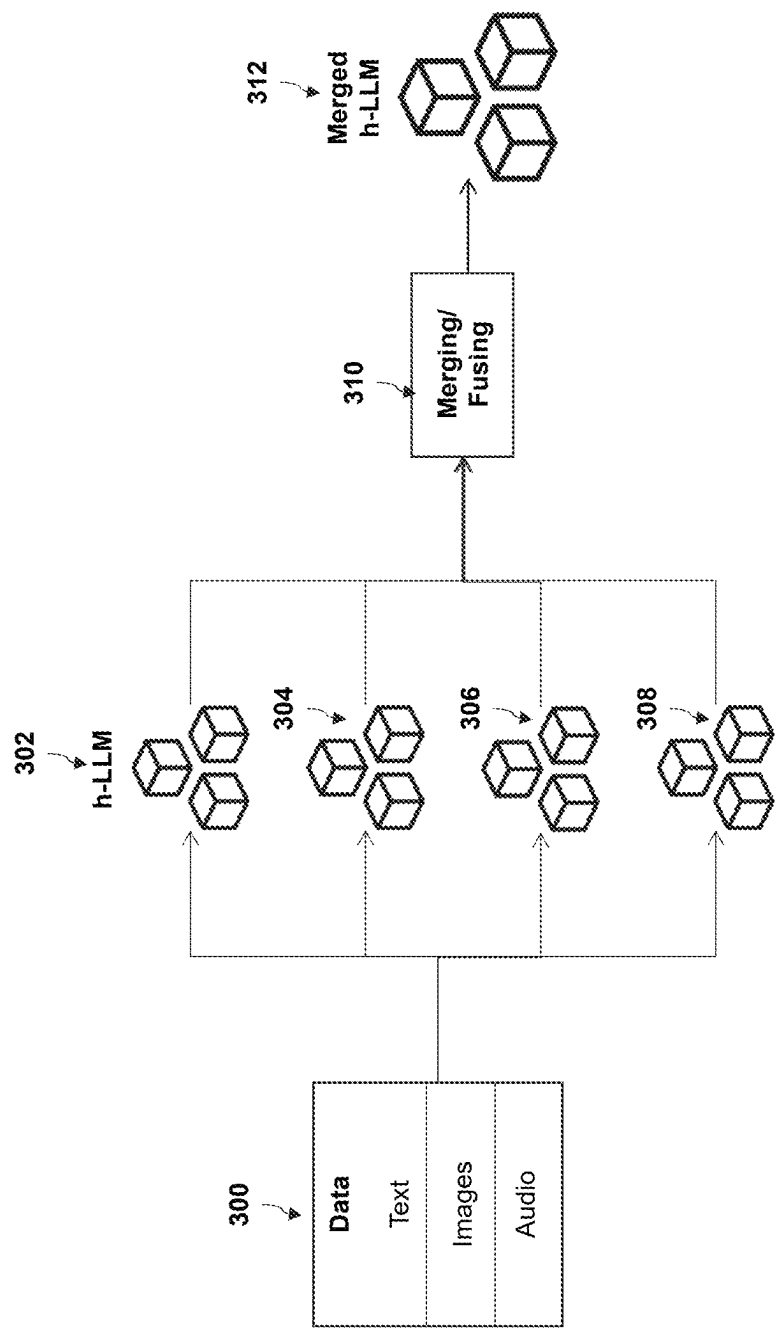
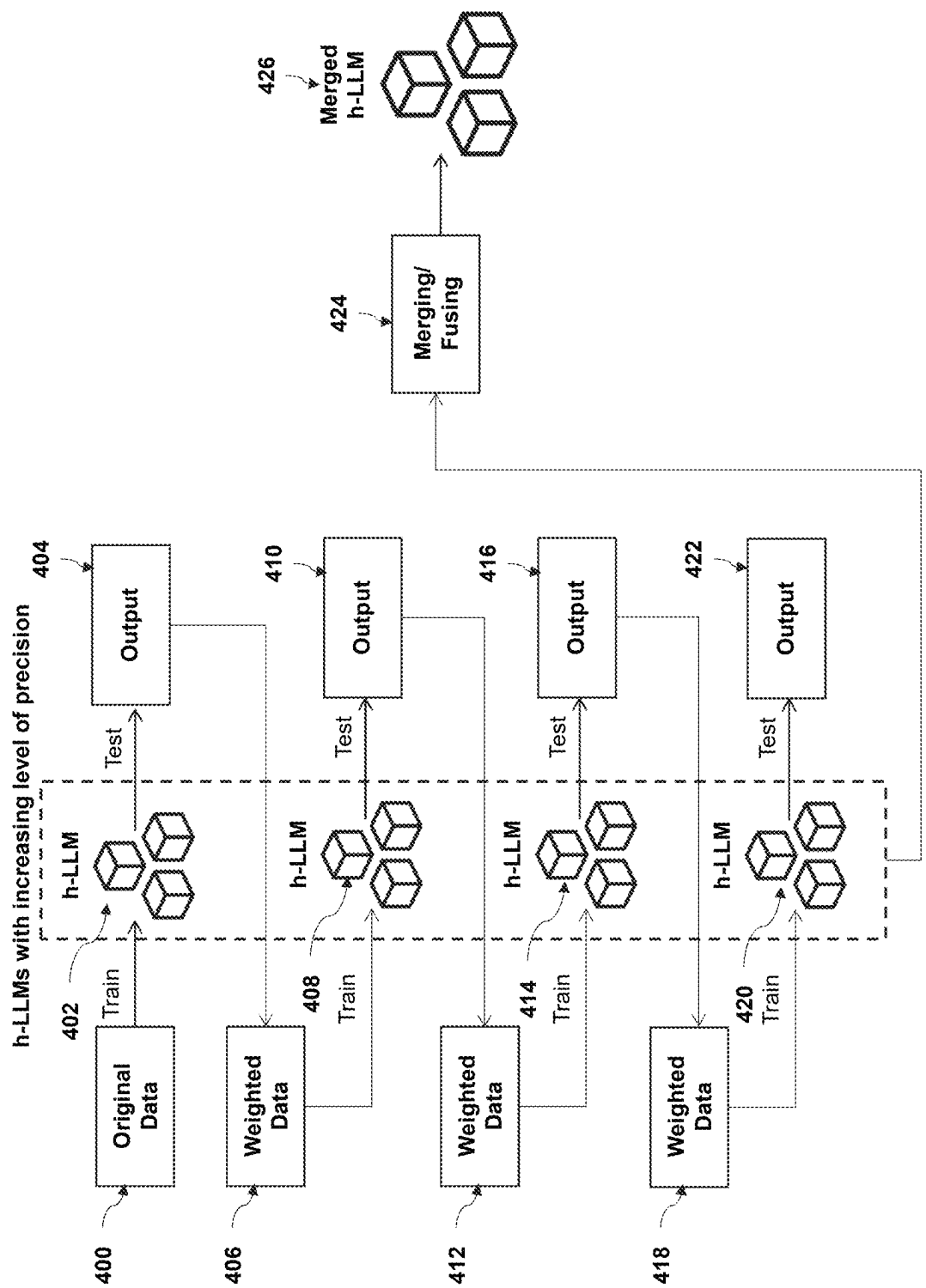


FIG. 4



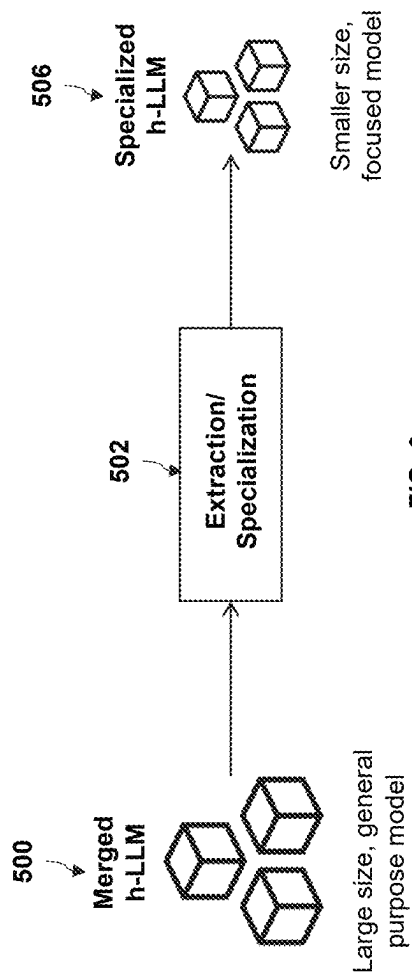


FIG. 6

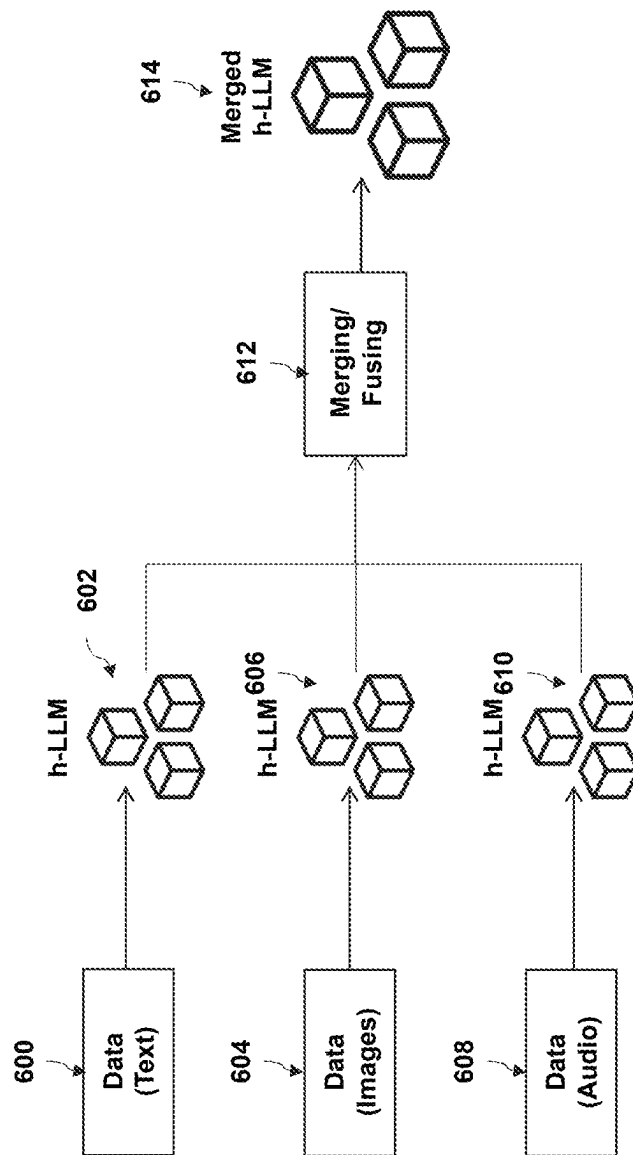


FIG. 7

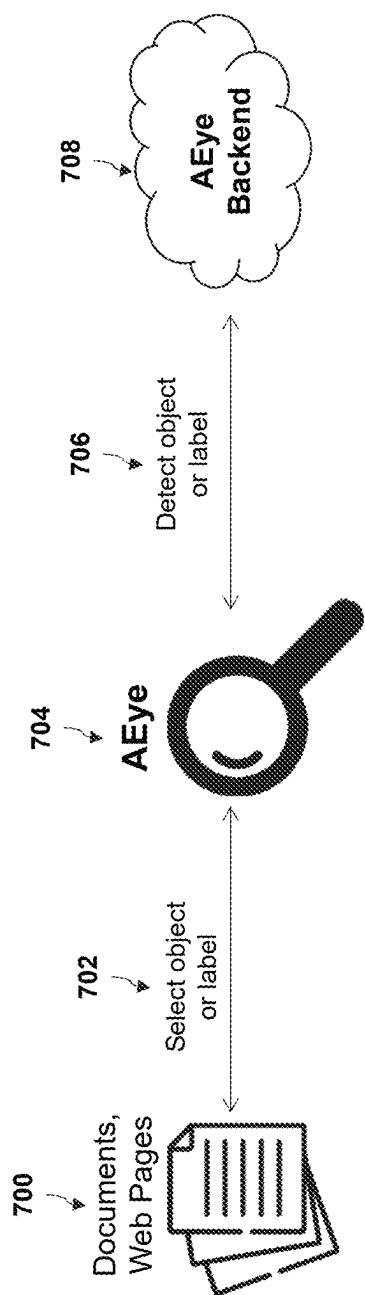


FIG. 8

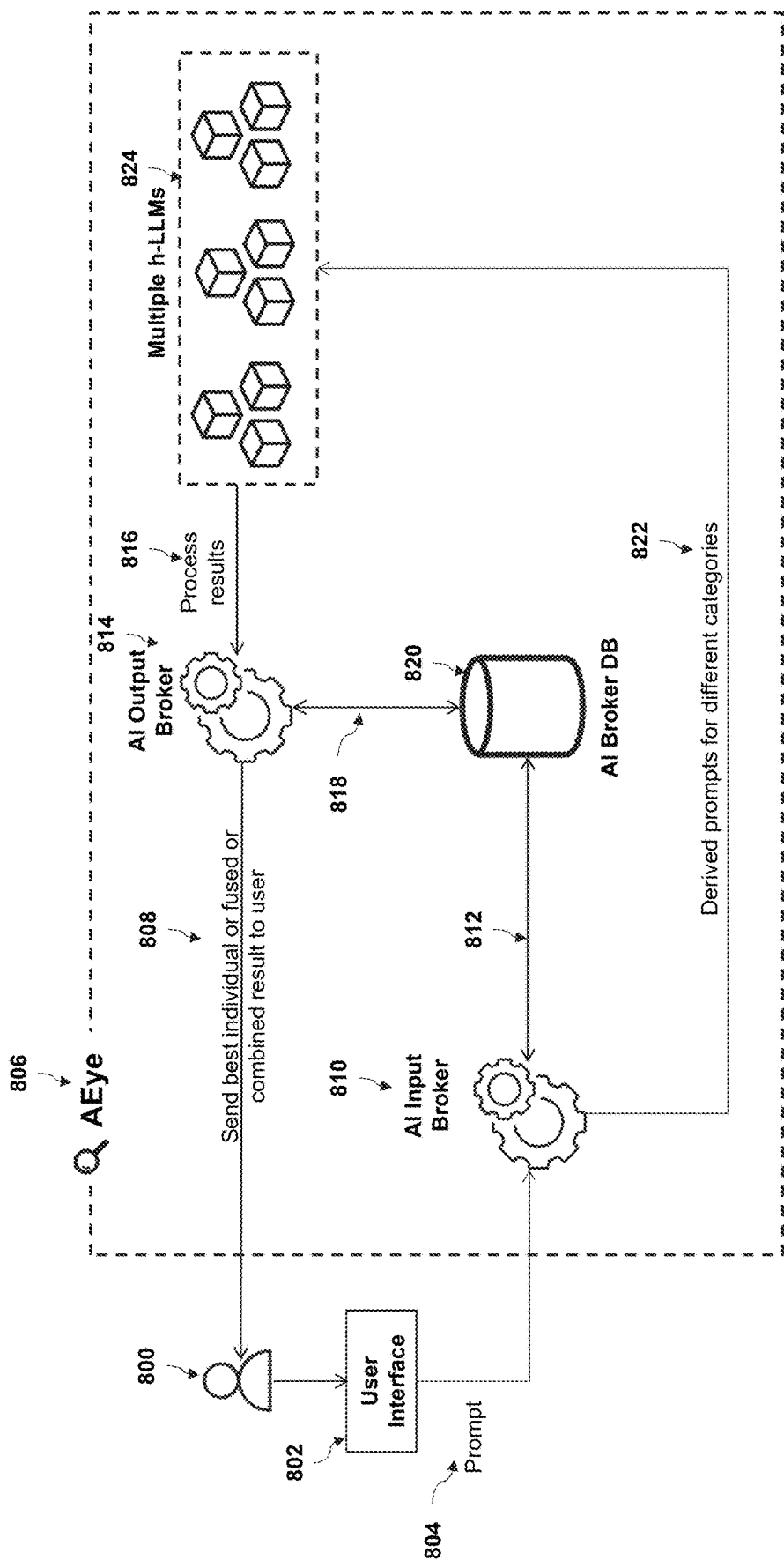


FIG. 9

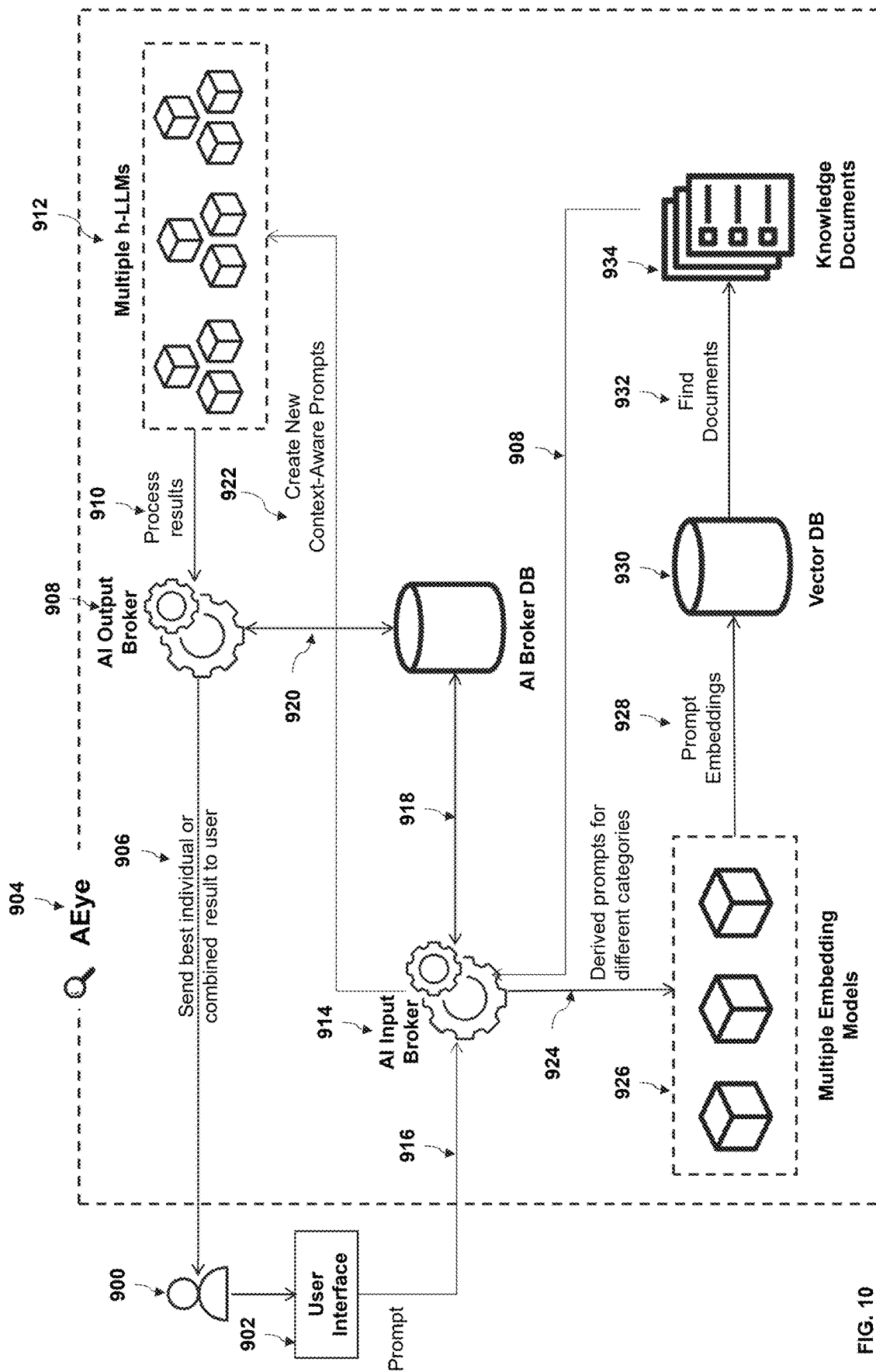


FIG. 10

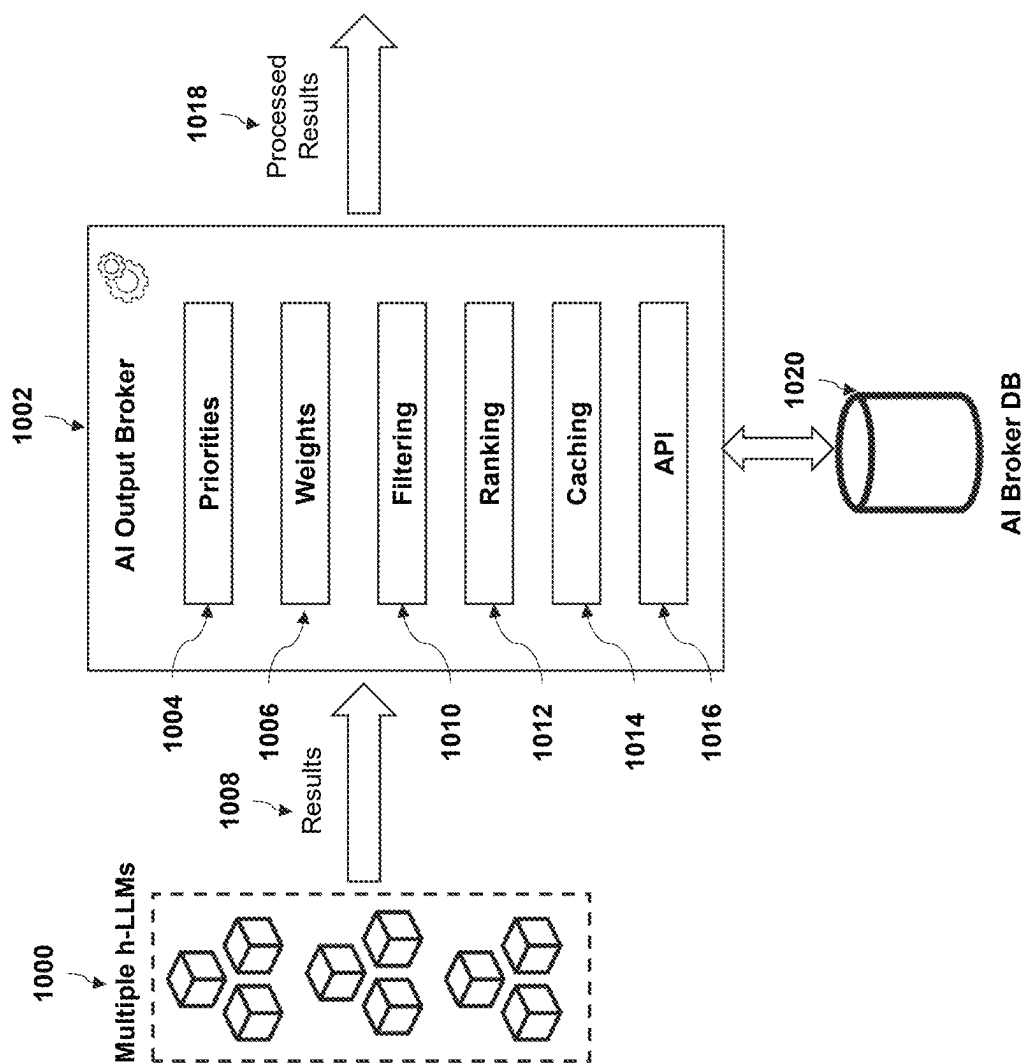


FIG. 11

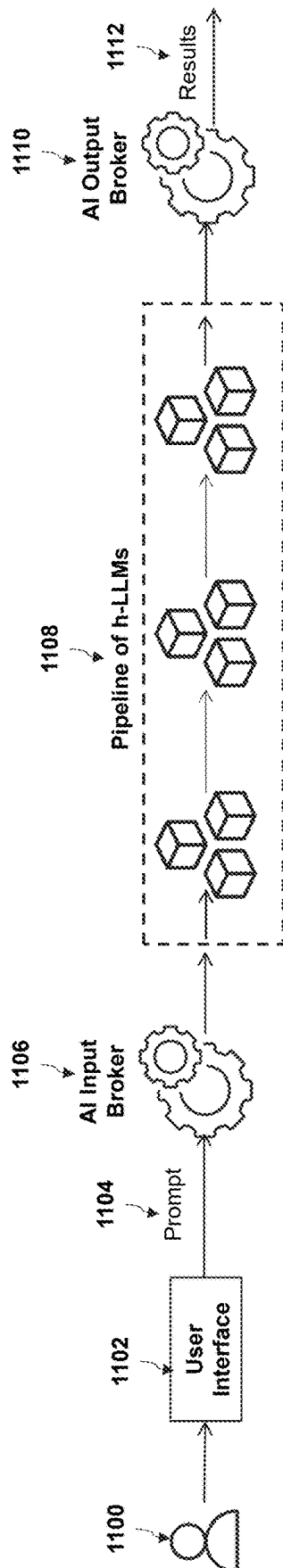


FIG. 12

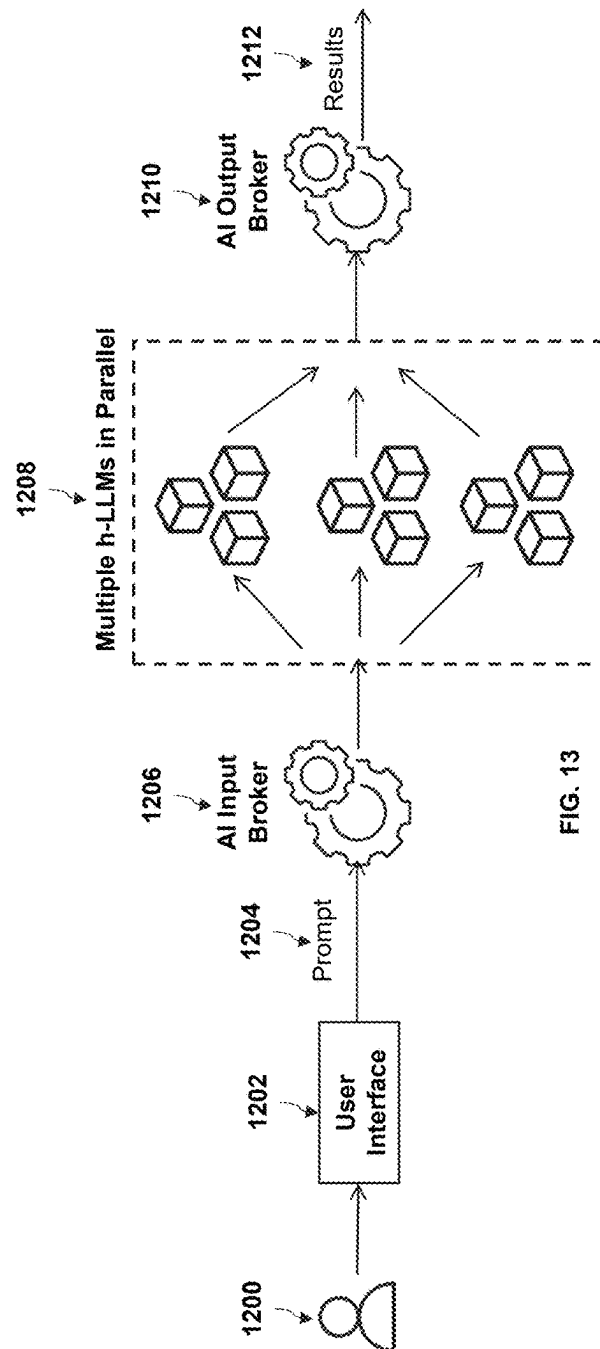


FIG. 13

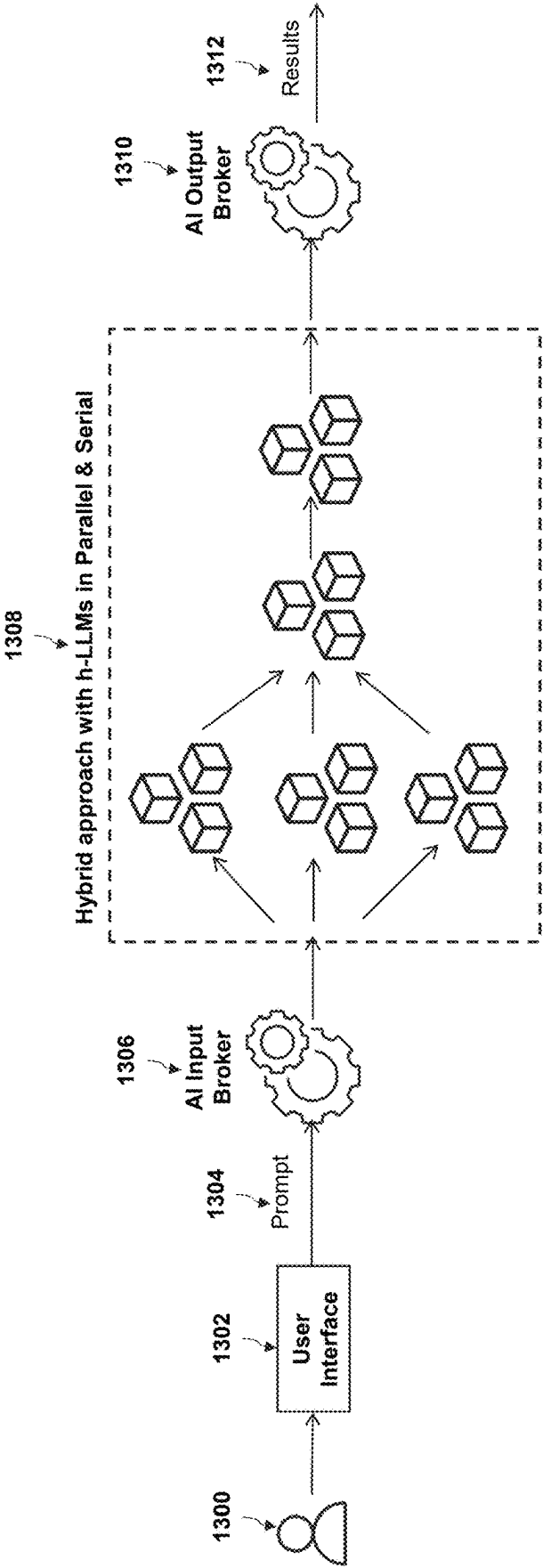


FIG. 14

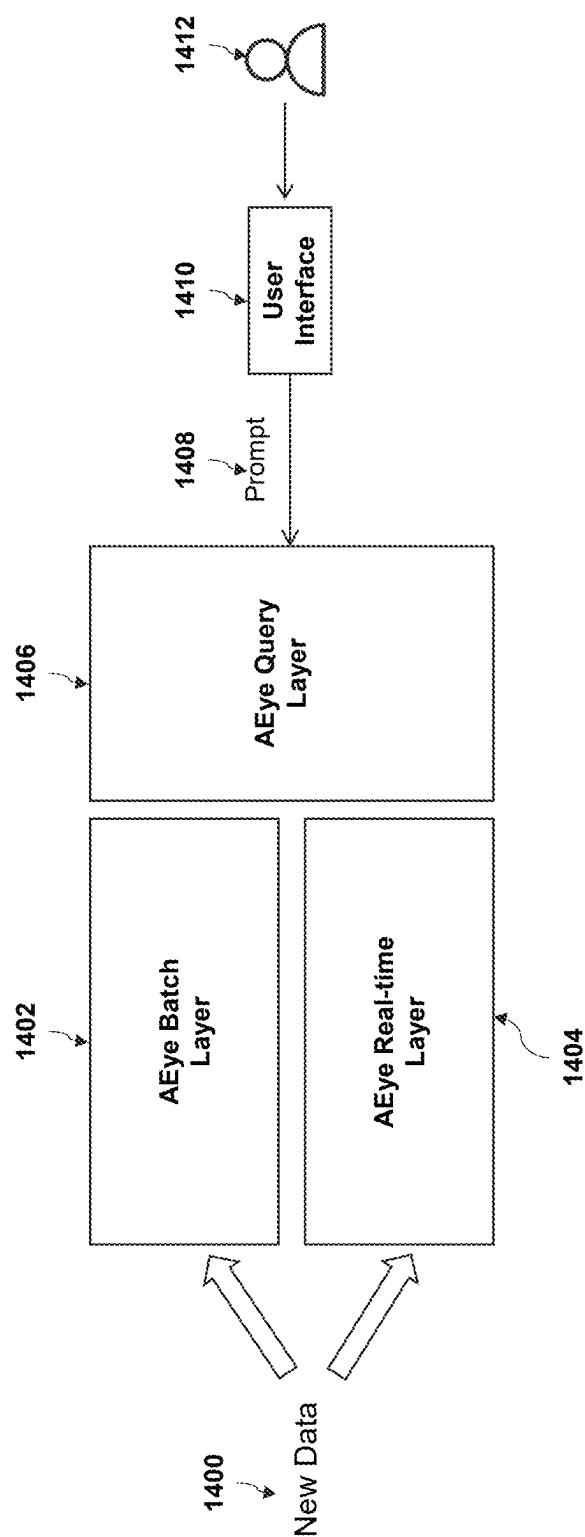


FIG. 15

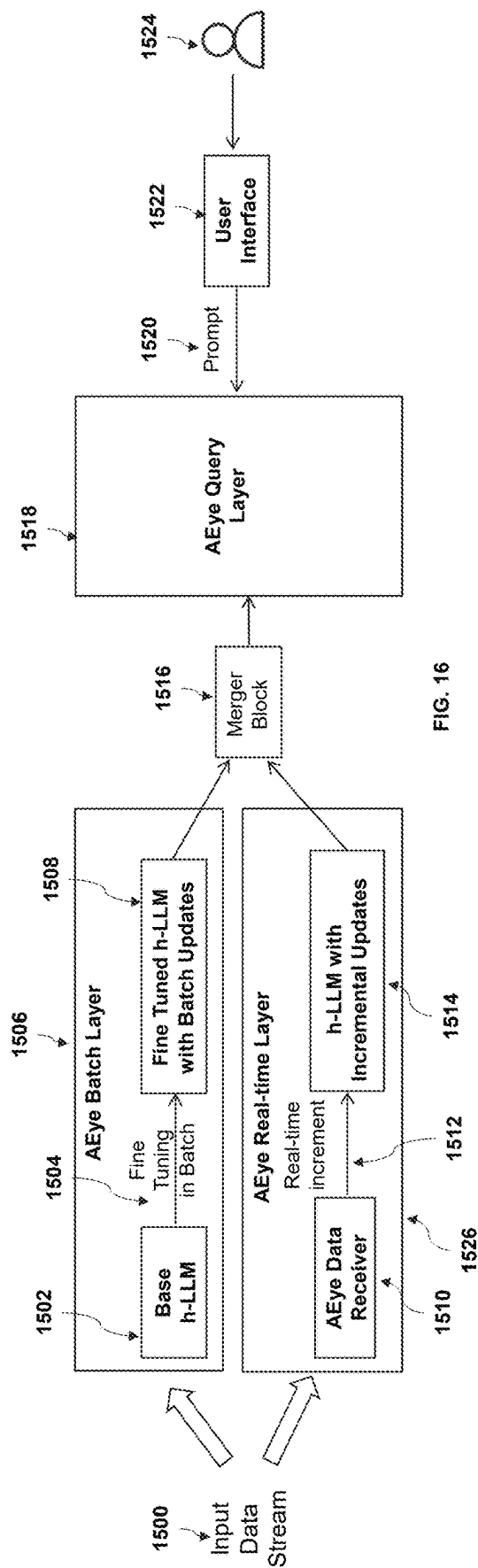


FIG. 16

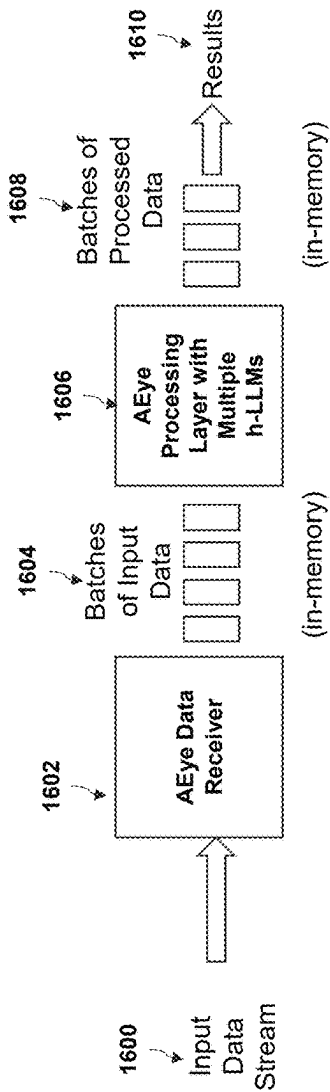


FIG. 17

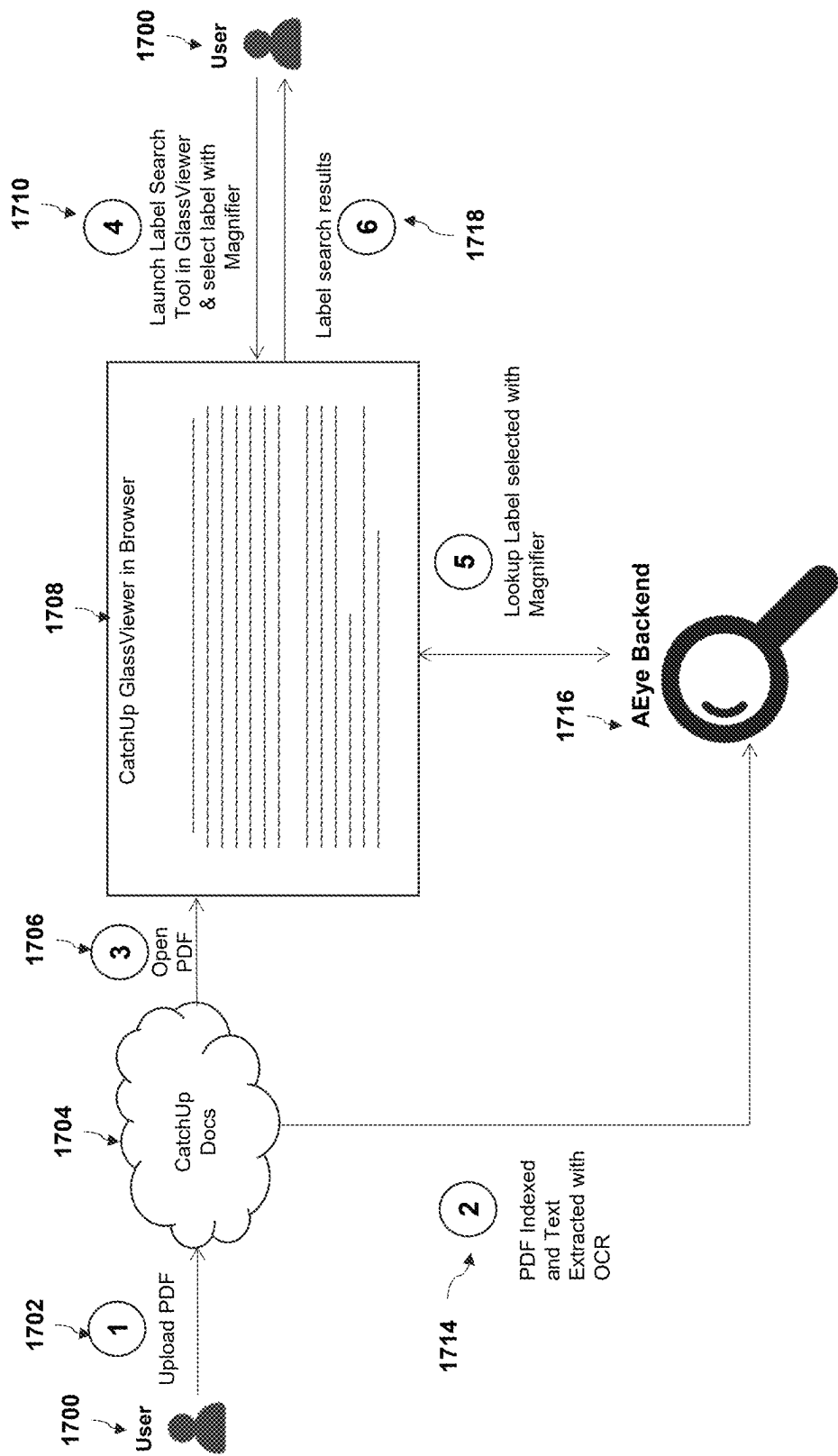


FIG. 18

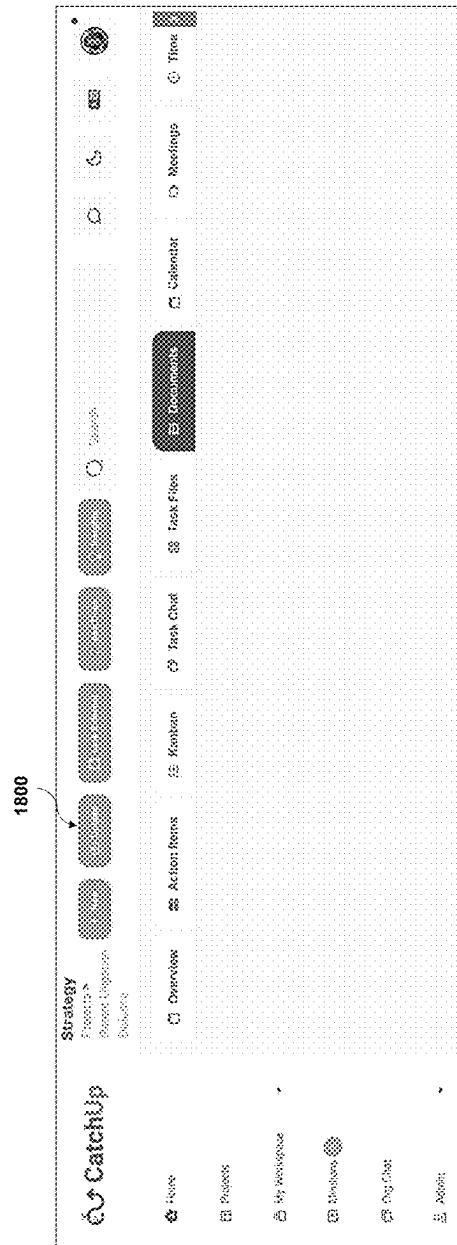
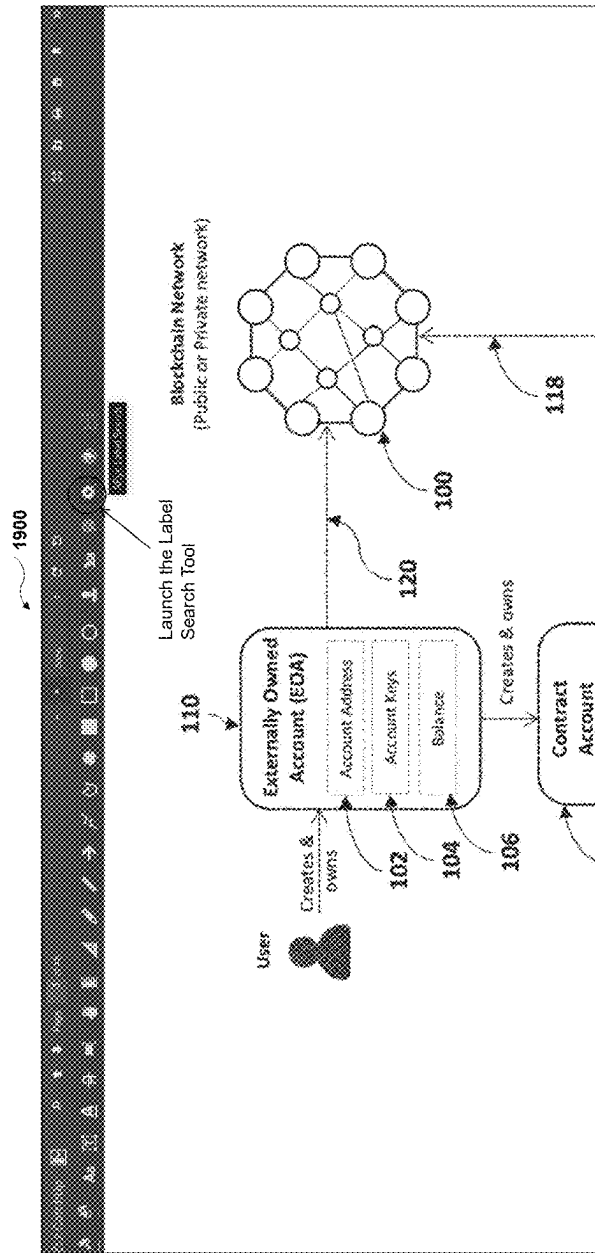
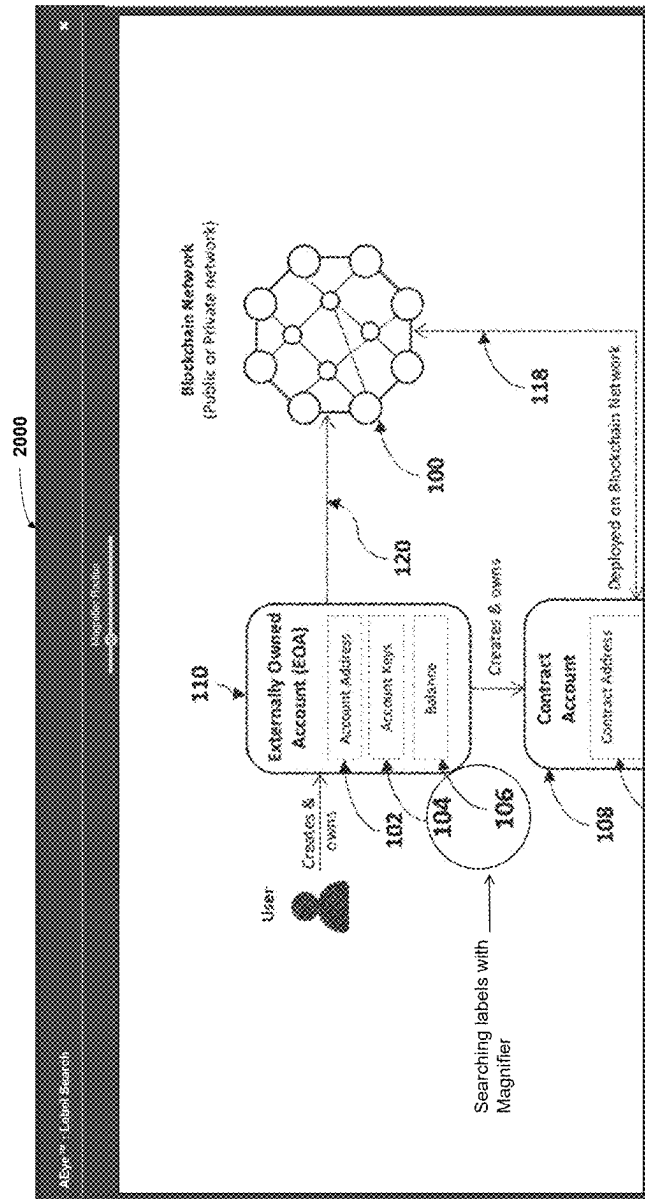


FIG. 19





2100

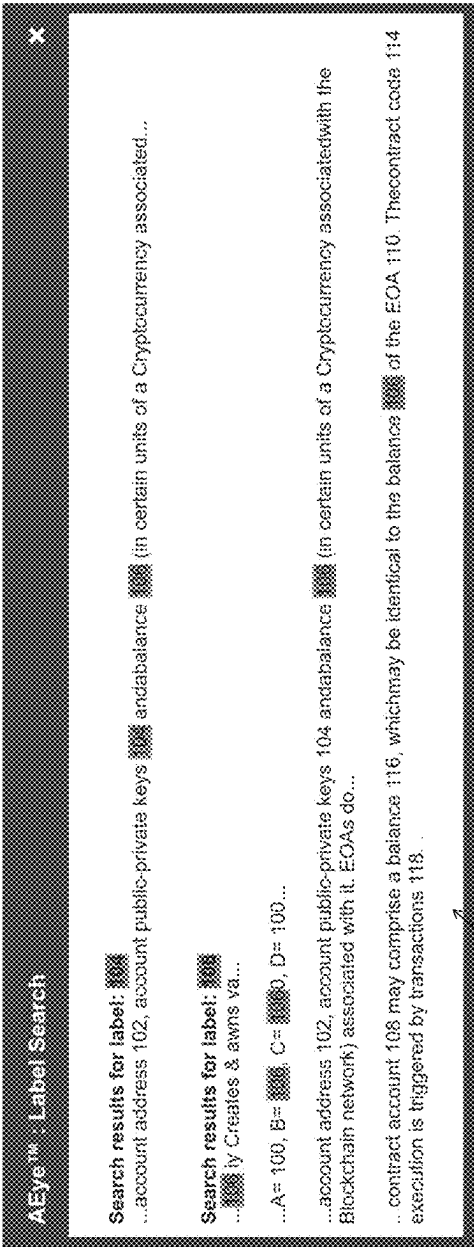


FIG. 22

Search results for selected labels from text specification in database. Alternatively generated results may also be displayed from additional h-LLMs drawn from other data and model sources.

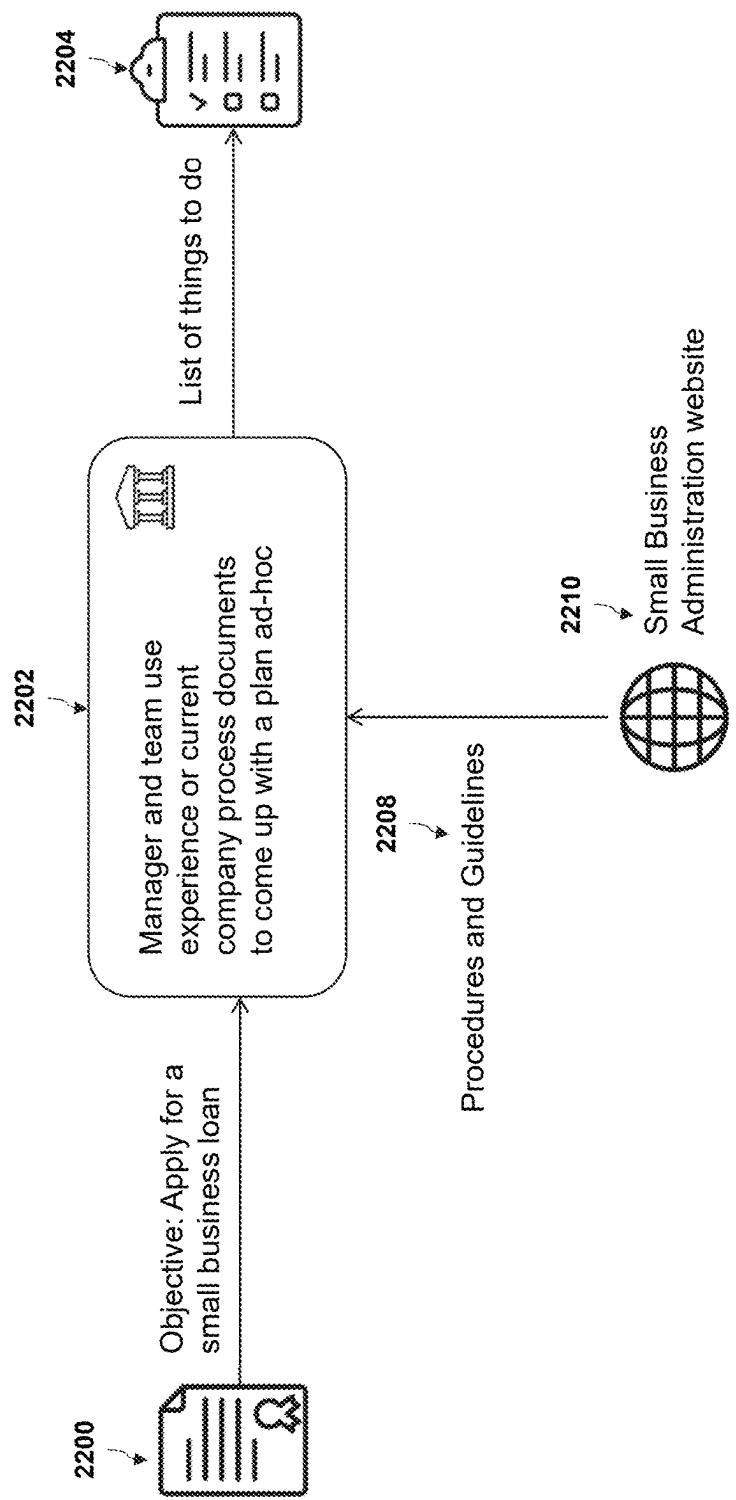


FIG. 23

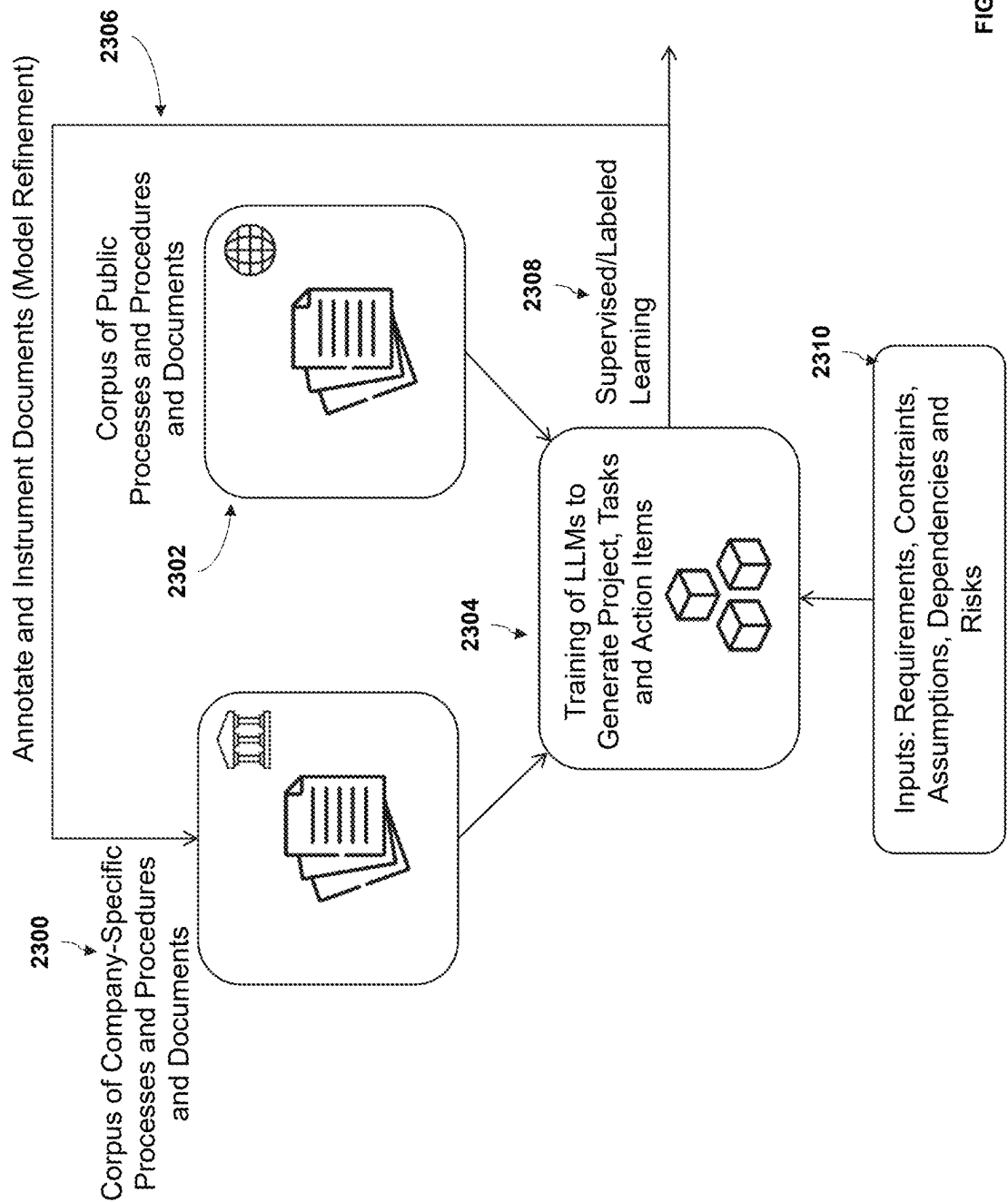


FIG. 24

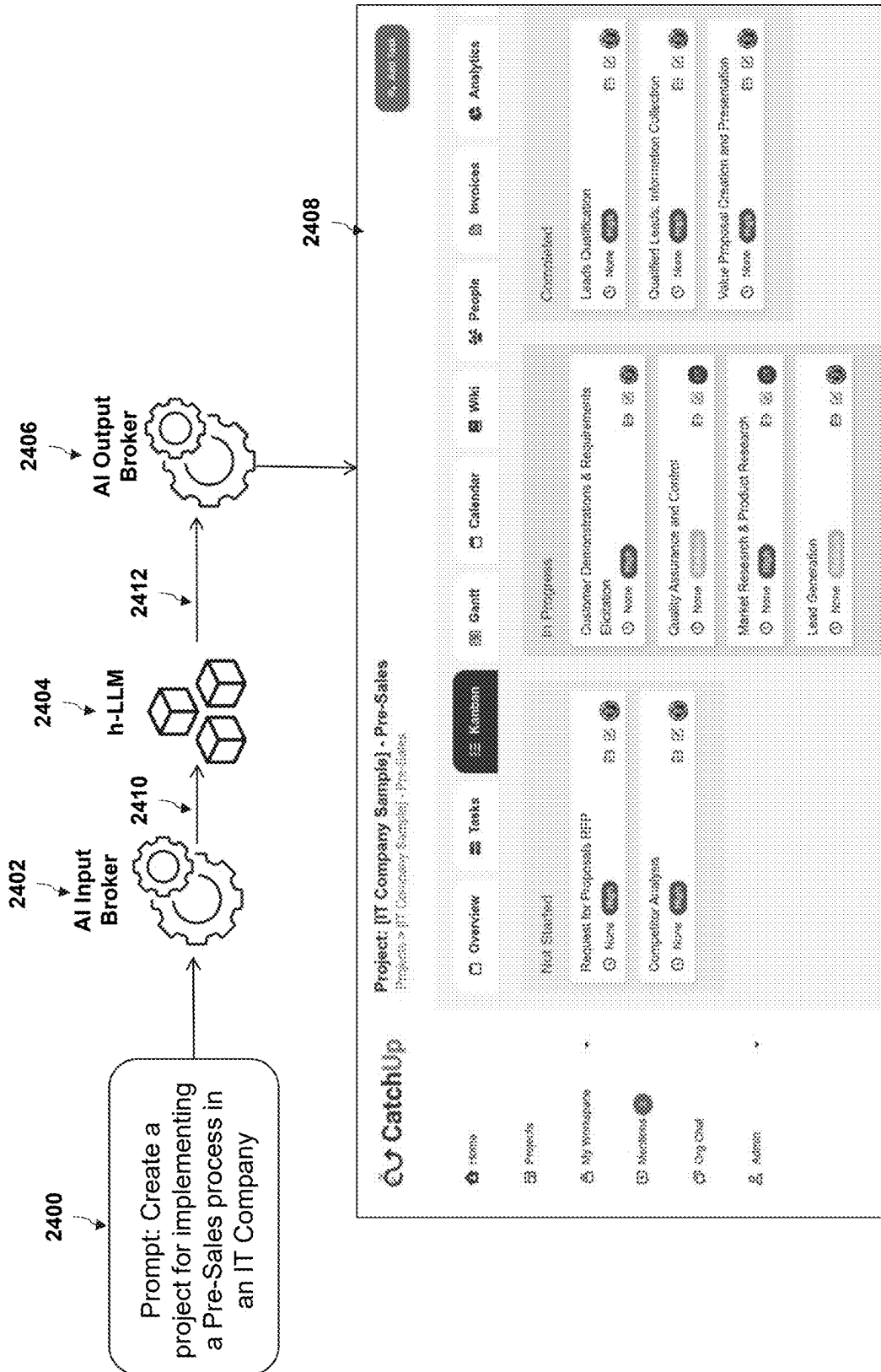


FIG. 25

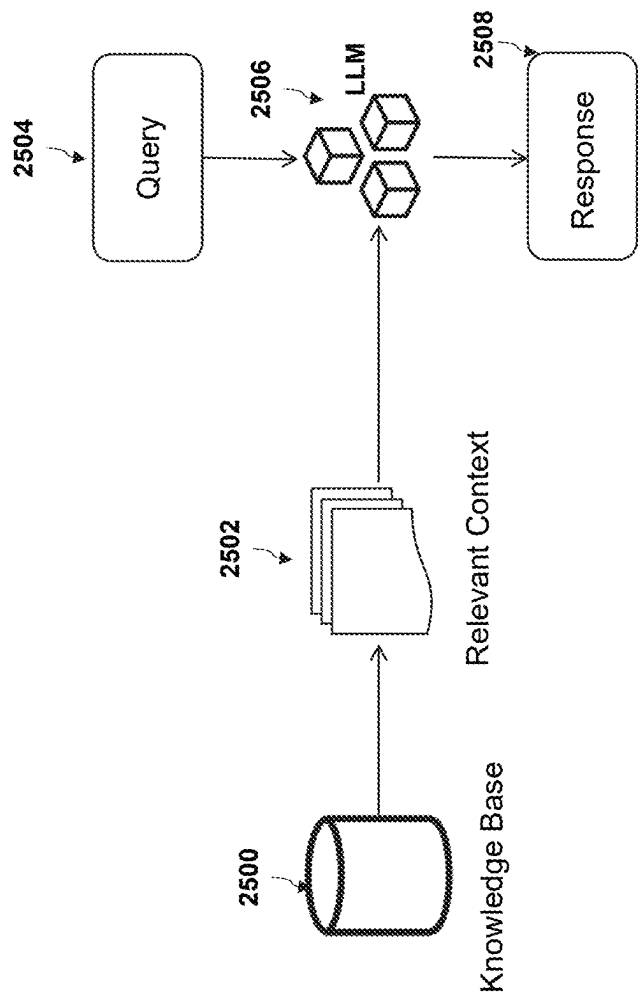


FIG. 26

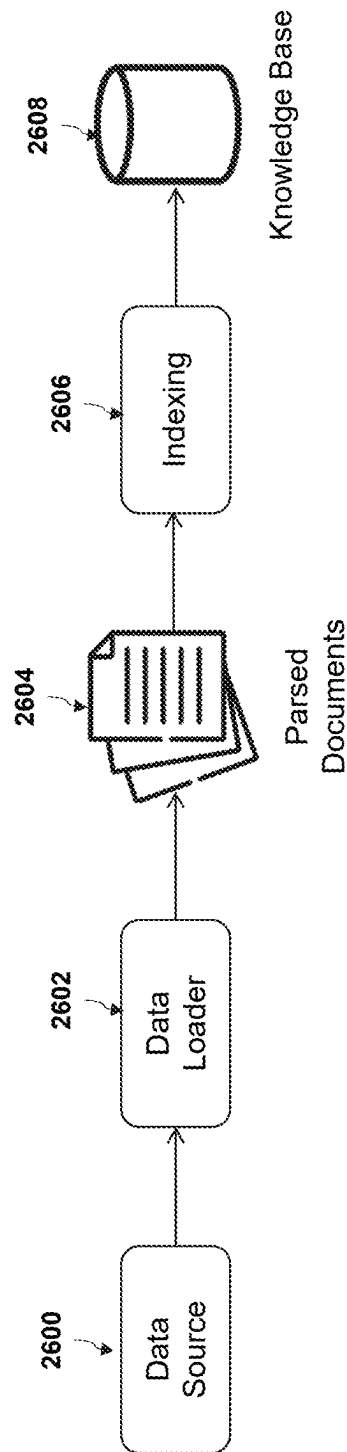


FIG. 27

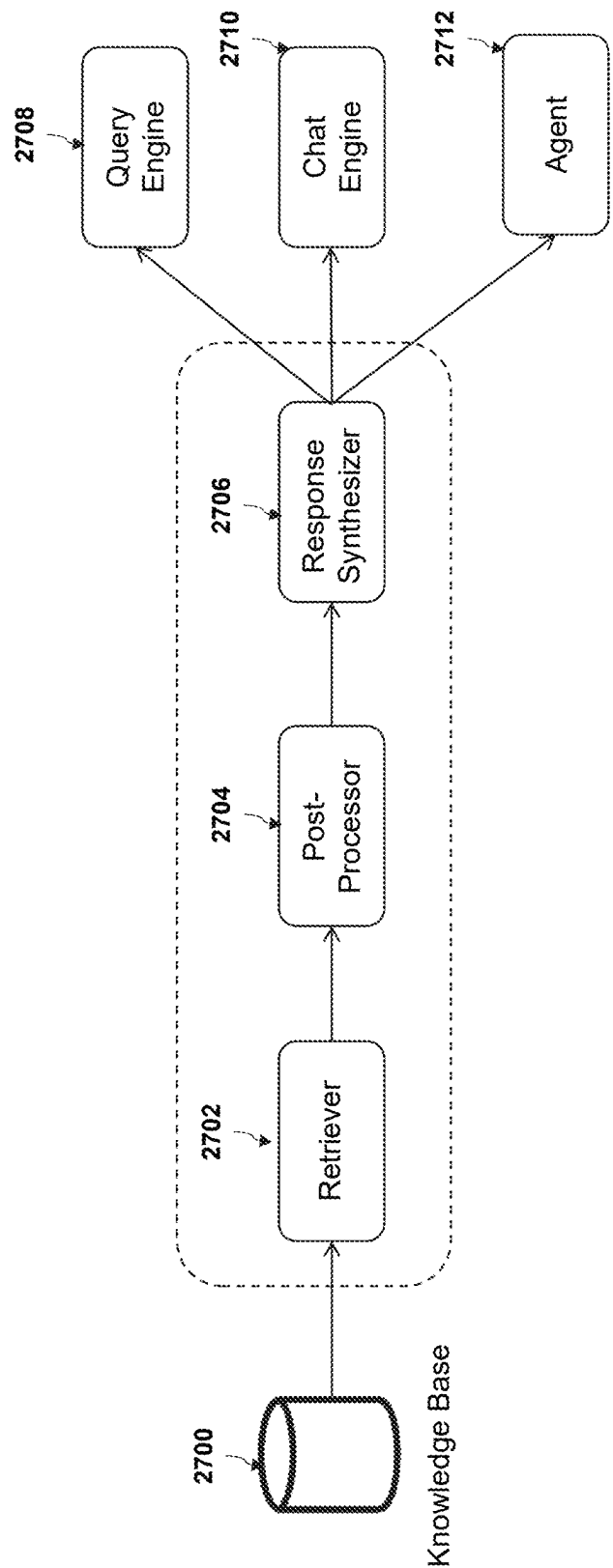


FIG. 28

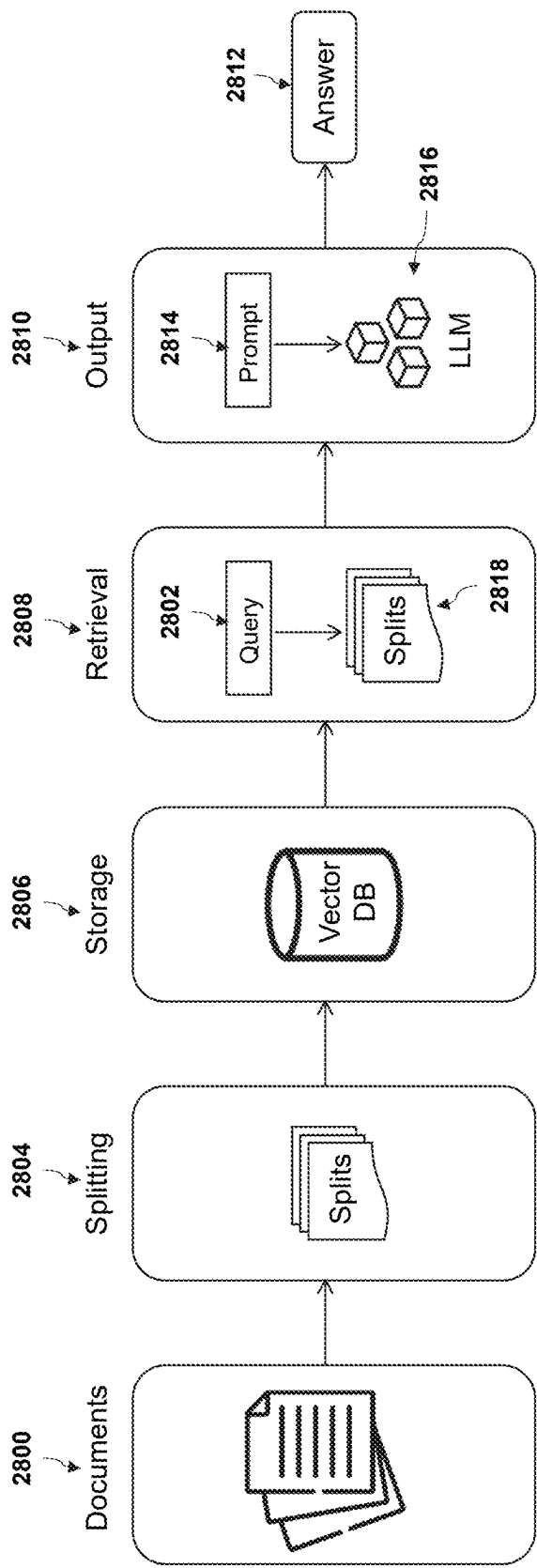


FIG. 29

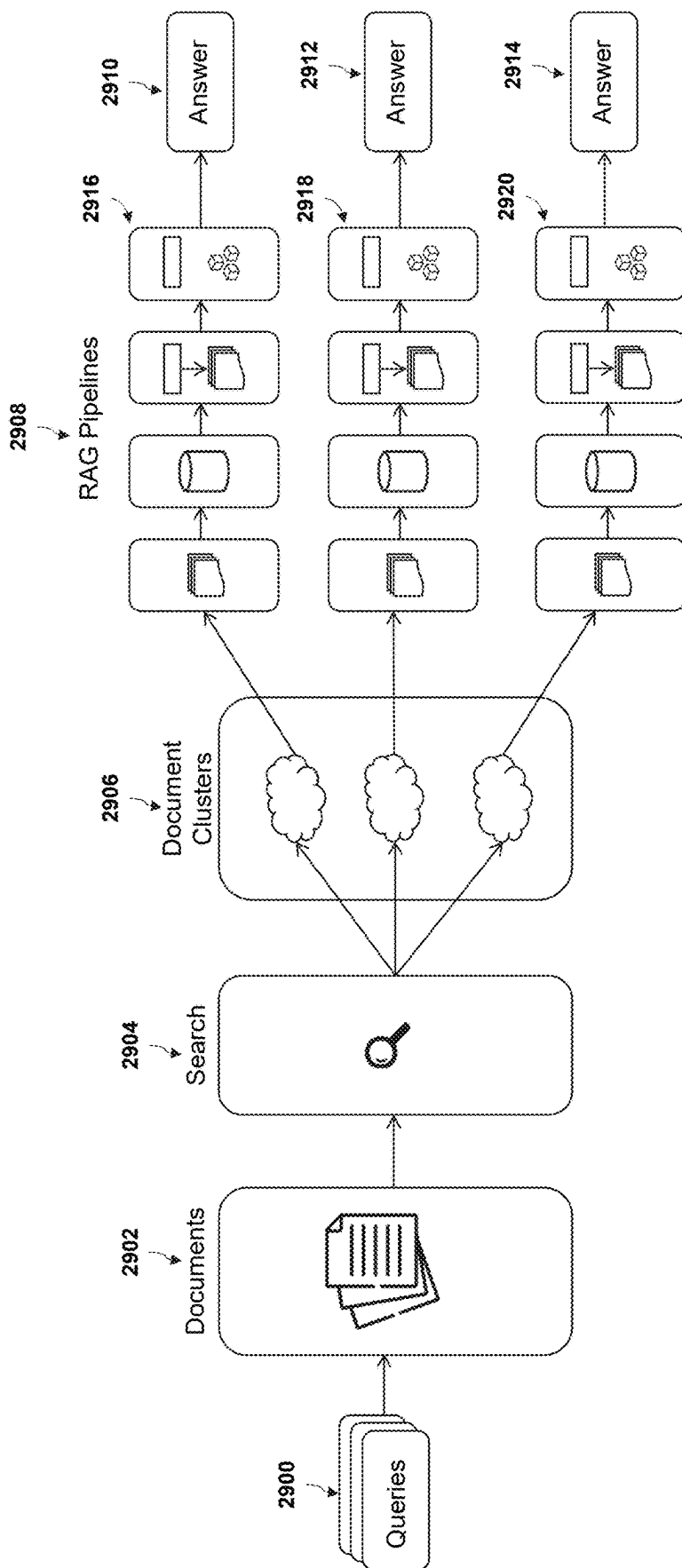


FIG. 30

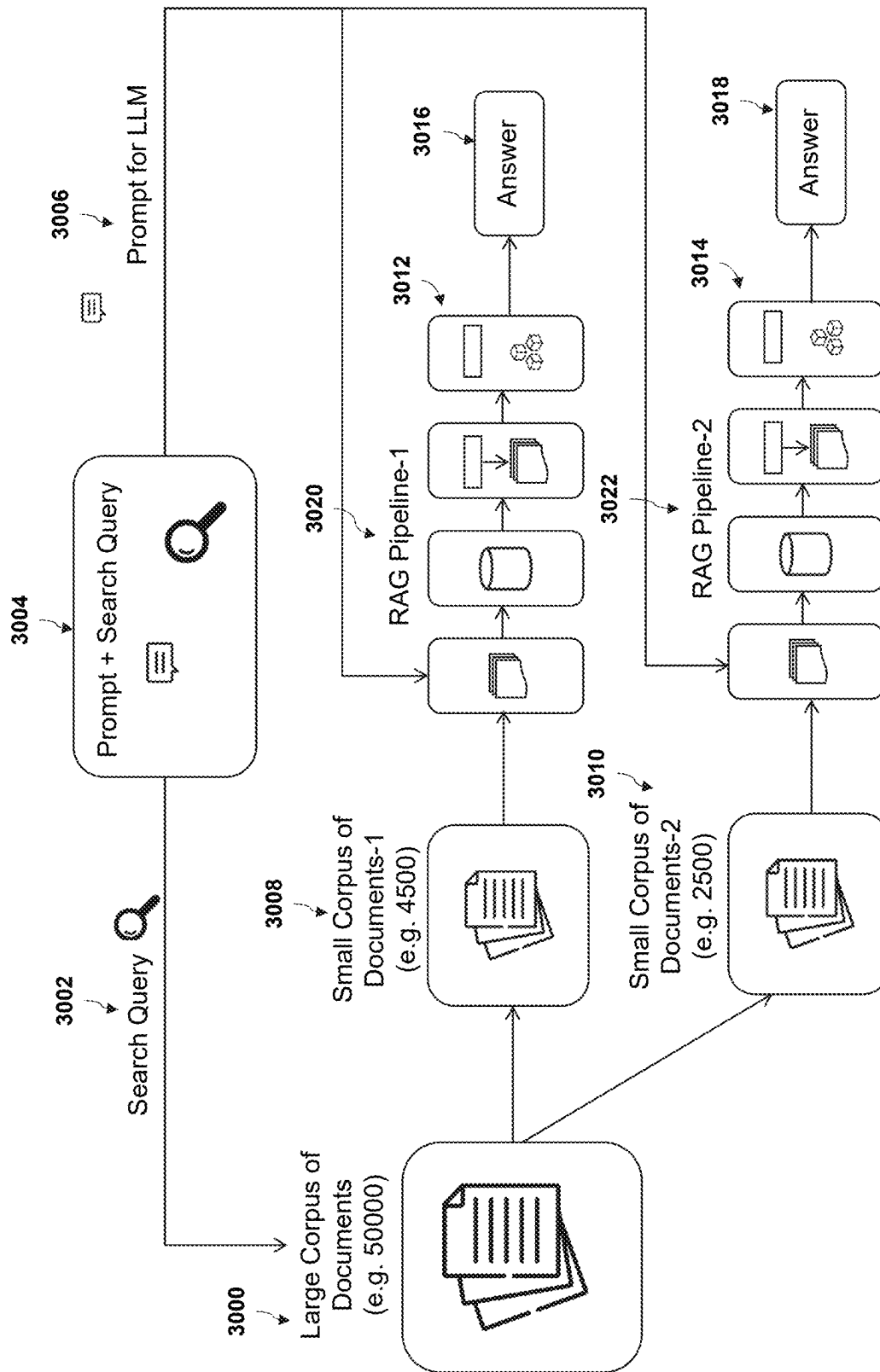


FIG. 31

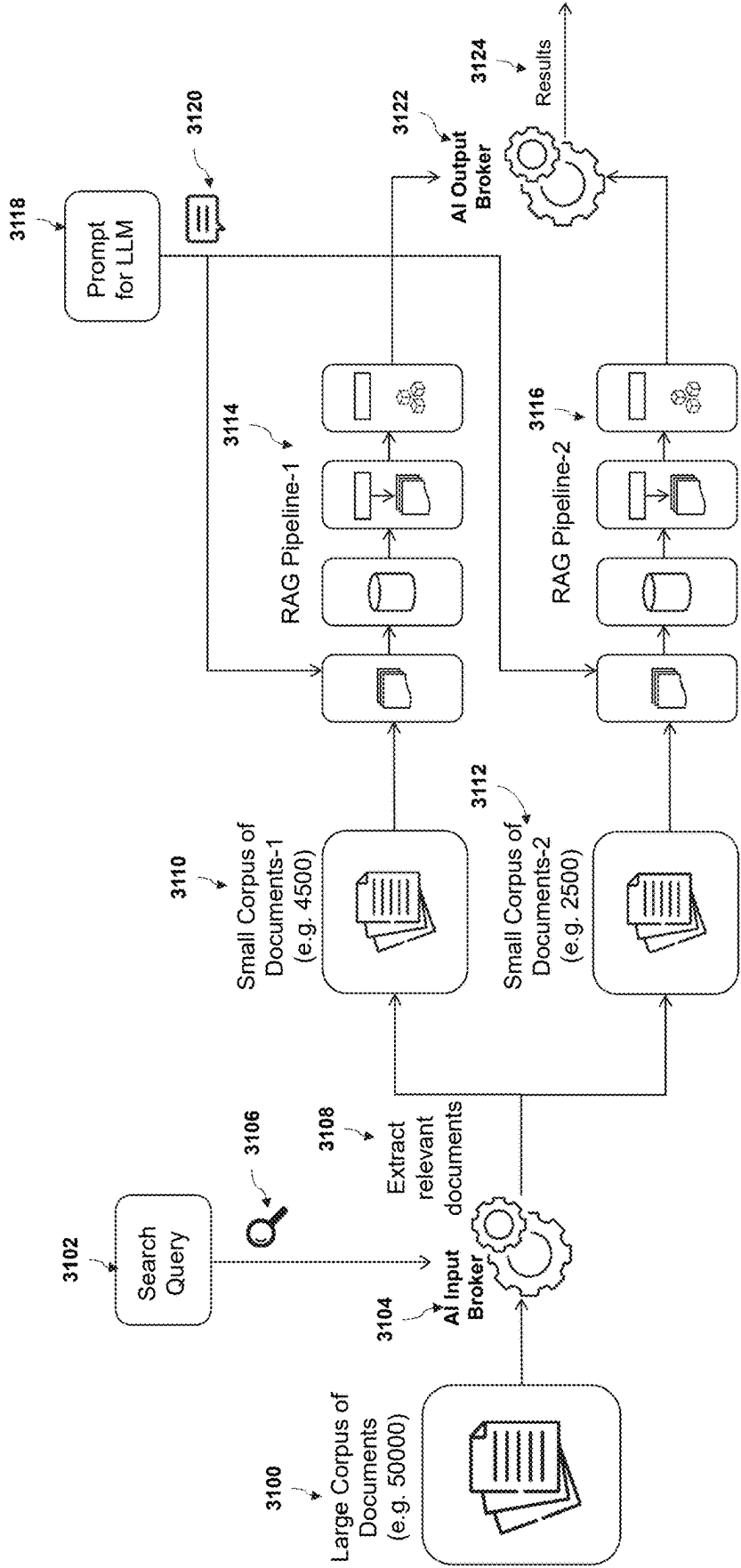


FIG. 32

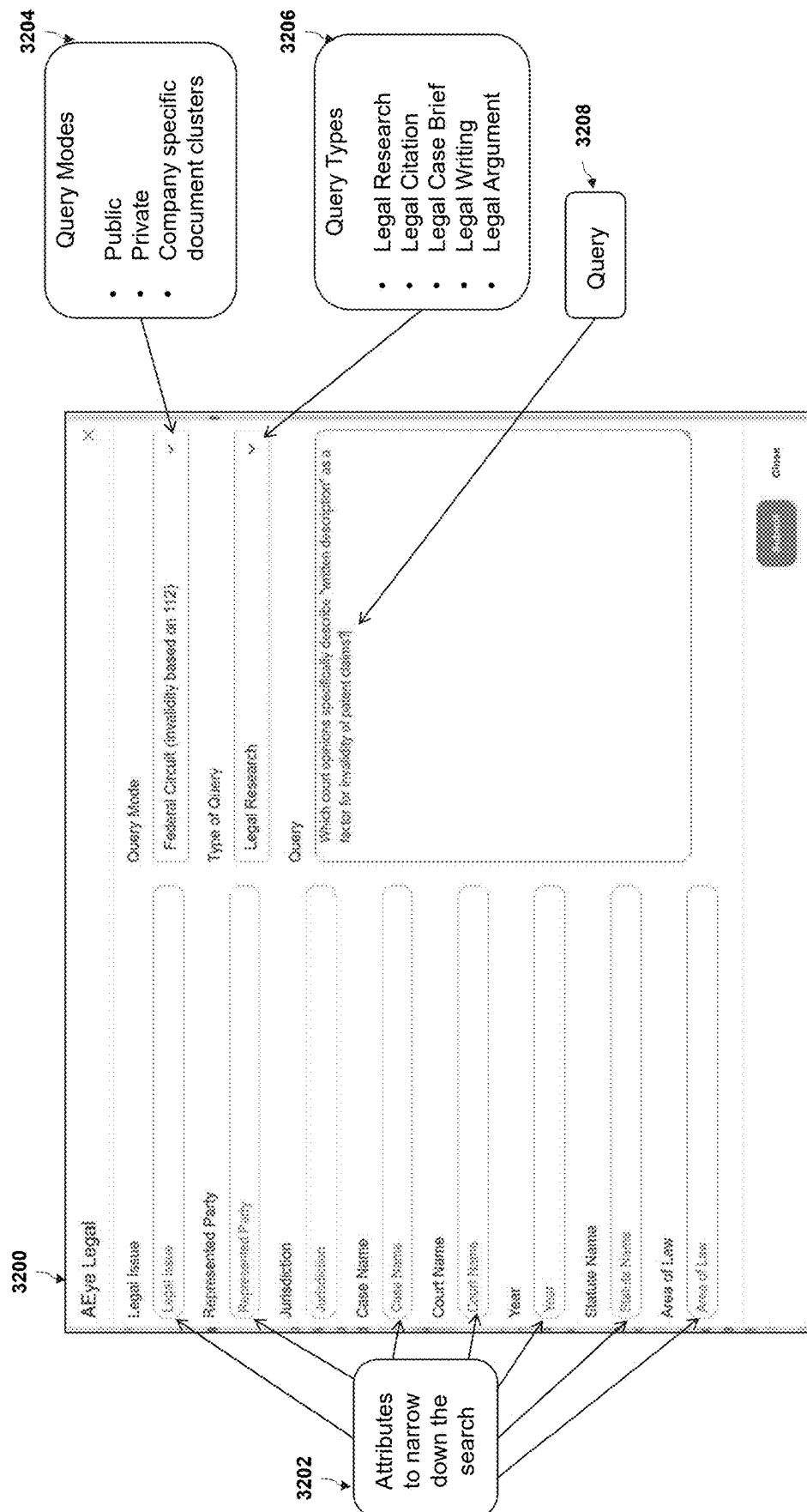
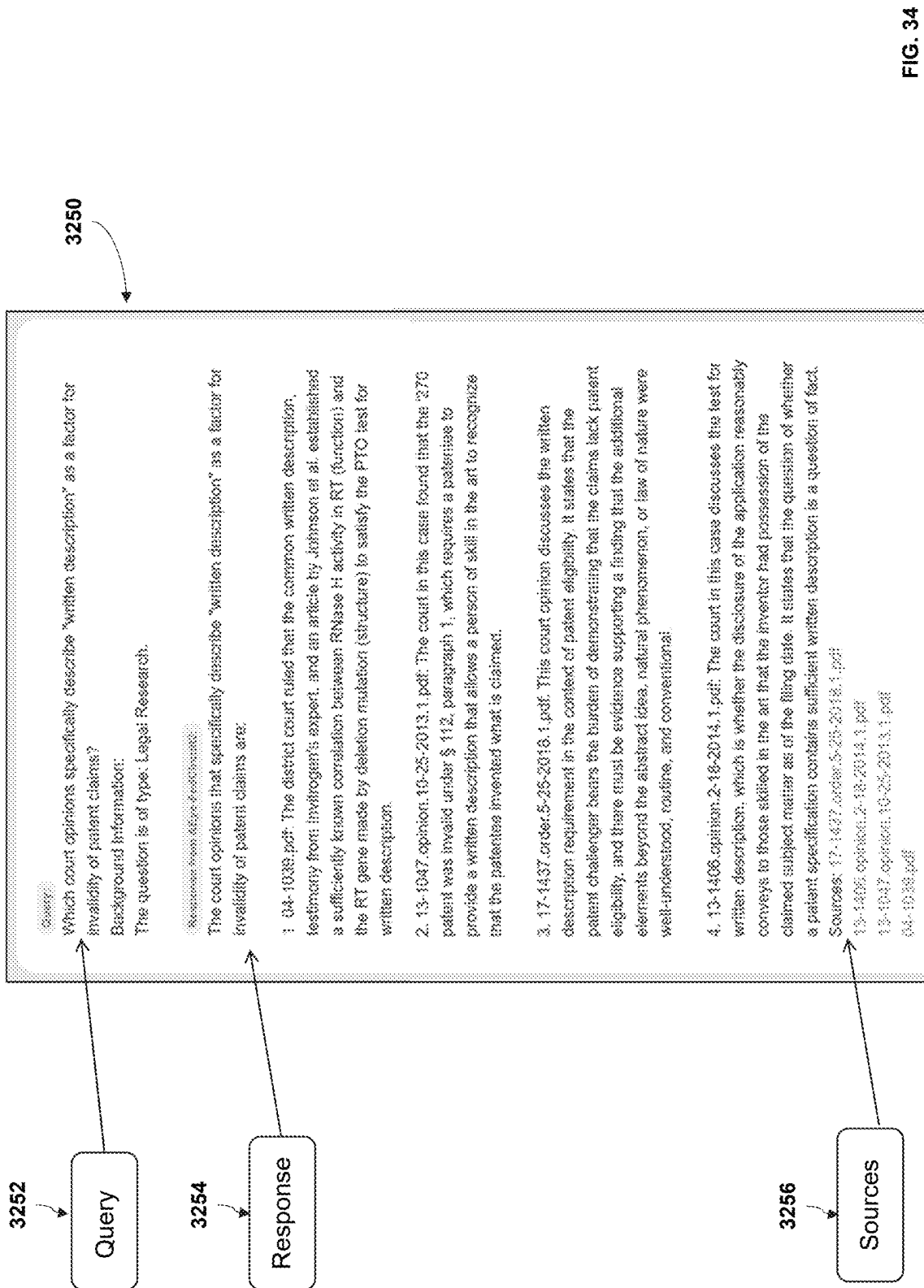
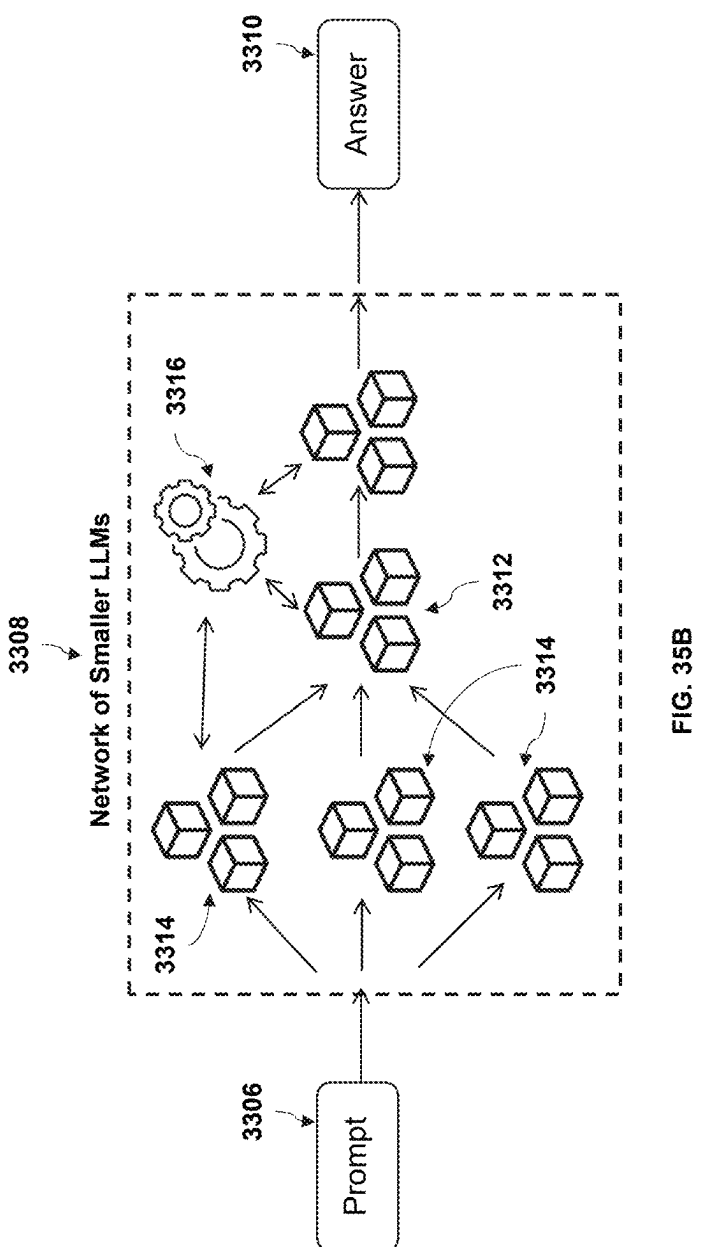
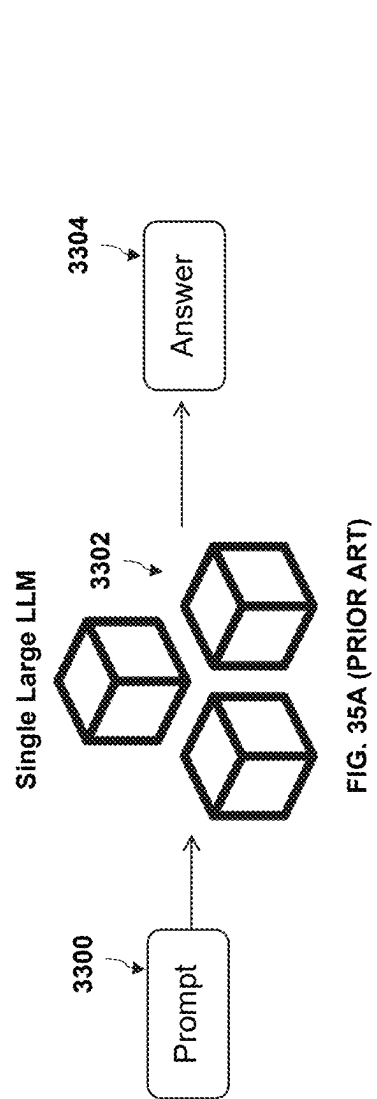


FIG. 33





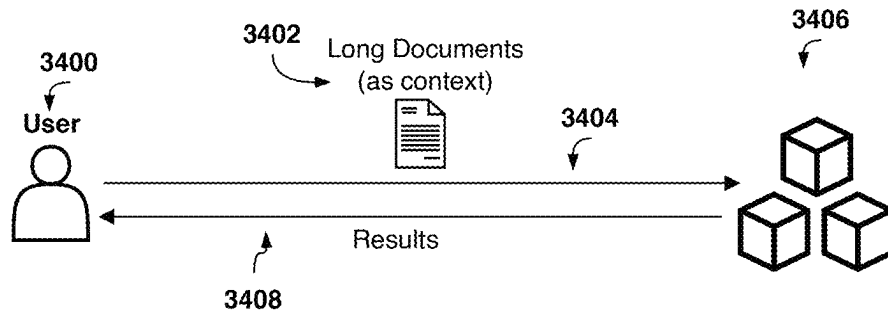


FIG. 36A (PRIOR ART)

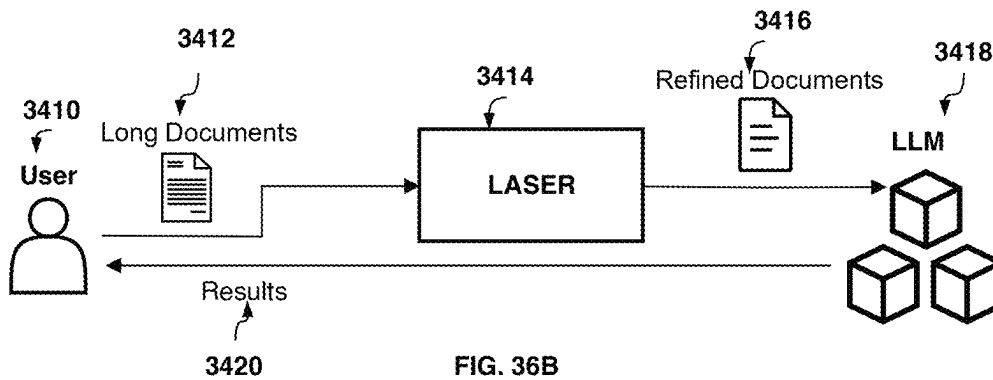


FIG. 36B

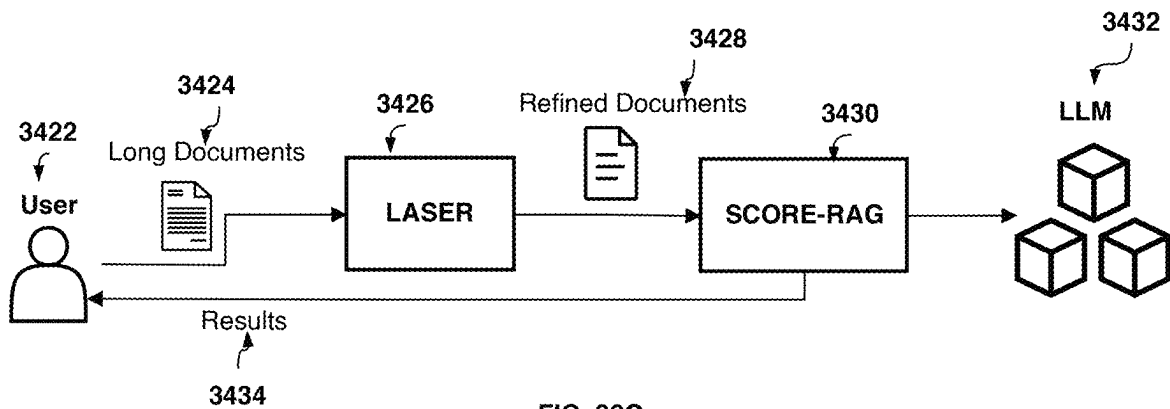


FIG. 36C

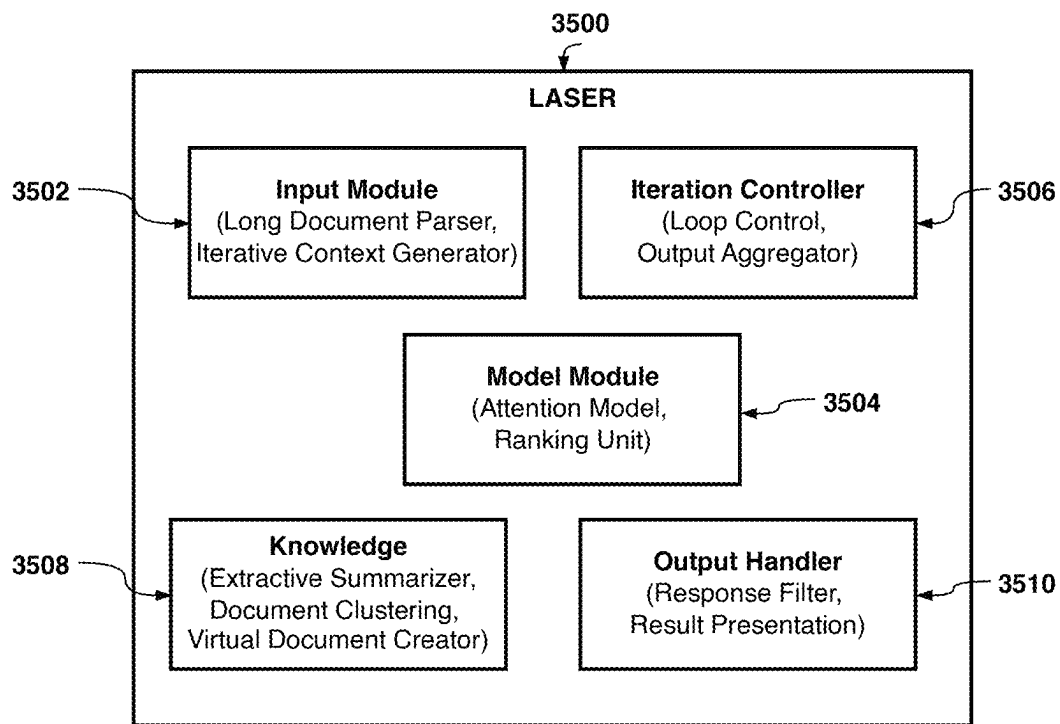


FIG. 37

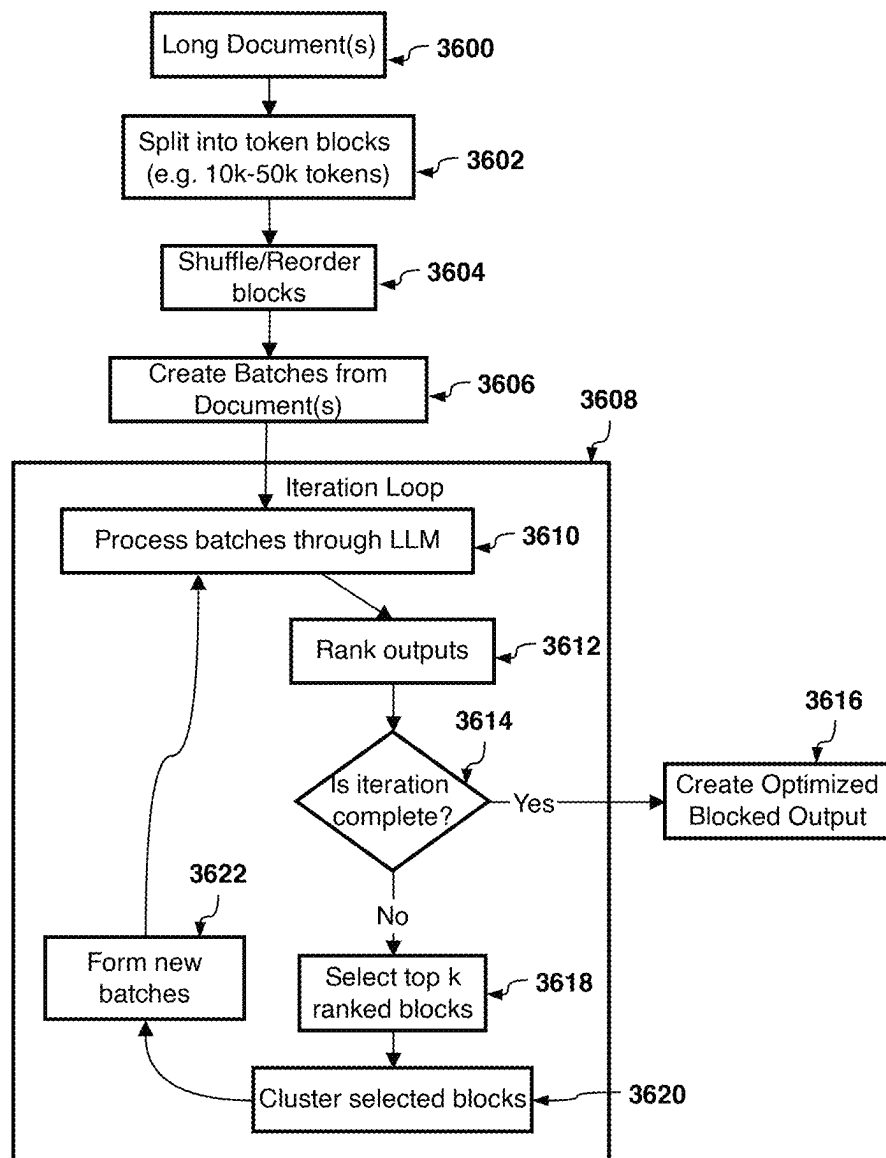


FIG. 38

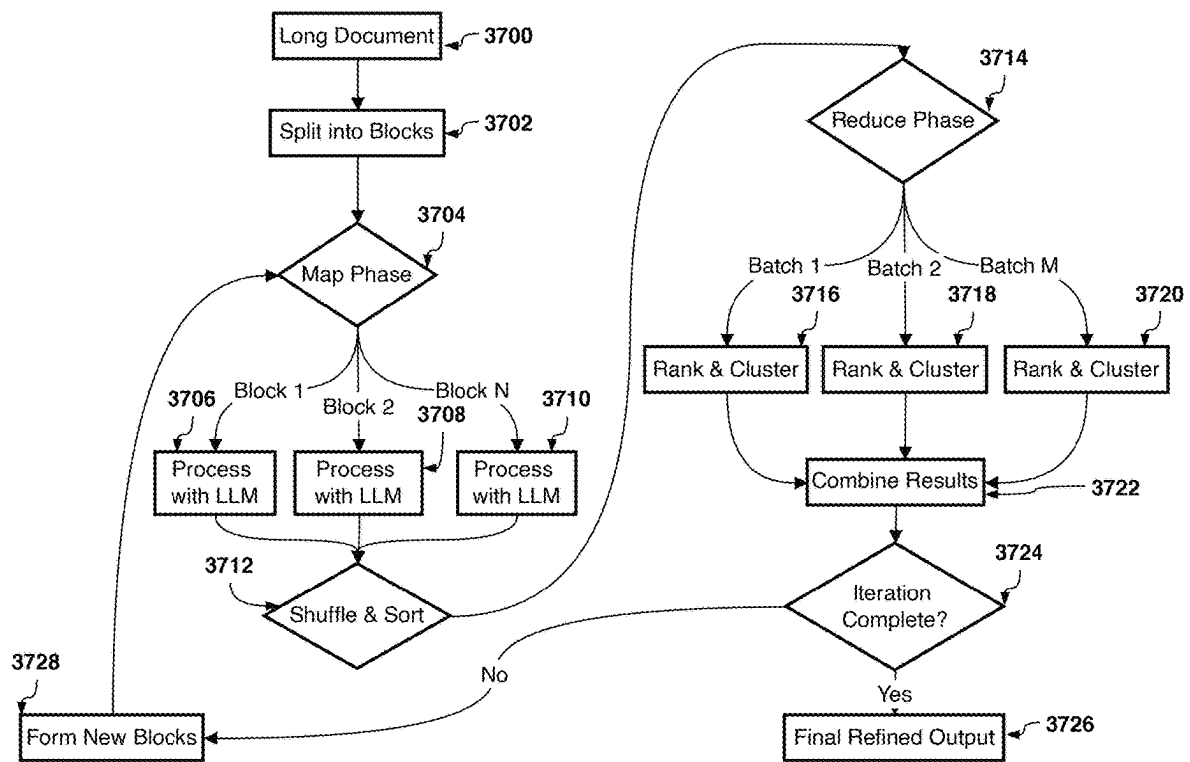


FIG. 39

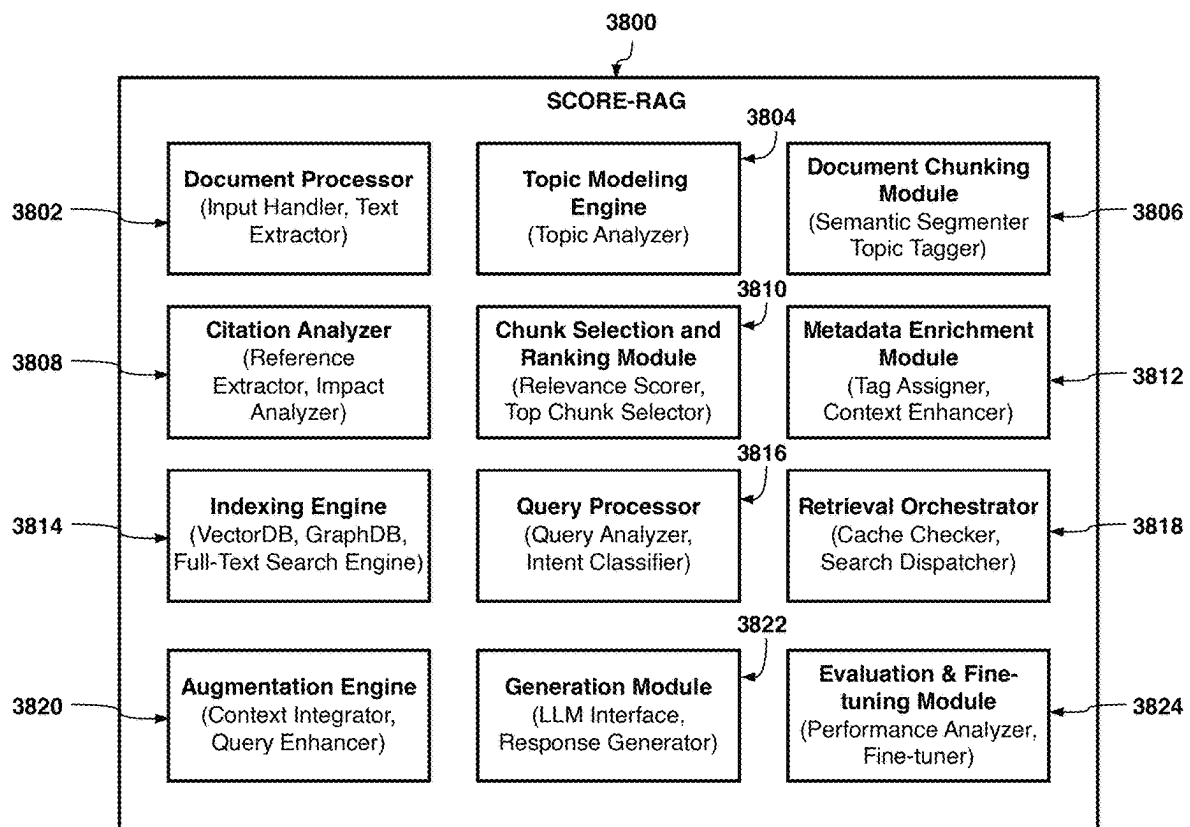


FIG. 40

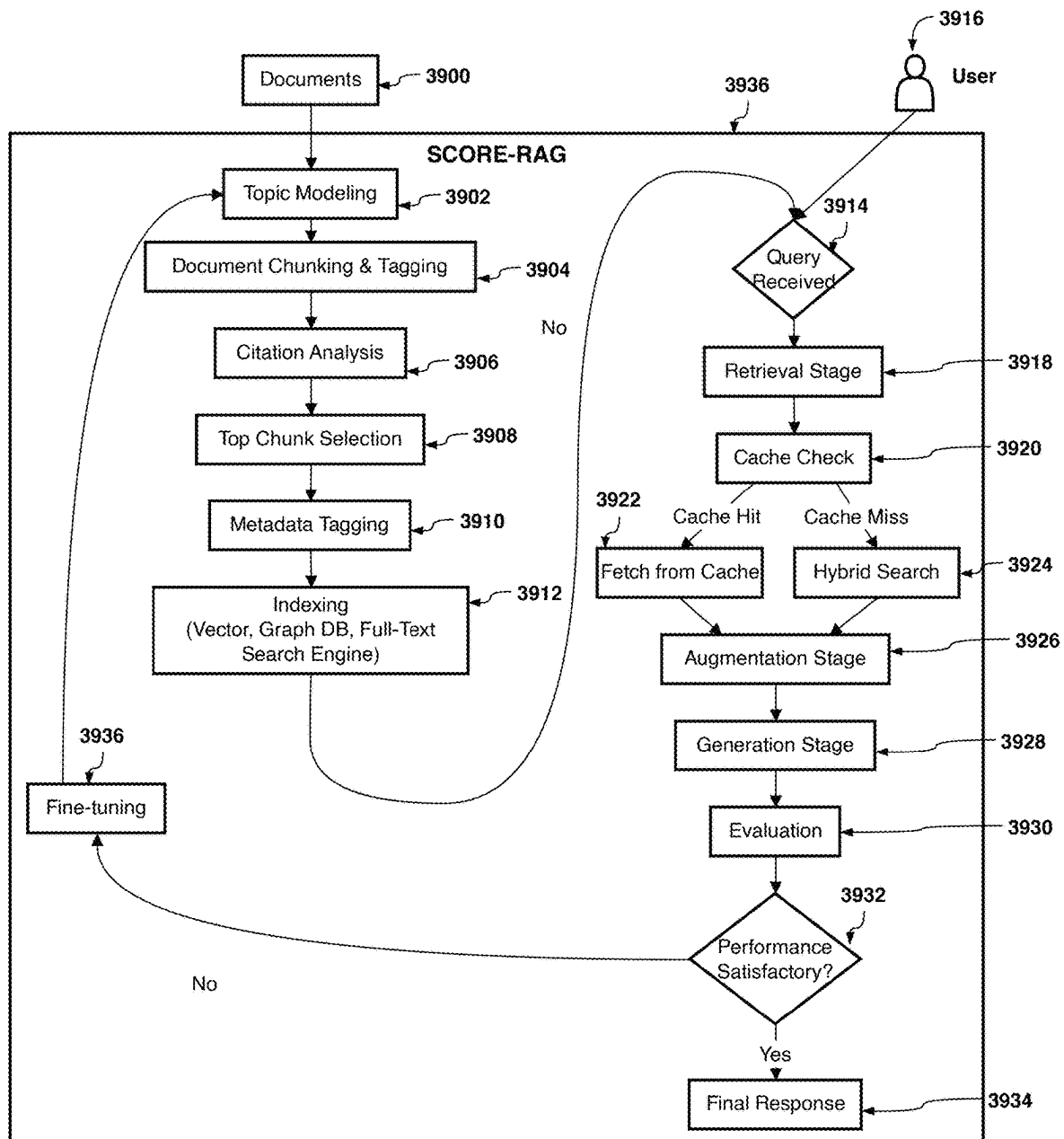


FIG. 41

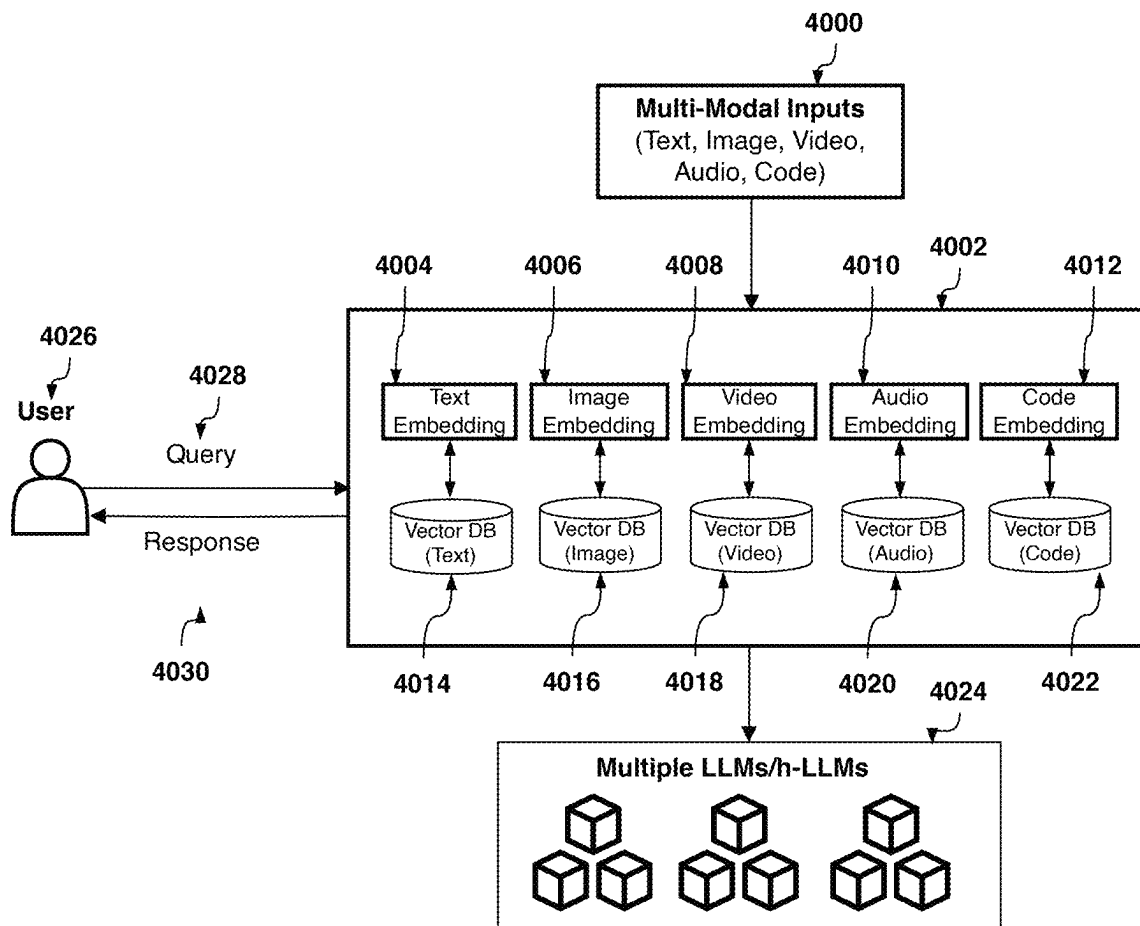


FIG. 42

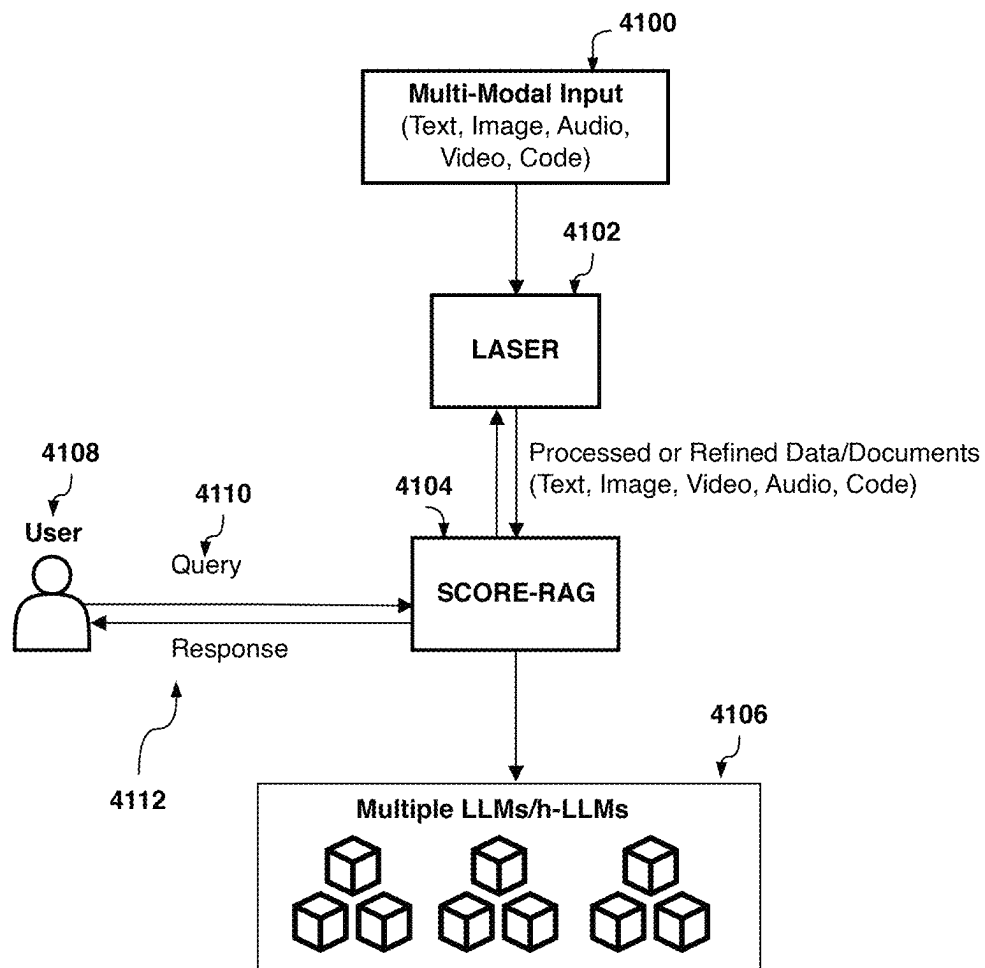


FIG. 43

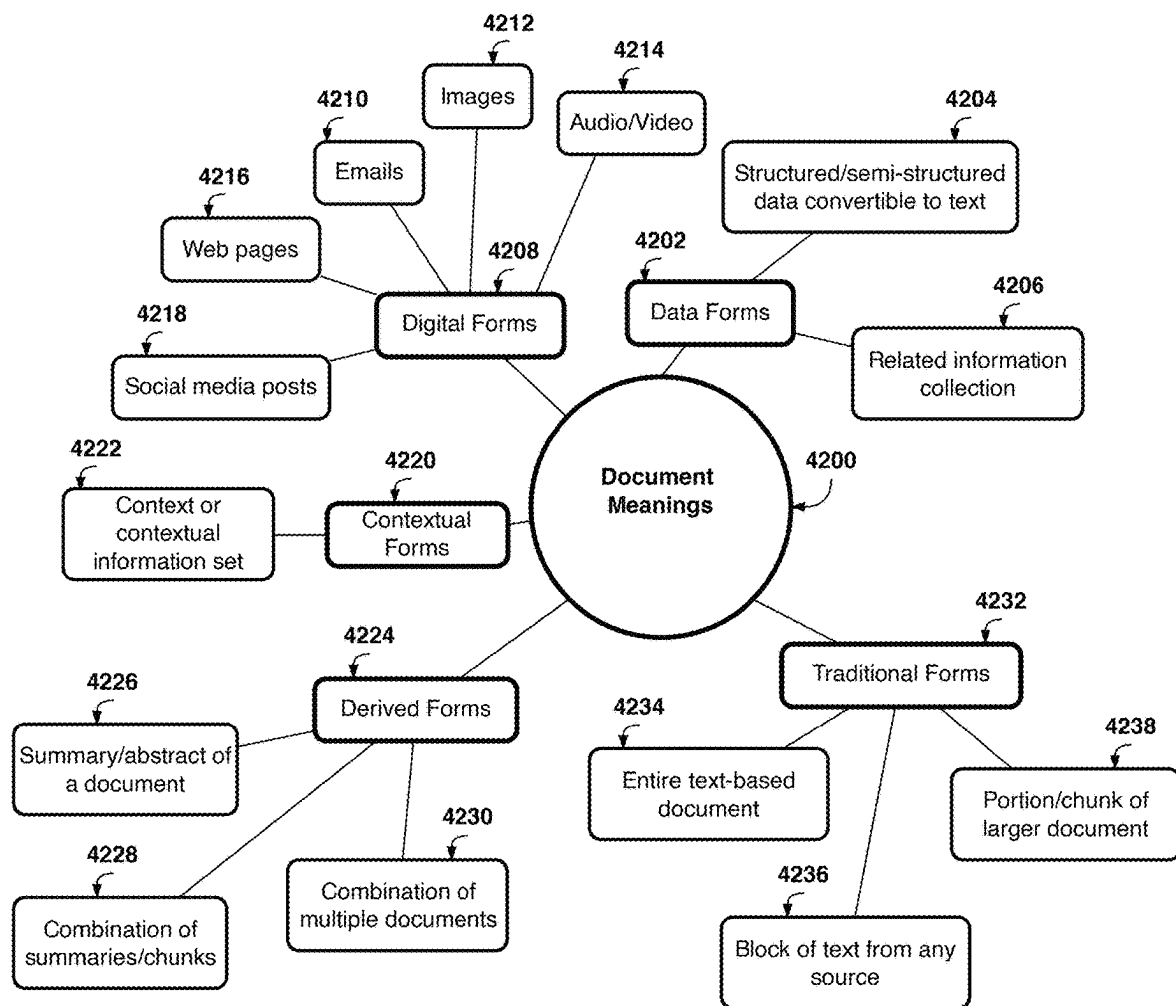


FIG. 44

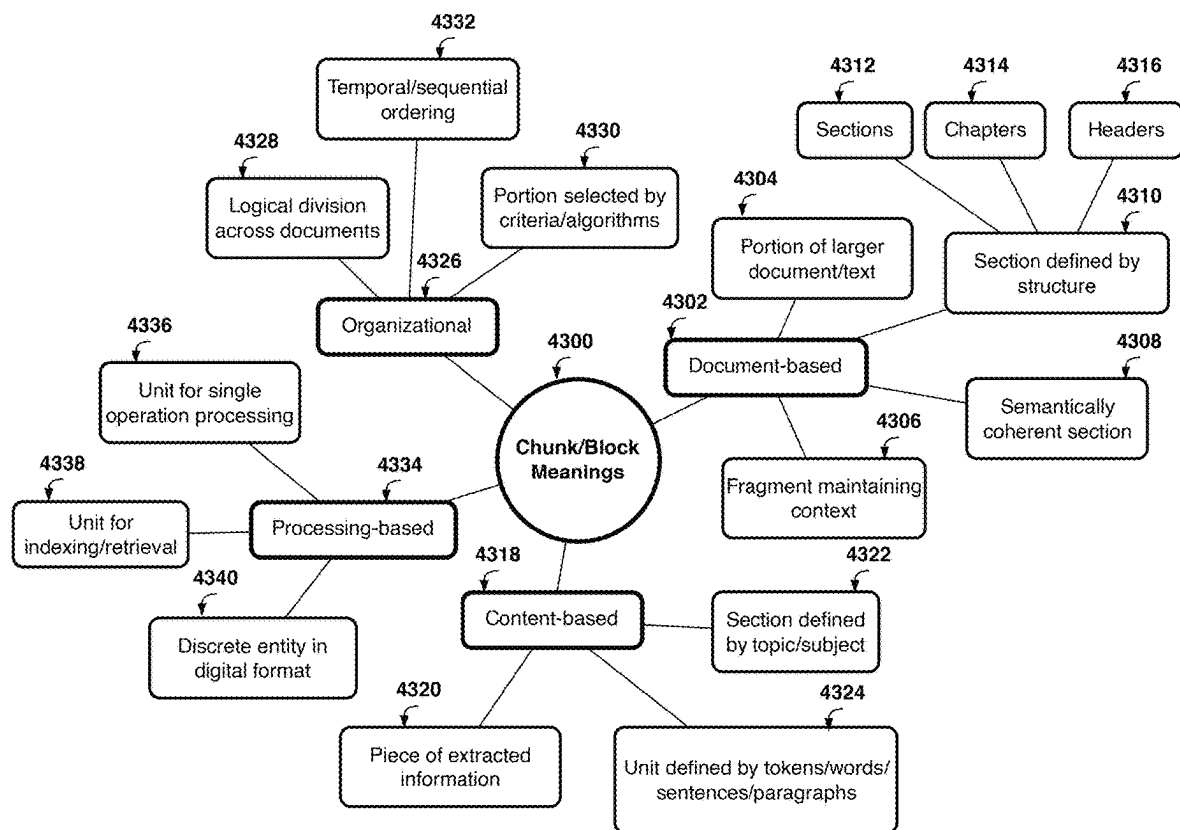


FIG. 45

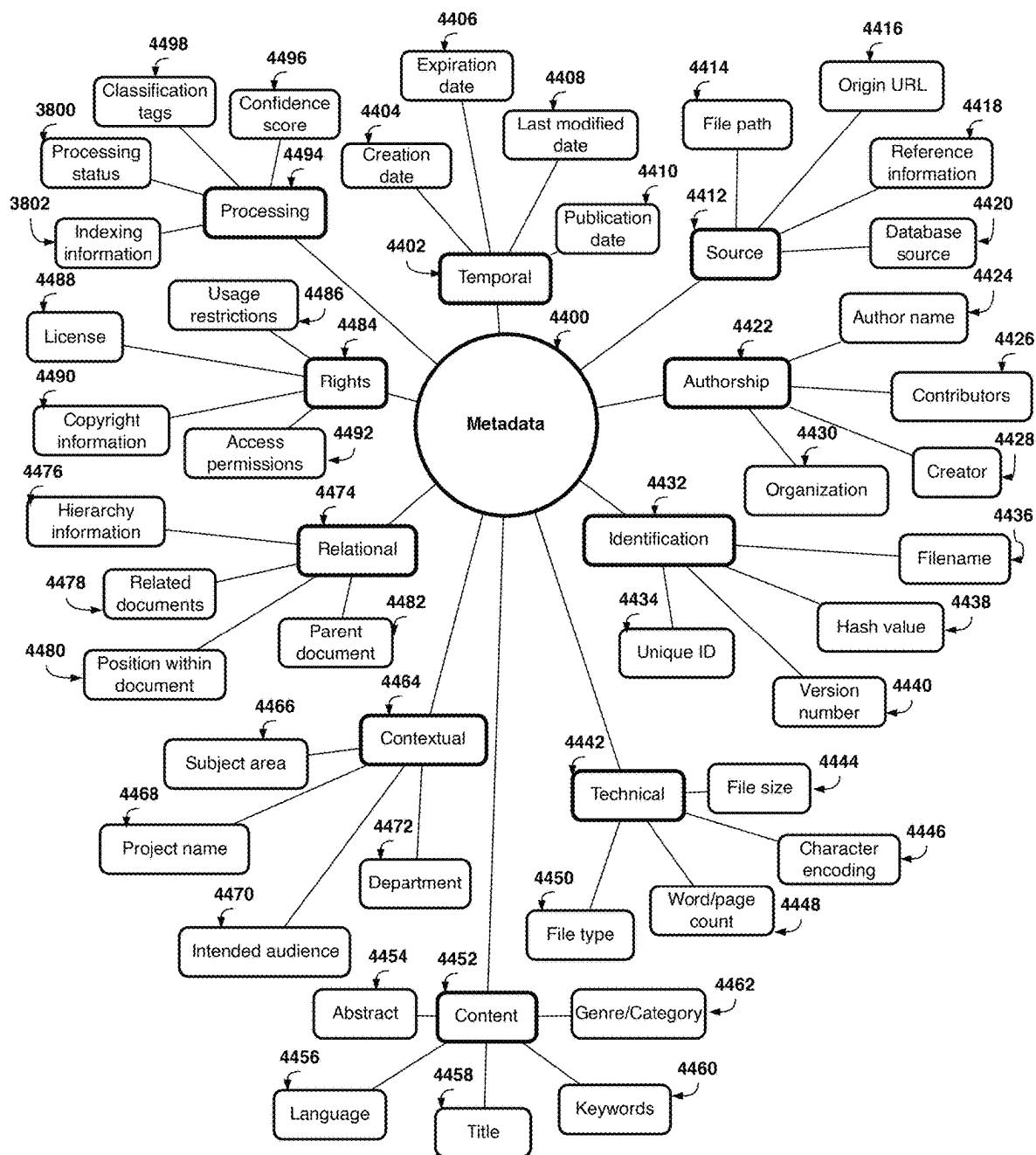


FIG. 46

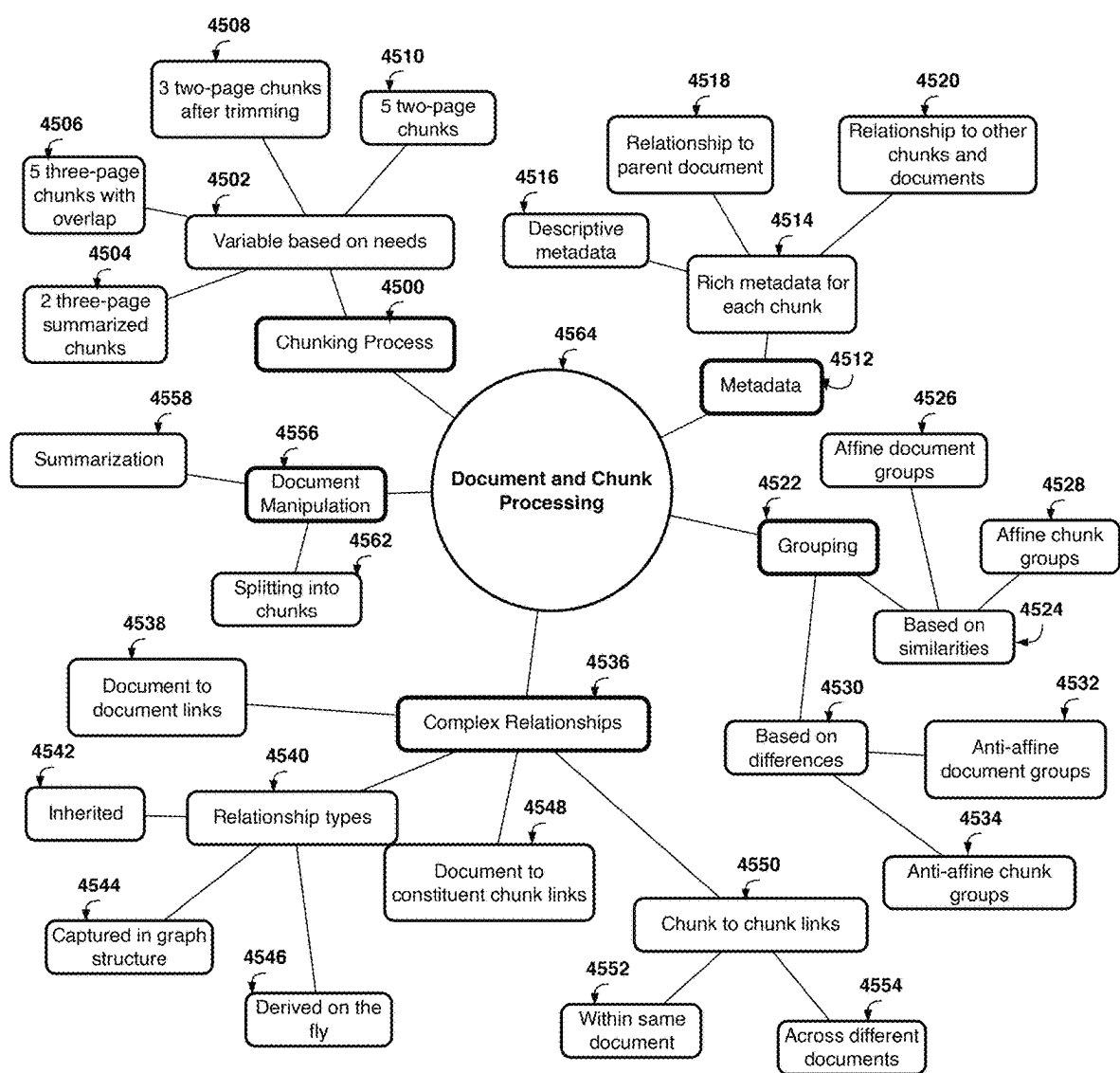


FIG. 47

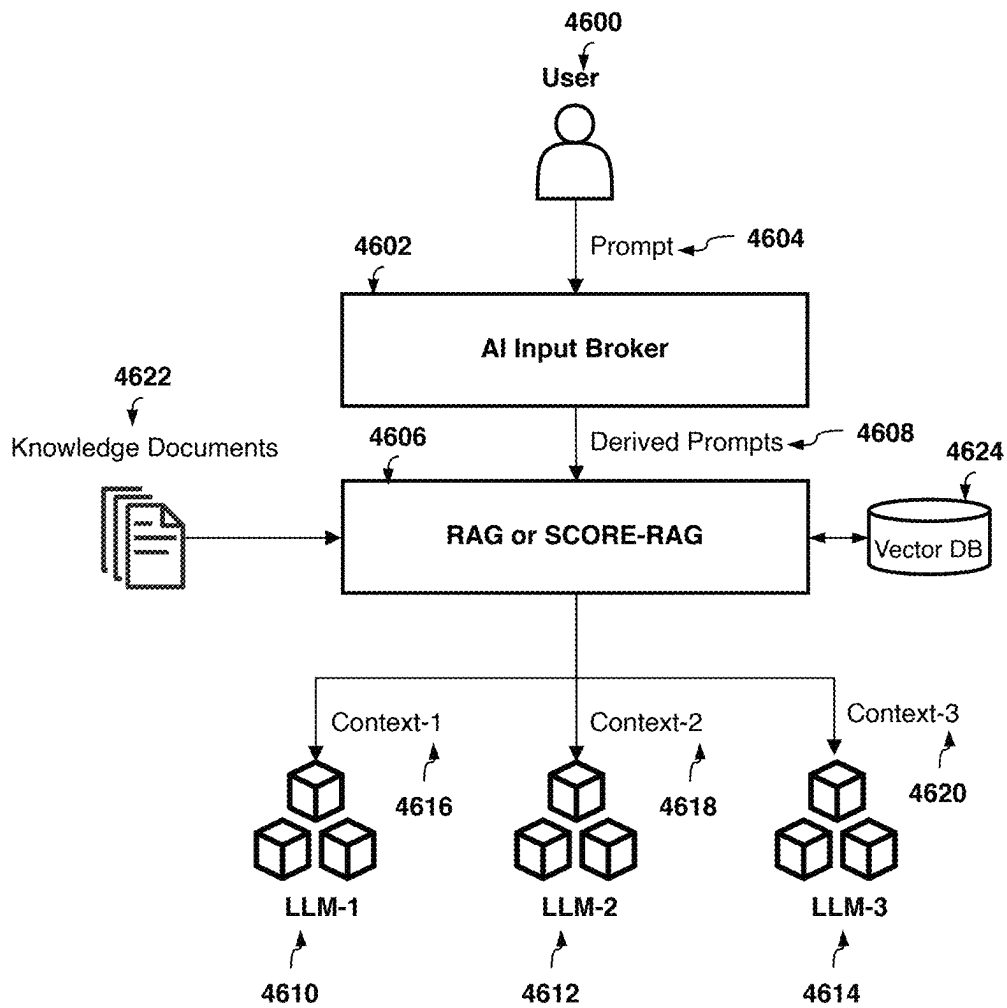


FIG. 48

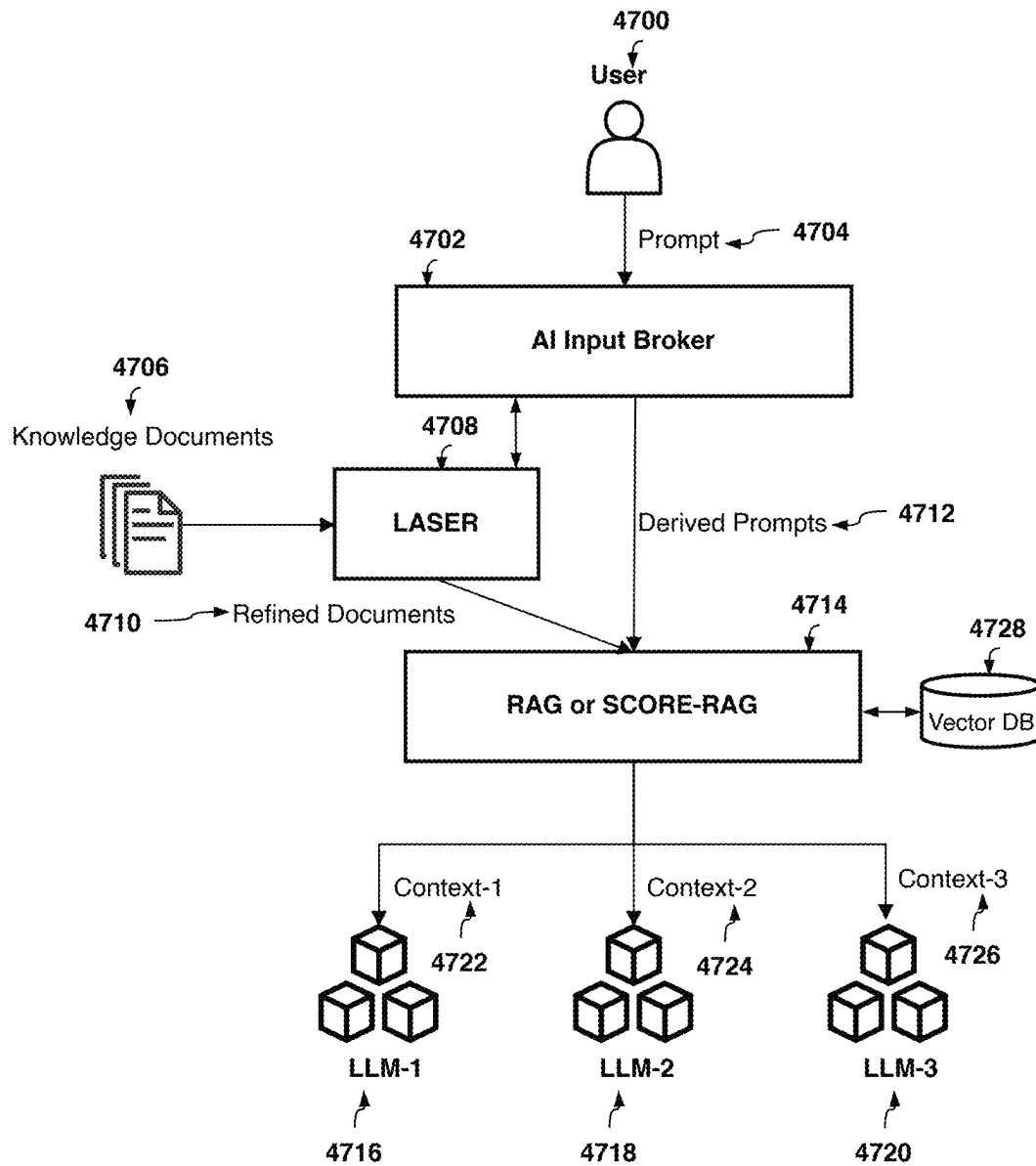


FIG. 49

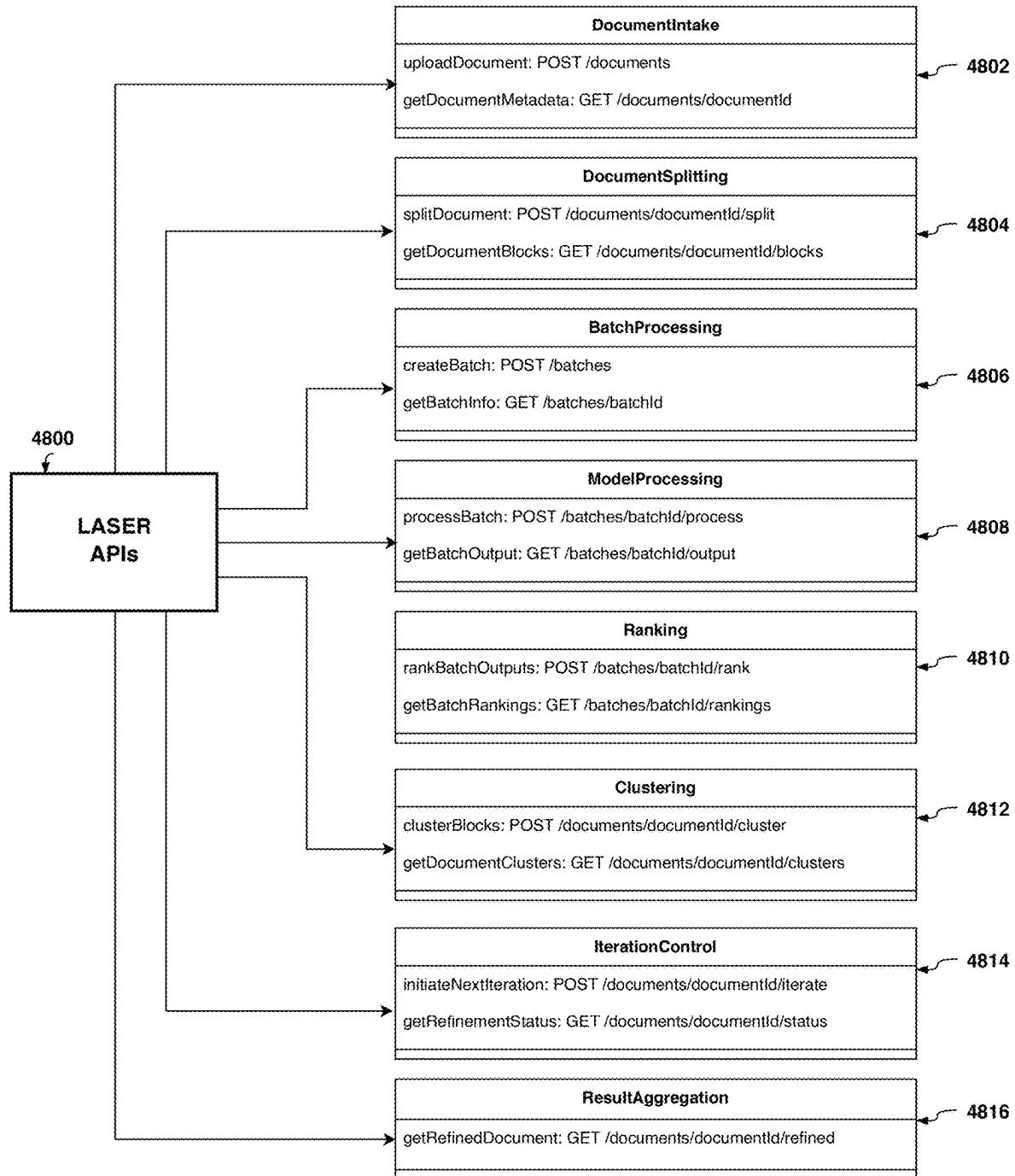


FIG. 50

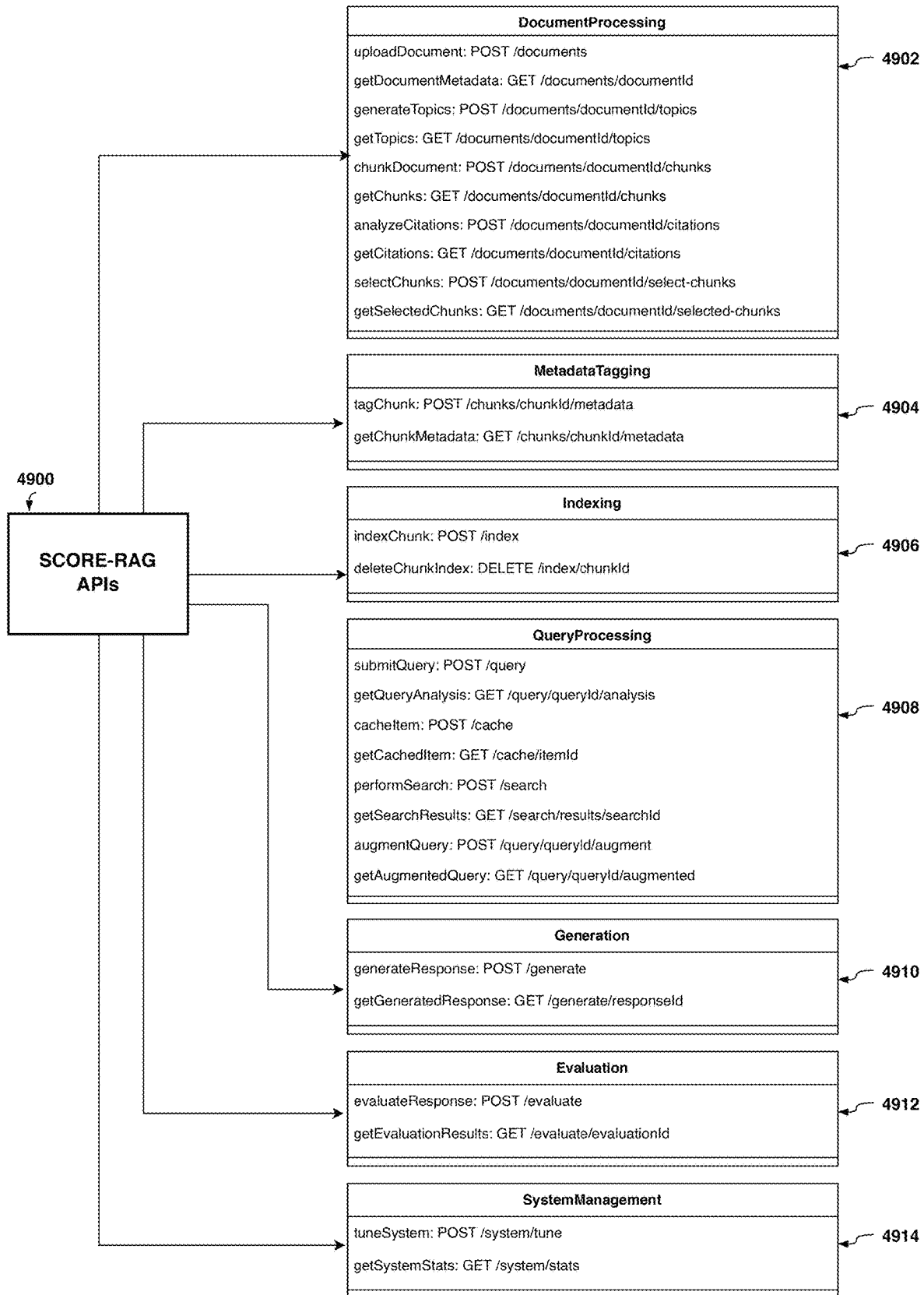


FIG. 51

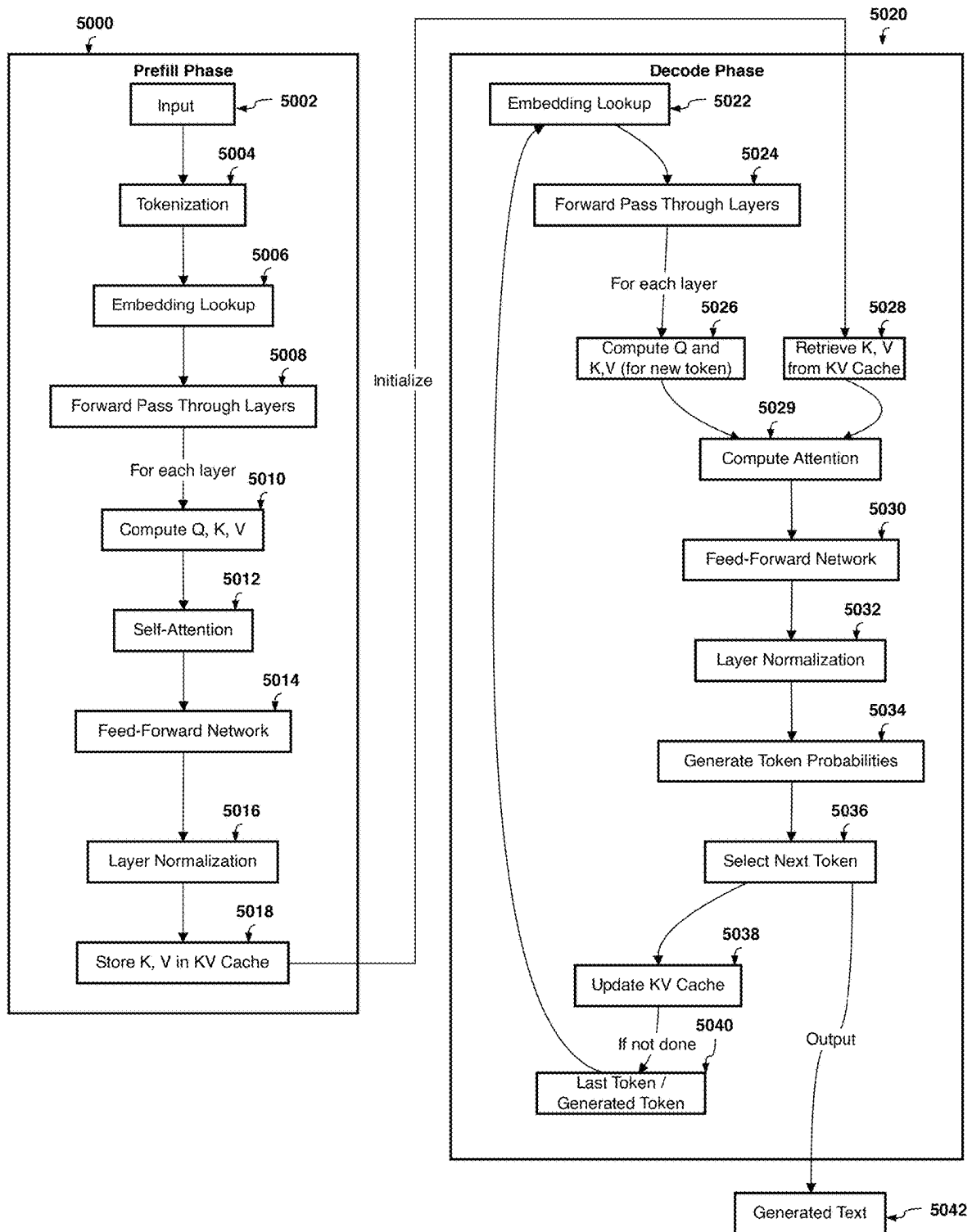


FIG. 52

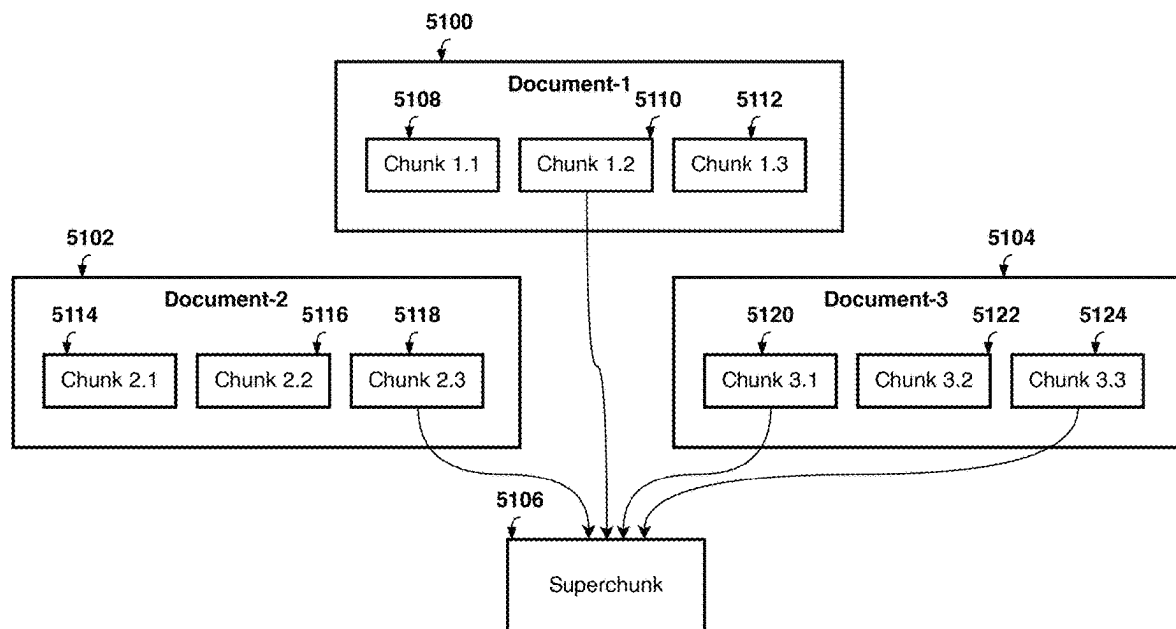


FIG. 53

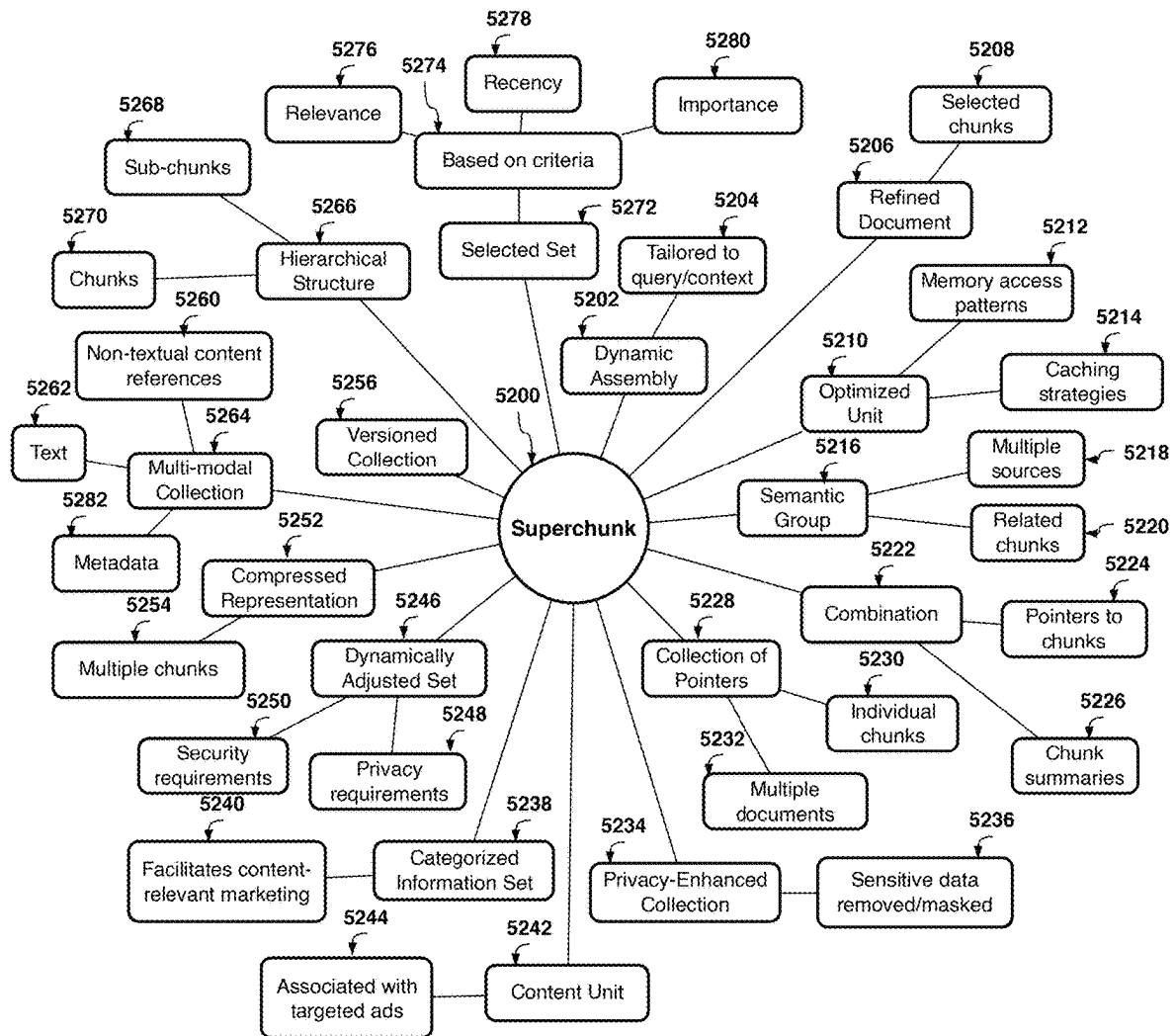


FIG. 54

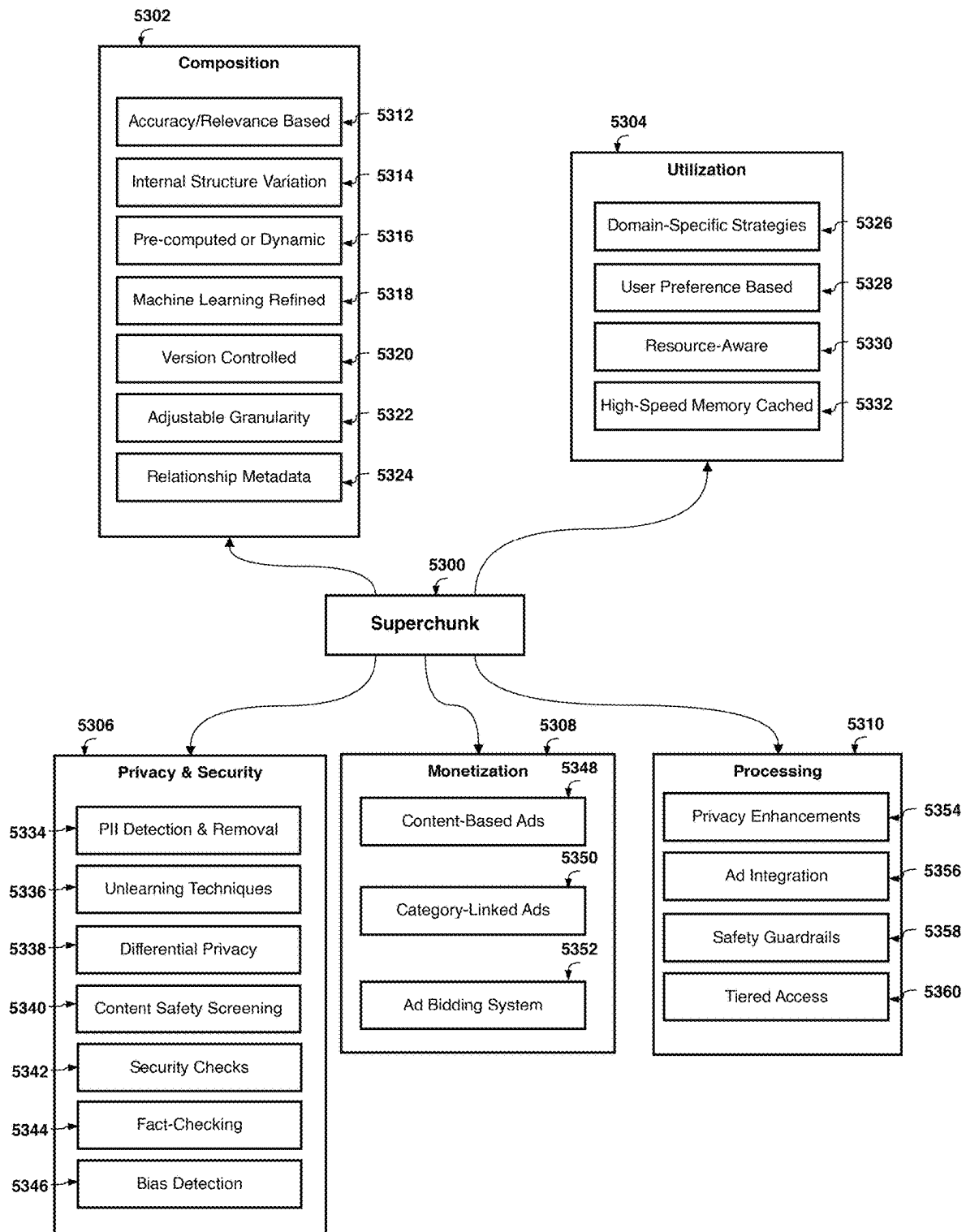


FIG. 55

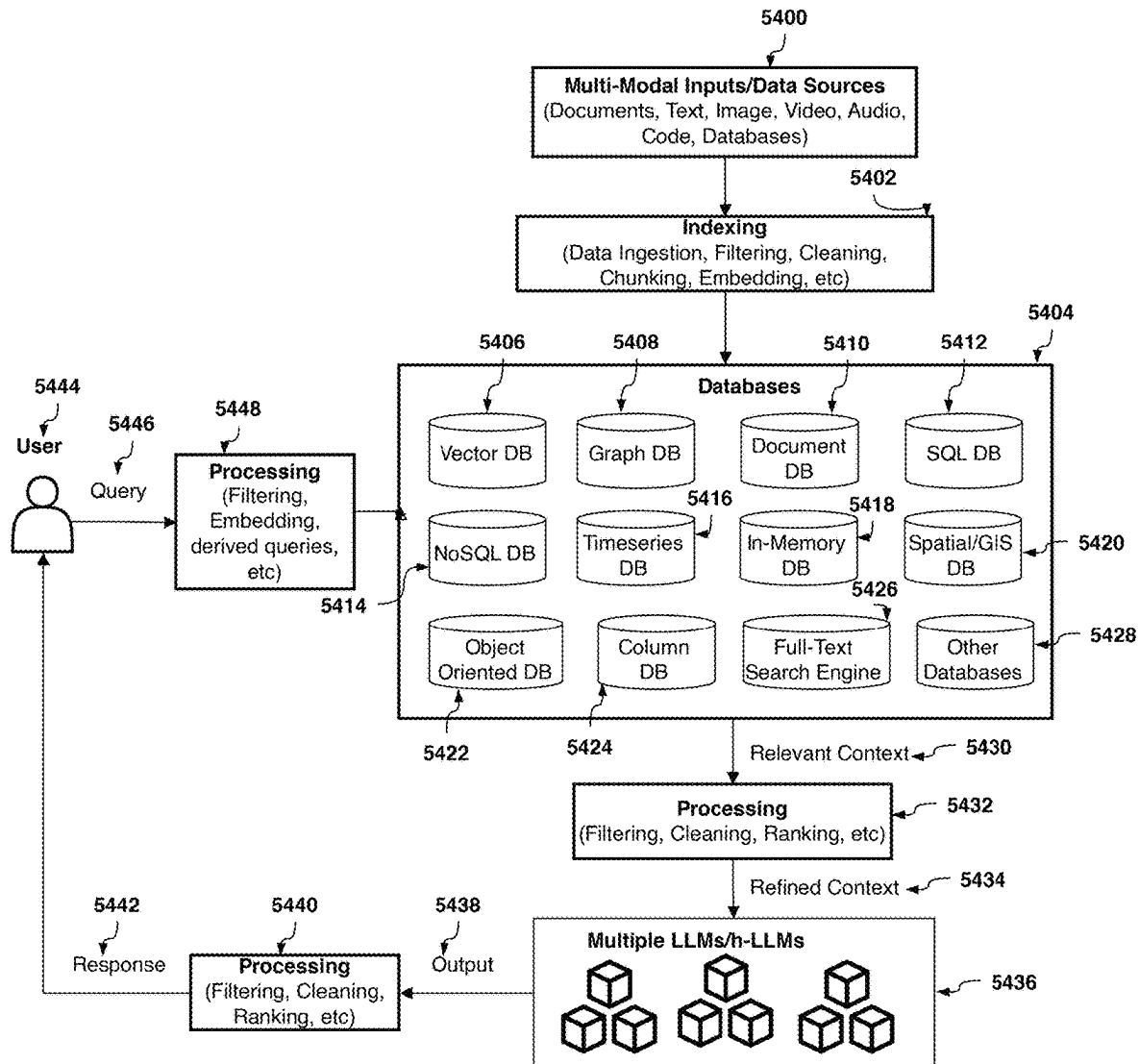


FIG. 56

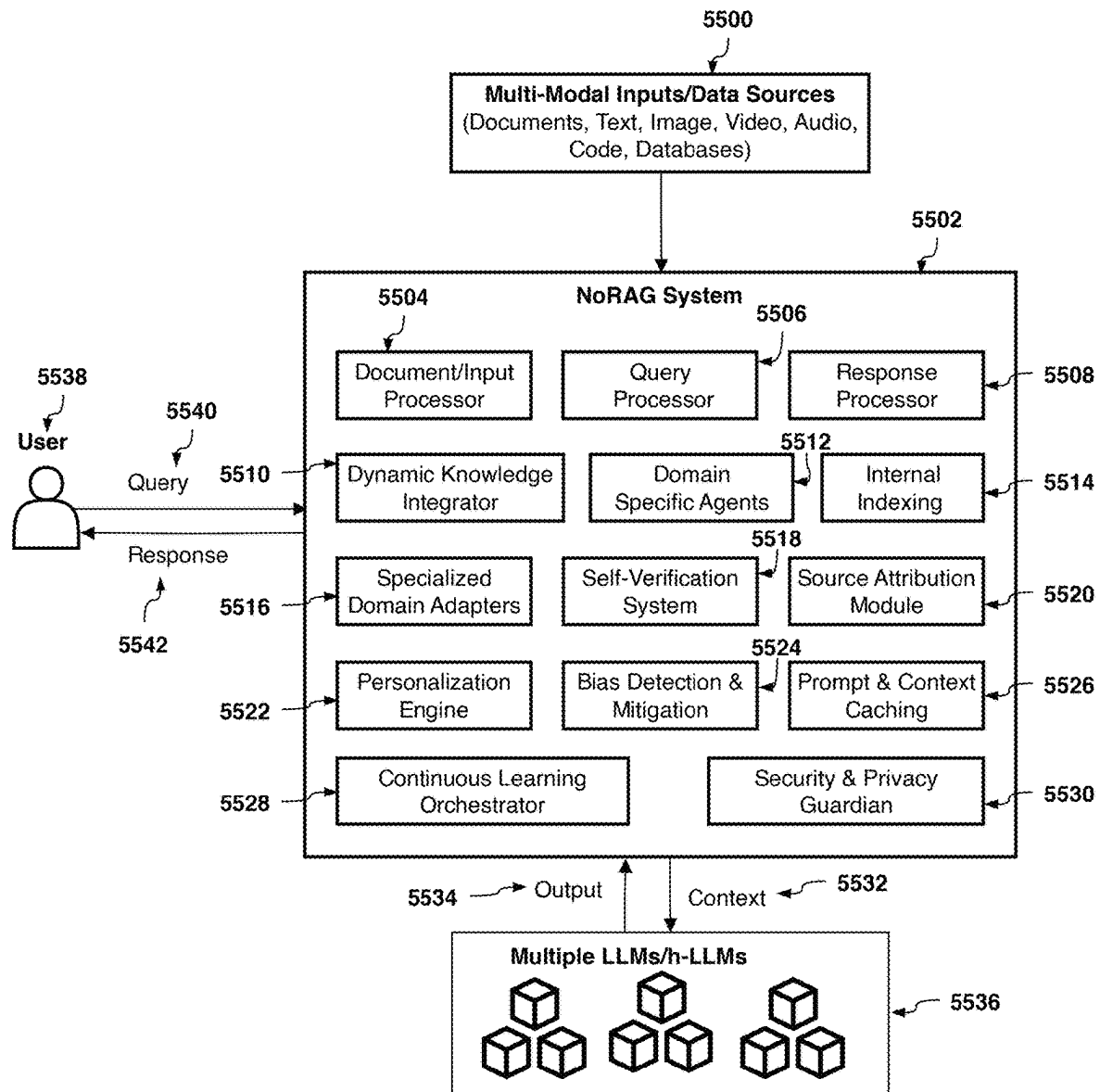


FIG. 57

**METHOD AND SYSTEM FOR OPTIMIZING
USE OF RETRIEVAL AUGMENTED
GENERATION PIPELINES IN GENERATIVE
ARTIFICIAL INTELLIGENCE
APPLICATIONS**

RELATED APPLICATIONS

This application claims priority under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application Ser. No. 63/535,118 filed on Aug. 29, 2023 and titled Networked LLMs and Focused LLMs, U.S. Provisional Patent Application Ser. No. 63/529,177 filed on Jul. 27, 2023 and titled Using LLMs to Create Projects and Tasks in an Optimized Way, U.S. Provisional Patent Application Ser. No. 63/534,974 filed on Aug. 28, 2023 and titled Using Prompts to Generate Search Queries for Context Generation in LLMs, U.S. Provisional Patent Application Ser. No. 63/647,092 filed on May 14, 2024 and titled Using LLMs to Influence Users and Organizations, U.S. Provisional Patent Application Ser. No. 63/607,112 filed on Dec. 7, 2023 and titled Long Document Attention Span Enhancement for LLMs, and U.S. Provisional Patent Application Ser. No. 63/607,647 filed on Dec. 8, 2023 and titled High-Level UI for Prompt Generation for LLMs, and is also a continuation-in-part application of and claims priority under 35 U.S.C. § 120 of U.S. patent application Ser. No. 18/744,199 filed on Jun. 14, 2024 and titled Method and System for Protecting and Removing Private Information Used in Large Language Models, which in turn claims priority under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application Ser. No. 63/551,548 filed on Feb. 9, 2024 and titled Generation of Synthetic Data for PII, U.S. Provisional Patent Application Ser. No. 63/604,909 filed on Dec. 1, 2023 and titled Guardian-Preventing Privacy Attacks on LLMs, which in turn claims priority under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application Ser. No. 63/602,675 filed on Nov. 27, 2023 and titled Object detection combined with LLMs, and is a continuation-in-part application of and claims priority under 35 U.S.C. § 120 of U.S. patent application Ser. No. 18/406,906, now U.S. Pat. No. 12,158,904, issued Dec. 3, 2024 filed on Jan. 8, 2024 and titled Method and System for Protecting and Removing Private Information Used in Large Language Models, which in turn claims priority under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application Ser. No. 63/602,675 filed on Nov. 27, 2023 and titled Object detection combined with LLMs, and U.S. Provisional Patent Application Ser. No. 63/604,910 filed on Dec. 1, 2023 and titled Targeted Forgetting in LLMs-Details, and is a continuation-in-part application of and claims priority under 35 U.S.C. § 120 of U.S. Patent Application Ser. No. 18/470,487, now U.S. Pat. No. 12,147,461, issued Nov. 19, 2024 filed on Sep. 20, 2023 and titled Method and System for Multi-Level Artificial Intelligence Supercomputer Design, which in turn is a continuation of U.S. patent application Ser. No. 18/348,692, now U.S. Pat. No. 12,001,462, issued Jun. 4, 2024 filed on Jul. 7, 2023 and titled Method and System for Multi-Level Artificial Intelligence Supercomputer Design, which in turn claims priority under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application Ser. No. 63/463,913 filed on May 4, 2023 and titled New Tools for Document Analysis in CatchUp and U.S. Provisional Patent Application Ser. No. 63/469,571 filed on May 30, 2023 and titled Multilevel AI PSupercomputer Design. The contents of these applications are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention primarily relates to artificial intelligence and large language models (LLMs) for generative AI applications.

BACKGROUND

Large Language Models (LLMs) are generative Artificial Intelligence (AI) models which are trained on limited amounts of data and can perform language processing tasks (with multimodal inputs—text, and more recently, image inputs as in Microsoft's Kosmos-1) and generate human-like text (and associated multimedia material, like images, video and advertisements). LLMs have many parameters (from millions to billions). LLMs can capture complex patterns in language and produce text that closely resembles human language.

The high-level goal of an LLM is to predict the text (and other multimedia material) that is likely to come next in a sequence. The applicants recognize that LLMs are a type of generative AI that is in usually different from traditional machine learning and AI applications. LLM also stands for Learning with Limited Memory and implies that LLM's are closely tied to their training data and make decisions based on the limited amount of data. Both generative AI and LLM generate content, but LLM does it in a manner that improves computational and memory efficiency.

Traditional machine learning type algorithms focus on analysis, such as statistical regression or clustering, and are usually again different from Generative AI and LLMs, which focus on generating content. LLMs have immediate practical implication in generation of new content that matches associated or preceding/future content in an optimized manner, such as legal briefs or computer code, based on training with a limited amount of data, such as existing briefs or code, both from private and public sources. In this invention, we focus on LLM models as the primary focus of these improvements, though we do not disclaim other AI models, unless expressly done as part of the claims.

LLMs are created with complex architectures such as transformers, encoders and decoders. LLMs, typically, use a technique of natural language processing called Tokenization that involves splitting the input text (and images) and output texts into smaller units called tokens. Tokens can be words, characters, sub-words, or symbols, depending on the type and the size of the model. Tokenization helps to reduce the complexity of text data, making it easier for LLMs to process and understand data thus reducing the computational and memory costs. Another important component of an LLM is Embedding, which is a vector representation of the tokens. The Encoder, within the Transformer architecture, processes the input text and converts it into a sequence of vectors, called embeddings, that represent the meaning and context of each word. The Decoder, within the Transformer architecture, generates the output text by predicting the next word in the sequence, based on the embeddings and the previous words. LLMs use Attention mechanisms that allow the models to focus selectively on the most relevant parts of the input and output texts, depending on the context of the task at hand, thus capturing the long-range dependencies and relationships between words.

LLMs are designed to learn the complexity of the language by being pre-trained on vast amounts of text (and multimedia) data from sources such as Wikipedia, books, articles on the web, social media data and other sources. The training procedure can be decomposed into two stages:

1. Pre-training on a large amount of unlabeled plain text; and
2. Supervised fine-tuning

Through training on limited amounts of data, the models are able to learn the statistical relationships between words, phrases, and sentences and other multimedia content. The trained models can then be used for generative AI applications such as Question Answering, Instruction Following, Inferencing, for instance, where an input is given to the model in the form of a prompt and the model is able to generate coherent and contextually relevant responses based on the query in the prompt.

Popular LLM models include GPT (Generative Pre-trained Transformer), BERT (Bidirectional Encoder Representations from Transformers), BART (Bidirectional and Auto-Regressive Transformers) and PaLM (Pathways Language Model). See, for example, public domain websites, such as openai.com or bard.google.com for more information as to how a person of ordinary skill in the art may use these models. Public domain and company-specific LLMs, such as GPT4All, MiniGPT4, RMKV, BERT, MPT-7B, Kosmos-1 (which accepts image and multimodal inputs), YaLM, are also available for wide use, as for example, described in medium.datadriveninvestor.com/list-of-open-source-large-language-models-llms-4eac551bda2e.

Current AI generative models and LLMs require super-computing efforts to compute results and an efficient way to improve response times, accuracies, and reduce computational load is required to improve both cost and scalability and expandability of existing AI models and their use.

LLMs face significant challenges when processing long documents, particularly in maintaining coherence and performing long-range reasoning. This limitation, often referred to as the “attention span problem,” causes a noticeable drop in performance as the length of the input context increases, typically above 10,000 to 50,000 tokens.

The attention span problem has substantial implications for real-world applications, especially in domains that frequently deal with lengthy and complex documents, such as legal, engineering, healthcare, and academic research. In these fields, the ability to comprehend and reason over extended contexts is crucial for tasks like document summarization, question answering, and information extraction.

Existing approaches to mitigate the attention span problem, such as sliding window techniques or hierarchical attention mechanisms, have shown limited success. They often struggle to maintain global coherence or fail to capture long-range dependencies effectively. As a result, there is a pressing need for innovative solutions that can enhance the attention span of LLMs and enable them to process long documents more effectively.

LLMs face inherent limitations due to their reliance on pre-trained knowledge. These include a fixed knowledge cutoff, potential for hallucination, and lack of specificity in responses. Retrieval-Augmented Generation (RAG) is a useful approach in AI that combines the strengths of LLMs with external knowledge retrieval. RAG addresses the limitations of LLMs by providing them with relevant, up-to-date information from a curated knowledge base. This approach grounds LLM outputs in retrieved facts, significantly reducing hallucinations while enabling more accurate and context-specific responses.

Existing RAG systems have shown promise in enhancing the performance of LLMs by providing relevant context from external knowledge sources. However, these systems face significant challenges in processing and retrieving information from long, complex documents. Current RAG

implementations often struggle with inefficient document chunking, leading to loss of context and semantic coherence. They typically rely on simplistic keyword-based retrieval methods, which fail to capture the nuanced graph-like relationships between concepts. Moreover, existing systems lack sophisticated mechanisms for dynamically adapting to different types of queries and documents, resulting in sub-optimal retrieval and generation performance. The inability to effectively handle large volumes of text, combined with inadequate context preservation and limited semantic understanding, hinders the widespread adoption of RAG systems in domains that deal with extensive and intricate textual information, such as legal, medical, engineering, and scientific research fields.

This background information is provided to reveal information believed by the applicant to be of possible relevance to the present invention. No admission is necessarily intended, nor should be construed that any of the preceding information constitutes prior art against the present invention.

SUMMARY OF THE INVENTION

With the above in mind, embodiments of the present invention are directed to a system and associated methods for multi-level generative AI and large language models (LLM) for generative AI applications, that utilize the following techniques:

Derived Requests: An initial level of generative AI software program, or AI broker, evaluates the incoming client request (maybe a conversational query or through an API, such as OpenAI API) and identifies its specific AI “characteristics” that may make it suitable for one or other or both or multiple AI language models and checks its “derived requests” categories to see if the query suits one of the “derived requests” categories and/or it can or should create a new request.

Multiple h-LLMs: If the new request does not assigned to one or more of the “derived requests” categories, it evaluates the request and selects one or more AI h-LLM model categories for its evaluation. An h-LLM is a family of models, such as GPT-4, that (in addition) have been trained according to a particular training set T1. A family of generative models, LLM1, trained with a data set T1, can be represented as h-LLM1, while a family of models, LLM2, trained with data set T2, can be represented as h-LLM12. Further, a family of models, LLM1, trained with a data set T3, can be represented as h-LLM35. The combination of models and their training sets (T1 could be a subset of T3, for example, or they can be different) may be used in our proposed invention and they are referred to as h-LLMs, throughout. A family of LLMs that operate at a lower arithmetic precision, on computer CPUs or graphical processing units (GPUs, such as Nvidia’s H100), may also be called by a different identifier, e.g., h-LLM14, when trained with its corresponding data set.

Choosing h-LLMs with varying levels of accuracy: It further checks the workload of the AI h-LLM models in the one or more categories and its level of training and its accuracy-called its workload scores or its technical accuracy scores, or its business value metrics or a combination of these scores, and then assigns the request (or its derived form) to one or more of the AI h-LLM models within the selected AI h-LLM model categories.

Assigning weights to results: It then receives the results from the AI models in the AI h-LLM models categories and weights them to compute a result that could be returned to

the requester program, or it could resend the request back to the AI h-LLM models/categories hierarchy till it reaches a certain level of service level assurance.

Use of Local Database: It also updates a local database with the results of the request's path through its hierarchy and create an index of "derived requests" that may be used in future to select which set of "derived requests" an incoming request may fall into for further processing.

Distributed Architecture: The tasks may be implemented as containers within Kubernetes environment and a service mesh, such as Istio, may be used to instrument and parameterize the metrics and log collections, but not limited to these cloud models for implementation.

Efficient Search & Retrieval: Traditional online and offline approaches to cluster search are used to find the relevant subset of the documents being evaluated in Retrieval Augmented Generation (RAG) pipelines. Once this subset is retrieved then the traditional pipeline of LLMs operations are carried out as in LangChain and LlamaIndex. The cluster may be generated during time of the Query Prompt input (adds to the delay due to need to generate indexes) or could be used to select a subset of indexes in Vector Db in a quicker approach. Few important queries and prompts may be used to generate clusters (offline) and each new online query may be mapped to the best "cluster" that was pre-generated based on that query or similar queries.

Network of LLMs working together to replace a larger LLM: Currently a single large LLM is trained on all types of data and has large number of parameters (e.g. OpenAI GPT3.5 has 175 billion parameters and GPT-4 has over 1 trillion of parameters). A approach using a Network of LLMs is proposed which combines smaller LLMs (with 3B or 7B parameters, for example), preferably each focused on a specific type of result (cost estimation, profit estimation, expense estimation or prediction). The network of LLMs is used to provide a composite result that is easier to prompt for, easier to optimize and easier to "explain" how it works by having smaller focused LLMs trained on specialized training sets.

Embodiments of the present invention are directed to a system and associated methods for enhancing the attention span of Large Language Models (LLMs) when processing long documents. The system, long-document attention span enhancement through refinement ("LASER"), uses an iterative attention focusing technique that dynamically refines and condenses document and chunk context to improve model comprehension and coherence over extended inputs.

Other embodiments of the present invention are directed to enhancing Retrieval-Augmented Generation (RAG) through context-optimized retrieval techniques. The system, scored context-optimized retrieval enhancement for retrieval augmented generation ("SCORE-RAG"), addresses the limitations of existing RAG systems by incorporating advanced document (including chunk) processing, intelligent information retrieval, and adaptive response generation mechanisms.

In one embodiment, the present invention comprises a document processing system that includes an input module, a model module, an iteration controller, a knowledge module, and an output handler. The input module is configured to split long documents into manageable blocks (or chunks) and generate iterative contexts, while the model module contains an attention model for processing these contexts and a ranking unit for evaluating outputs.

Another embodiment of the invention involves a method for iterative attention focusing, which includes splitting a long document into blocks, batching these blocks, process-

ing them through an LLM, ranking the outputs, and then clustering and reforming new batches based on the highest-ranked content. This process is repeated iteratively, gradually condensing the document to its most relevant parts.

Another embodiment of the invention provides a mechanism for dynamically adjusting the attention focus of an LLM. This mechanism employs a ranking system that scores model outputs based on coherence and relevance, allowing the system to identify and prioritize the most important parts of a document or chunks across multiple processing cycles.

Another embodiment of the invention introduces a knowledge module that incorporates an extractive summarizer and a document clustering component. These elements work together to identify information and group related content, further enhancing the system's ability to distill and focus on critical parts of long documents and chunks.

Another embodiment of the invention comprises a document processor for handling various input formats, a topic modeling engine for semantic analysis, and an intelligent document chunking module that preserves contextual integrity. This embodiment also features a hybrid search module that combines keyword-based and vector similarity search methods for improved retrieval accuracy.

Another embodiment of the invention involves a method for dynamically processing and indexing documents. This method employs a citation analyzer to assess the importance of different text segments, a chunk selection and ranking module to identify the most relevant portions of a document, and a metadata enrichment module to enhance the contextual information associated with each text chunk. The method further includes an adaptive indexing process that optimizes storage and retrieval of processed information.

Another embodiment of the invention provides a mechanism for query augmentation and response generation. This mechanism utilizes a Query Processor to analyze and classify user inputs, an Augmentation Engine to integrate retrieved context with the original query, and a Generation Module that interfaces with LLMs to produce coherent and relevant responses.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an illustration of the training process for creating multiple specialized large language models for specific tasks/categories, according to an embodiment of the present invention.

FIG. 2 is an illustration of h-LLMs trained with different training sets, according to an embodiment of the invention.

FIG. 3 is an illustration of the process for generating synthetic data from multiple h-LLMs and using it for model refinement, according to an embodiment of the invention.

FIG. 4 is an illustration of a "bagging" approach where multiple h-LLMs with lower precision and accuracy are merged/fused to create a merged h-LLM with higher precision and accuracy, according to an embodiment of the invention.

FIG. 5 is an illustration of a "boosting" approach where multiple h-LLMs of increasing precision and accuracy are created in a sequential manner and then merged/fused to create a merged h-LLM, according to an embodiment of the invention.

FIG. 6 is an illustration of creating a smaller and more specialized h-LLM through extraction/specialization process from a larger h-LLM, according to an embodiment of the invention.

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FIG. 7 is an illustration of combining h-LLMs trained with text, image and audio data to create a merged h-LLM, according to an embodiment of the invention.

FIG. 8 is an exemplary illustration of an application of using AI models for detecting labels in PDF files, according to an embodiment of the invention.

FIG. 9 is an illustration of generating derived prompts for different categories and using them with multiple h-LLMs to generate the best results, according to an embodiment of the present invention.

FIG. 10 is an illustration of using multiple h-LLMs to answer questions from specific input documents, according to an embodiment of the present invention.

FIG. 11 is an illustration of an AI Broker for processing results from multiple h-LLMs, according to an embodiment of the present invention.

FIG. 12 is an illustration of the combining h-LLMs in series, according to an embodiment of the present invention.

FIG. 13 is an illustration of combining h-LLMs in parallel, according to an embodiment of the present invention.

FIG. 14 is an illustration of a hybrid approach of combining h-LLMs in series and parallel, according to an embodiment of the present invention.

FIG. 15 is an illustration of the lambda architecture for h-LLMs, according to an embodiment of the present invention.

FIG. 16 is an illustration of batch and real-time processing architecture for h-LLMs, according to an embodiment of the present invention.

FIG. 17 is an illustration of an in-memory processing architecture for h-LLMs, according to an embodiment of the present invention.

FIG. 18 is an illustration of the architecture of PDF label search tool with CatchUp GlassViewer, according to an embodiment of the invention.

FIG. 19 is an exemplary interface of the CatchUp platform showing the document management system, according to an embodiment of the invention.

FIG. 20 is an exemplary interface of the CatchUp platform showing the PDF viewer (GlassViewer), according to an embodiment of the invention.

FIG. 21 is an exemplary interface of the CatchUp platform showing a magnifier tool within the GlassViewer for searching labels, according to an embodiment of the invention.

FIG. 22 is an exemplary interface of the CatchUp platform showing label search results within GlassViewer, according to an embodiment of the invention.

FIG. 23 is an illustration of the process used by organizations to decide the list of tasks for a project, according to an embodiment of the invention.

FIG. 24 is an illustration of the process of training LLMs to generate project, tasks and action items, according to an embodiment of the invention.

FIG. 25 is an illustration of the process of creating a project within CatchUp using a prompt, according to an embodiment of the invention.

FIG. 26 is an illustration of the process of Retrieval Augmented Generation (RAG), according to an embodiment of the invention.

FIG. 27 is an illustration of the process of indexing documents RAG, according to an embodiment of the invention.

FIG. 28 is an illustration of the process of querying documents in RAG, according to an embodiment of the invention.

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FIG. 29 is an illustration of the Retrieval Augmented Generation RAG pipeline, according to an embodiment of the invention.

FIG. 30 is an illustration of an approach that uses multiple RAG pipelines with clusters of documents, according to an embodiment of the invention.

FIG. 31 is an illustration of an approach that uses a prompt augmented with a search query, according to an embodiment of the invention.

FIG. 32 is an illustration of an approach that uses a search query to extract relevant context for RAG pipelines, according to an embodiment of the invention.

FIG. 33 is an exemplary interface of the CatchUp platform for RAG, according to an embodiment of the invention.

FIG. 34 is an exemplary interface of the CatchUp platform showing response a RAG pipeline, according to an embodiment of the invention.

FIG. 35A is an illustration of a prior art embodiments of using a single LLM.

FIG. 35B is an illustration of using a network of LLMs, according to an embodiment of the invention.

FIGS. 36A-C is a comparison of the current (36A) and proposed (36B, 36C) approaches of a long-document attention span enhancement through refinement (LASER) system and a scored context-optimized retrieval enhancement for retrieval augmented generation (SCORE-RAG) system according to an embodiment of the invention.

FIG. 37 is an illustration of the high-level architecture of a LASER system according to an embodiment of the invention.

FIG. 38 is a flow chart illustrating a method performed by a LASER system by which long documents are processed and refined to enhance the attention span of LLMs according to an embodiment of the invention.

FIG. 39 is a flow chart illustrating a method of adapting a MapReduce model for the LASER system according to an embodiment of the invention.

FIG. 40 is an illustration of the architecture of a SCORE-RAG system according to an embodiment of the invention.

FIG. 41 is a flow chart illustrating a method of a SCORE-RAG system according to an embodiment of the invention.

FIG. 42 is an illustration of an implementation of a SCORE-RAG system according to an embodiment of the invention for multi-modal RAG.

FIG. 43 is an illustration of an implementation of a LASER system and a SCORE-RAG system according to an embodiment of the invention for multi-modal RAG.

FIG. 44 is an illustration of different meanings of “document” in the context of the present invention.

FIG. 45 is an illustration of different meanings of “chunk” and “block” in the context of the present invention.

FIG. 46 is an illustration of different meanings of “metadata” in the context of the present invention.

FIG. 47 is an illustration of different meanings of “document processing” or “chunk/block processing” in the context of the present invention.

FIG. 48 is an illustration of an approach of generating derived prompts and using them with RAG or SCORE-RAG systems to generate improved results according to an embodiment of the invention.

FIG. 49 is an illustration of an approach of generating derived prompts and using LASER and RAG or SCORE-RAG systems to generate improved results according to an embodiment of the invention.

FIG. 50 is an illustration of an exemplary set of APIs of a LASER system according to an embodiment of the invention.

FIG. 51 is an illustration of an exemplary set of APIs of the SCORE-RAG system, according to an embodiment of the invention.

FIG. 52 is a flowchart illustrating methods of performing prefill and decode phases of an LLM inference process according to an embodiment of the invention.

FIG. 53 is an illustration depicting the formation of a superchunk from several chunks from discrete documents according to an embodiment of the invention.

FIG. 54 is an illustration of different meanings of “superchunk” in the context of the present invention.

FIG. 55 is an illustration of characteristics of superchunks according to an embodiment of the invention.

FIG. 56 is an illustration of the architecture of a Hybrid-RAG system according to an embodiment of the invention.

FIG. 57 is an illustration of an architecture of a NoRAG system according to an embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. Those of ordinary skill in the art realize that the following descriptions of the embodiments of the present invention are illustrative and are not intended to be limiting in any way. Other embodiments of the present invention will readily suggest themselves to such skilled people having the benefit of this disclosure. Like numbers refer to like elements throughout.

Although the following detailed description contains many specifics for the purposes of illustration, anyone of ordinary skill in the art will appreciate that many variations and alterations to the following details are within the scope of the invention. Accordingly, the following embodiments of the invention are set forth without any loss of generality to, and without imposing limitations upon, the claimed invention.

In this detailed description of the present invention, a person skilled in the art should note that directional terms, such as “above,” “below,” “upper,” “lower,” and other like terms are used for the convenience of the reader in reference to the drawings. Also, a person skilled in the art should notice this description may contain other terminology to convey position, orientation, and direction without departing from the principles of the present invention.

Furthermore, in this detailed description, a person skilled in the art should note that quantitative qualifying terms such as “generally,” “substantially,” “mostly,” and other terms are used, in general, to mean that the referred to object, characteristic, or quality constitutes a majority of the subject of the reference. The meaning of any of these terms is dependent upon the context within which it is used, and the meaning may be expressly modified.

Referring now to FIG. 1 is an illustration of the training process for creating multiple specialized large language models for specific tasks/categories, is described in more detail. Data 100 (such as text, images, and audio) is used to

pre-train a model in a process called unsupervised pre-training 102 which generates a base h-LLM model 104. The pre-training process is referred to as unsupervised as unlabeled data is used at this step. The base h-LLM model 104 is then fine-tuned in a process called supervised fine-tuning 106. The fine-tuning process uses smaller labeled data sets. The base h-LLM model 104 is fine-tuned to generate multiple h-LLM models which are specialized to perform specific tasks such as Question Answering, Information Extraction, Sentiment Analysis, Image Captioning, Object Recognition, Instruction Following, Classification, Inferencing, and Sentence Similarity, for instance.

Referring now to FIG. 2 is an illustration of h-LLMs trained with different training sets, is described in more detail. As used in this specification h-LLM usually refers to a family of LLMs, such as those used in Google’s Bard or OpenAI’s GPT-4, that have been trained on a particular training set T. Therefore, the same family of LLMs (e.g., GPT) if trained on a different training set, T1, as opposed to GPT trained on training set T2 could be differentiated as a separate h-LLM. The training sets can be private within an organization or public datasets.

For example, as shown in FIG. 2, h-LLM-1 152 is trained with training set-1 150, h-LLM-2 156 is trained with training set-2 154, h-LLM-3 160 is trained with training set-3 158, and h-LLM-3_4 164 is trained with training set-3 158 and training set-4 162.

An h-LLM can be described as a combination of LLM families and the training dataset used as follows:

h-LLM=LLM family (X) trained with Training Set (Y)

For example,

h-LLM_1=PaLM-2 may be trained with training set T12

h-LLM_2=PaLM-2 may be trained with training set T12+T45

h-LLM_3=GPT-4 may be trained with Training Set T65

h-LLM_4=GPT-4 may be trained with ANY data set

Referring now to FIG. 3, an illustration of the process for generating synthetic data from multiple h-LLMs and using it for model refinement, is described in more detail. Data 200 is used to train a base h-LLM model 204 using unsupervised pre-training 202 which is then fine-tuned in a supervised fine-tuning process 206 to generate multiple h-LLMs specialized for specific tasks or categories 208. Each of these h-LLMs 208 are used to generate synthetic data 210 which is then fed back to the models in feedback loop 212 through a process called model refinement 214.

Referring now to FIG. 4 is an illustration of a bagging approach, that has some similarity to what was originally used in the context of machine learning models in a different way (for analytics as opposed to generative AI applications, such as LLMs) that are described in this invention, where multiple h-LLMs with lower precision and accuracy are merged/fused to create a merged h-LLM with higher precision and accuracy, is described in more detail. Bagging is a machine learning technique which improves the stability and accuracy of machine learning models. Using the input data 300, multiple subsets of the data are created which are used to train multiple h-LLMs (302, 304, 306, 308) in parallel. These models are then combined in a process called merging or fusing 310 to create a merged h-LLM 312.

Referring now to FIG. 5 is an illustration a boosting approach, that has some similarities to that originally used in the context of machine learning models in a different way (for analytics as opposed to generative AI applications used in this invention) where multiple h-LLMs of increasing precision and accuracy are created in a sequential manner and then merged/fused to create a merged h-LLM, is

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described in more detail. Boosting is a machine learning technique that involves creating a stronger and more accurate model from a number of weaker models. The original data **400** is used to train an h-LLM **402**. The h-LLM **402** is tested and the output **404** is assigned weights to generate weighted data **406**. The weighted data **406** is then used to train h-LLM **408**. The same process is then repeated and h-LLMs **414** and **420** are generated in a sequence. The h-LLMs **402**, **408**, **414** and **420** are then combined in a process called merging or fusing **424** to create a merged h-LLM **426**.

Referring now to FIG. **6** is an illustration of creating a smaller and more specialized h-LLM through extraction/specialization process from a larger h-LLM, is described in more detail. The extraction/specialization process **502** extracts the specific knowledge required for a task from a big, general-purpose model, and creates a smaller h-LLM **506**. For example, a specific task can be sentiment analysis of input text, for which a smaller model **506** is more efficient as compared to a large, general-purpose model.

Referring now to FIG. **7** is an illustration of combining h-LLMs trained with text, image and audio data to create a merged h-LLM, is described in more detail. Text data **600** is used to train h-LLM **602**, image data **604** is used to train h-LLM **606** and audio data **608** is used to train h-LLM **610**. The h-LLMs **602**, **604**, **608** are combined in a process called merging/fusing to create a merged h-LLM **614**.

Referring now to FIG. **8** is an exemplary illustration of an application of using AI models for detecting labels in PDF files, is described in more detail. Patent documents (such as PDF files) have figures in which various entities/blocks/items are labeled using numeric labels (for instance **110**, **120** and so on). These labels are referenced and described in the patent text specification. When reviewing multiple documents, readers find it difficult to quickly lookup the labels mentioned in the figures (and what they refer to) from the text, as they need to go back and forth between a figure and the text in the specification. A novel PDF Label search solution is offered within CatchUp which allows quick lookup of labels in a figure using an innovative "AI Magnifier" approach. The user can select one or more labels using the Magnifier tool in the CatchUp GlassViewer (a PDF viewer tool within CatchUp that has annotation and other AI features). When one or more labels are selected using the Magnifier tool, the labels are searched within the PDF and the search results are returned. The PDF Label Search tool is built upon a novel AI Magnifier technology (which we refer to as AEye). AEye serves as a gateway to the world of Artificial Intelligence (AI) for documents and web pages. AEye can be used for a wide range of applications such as detecting objects in images, labels in documents, for instance. Documents or web pages **700** can be searched using an AEye application **704** which detects objects or labels utilizing an AEye backend **708**.

Referring now to FIG. **9** is an illustration of generating derived prompts for different categories and using them with multiple h-LLMs to generate the best results, is described in more detail. User **800** enters a prompt in user interface **802**. The prompt is sent to the AI Input Broker **810** which generates multiple derived prompts for different categories. The derived prompts **822** are sent multiple h-LLMs **824** which produce the results. The results **816** are sent to the AI Output Broker **814** which processes the results and performs tasks such as filtering, ranking, weighting, assigning priorities, and then sends the best results to the user **800**. The h-LLMs **824** can have varying levels of accuracy, and optimized for different tasks such as Question Answering,

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Information Extraction, Sentiment Analysis, Image Captioning, Object Recognition, Instruction Following, Classification, Inferencing, and Sentence Similarity, for instance. The AI Output Broker **814** computes various scores and assigns weights for ranking the results. The results may be sent back to the h-LLMs till a certain level of accuracy or service level assurance is reached. The AI Input Broker **810** and Output Broker **814** update a local AI Broker Database **820** with the results of the request's path through its hierarchy and create an index of "derived requests" that may be used in future to select which set of "derived requests" an incoming request may fall into for further processing.

Referring now to FIG. **10** is an illustration of using multiple h-LLMs to answer questions from specific input documents, is described in more detail. User **900** enters a prompt in user interface **902**. The prompt is sent to AI Input Broker **810** which generates multiple derived prompts for different categories **924**. The prompts are converted into embeddings using multiple embedding models **926**. The prompt embeddings **928** are sent to a vector database **930** which returns a list of knowledge documents **934** that are relevant to the prompt based on the similarity of their embeddings to the user's prompt. The knowledge documents **934** are sent to the AI Input Broker **810** which creates new context-aware prompts based on the user's initial prompt **916**, derived prompts **924** and the retrieved knowledge documents **934** as context and sends it to multiple h-LLMs **912**. The results produced by multiple h-LLMs are processed by the AI Output Broker **908** and the best result is sent to the user **900** along with citations from the knowledge documents **934**.

Referring now to FIG. **11** is an illustration of an AI Broker for processing results from multiple h-LLMs, is described in more detail. Results produced by multiple h-LLMs **1000** are sent to an AI Output Broker **1002** which performs tasks such as assigning priorities **1004** and weights **1006** to the results, filtering **1010**, ranking **1012** and caching **1014**. The AI Output Broker **1002** provides an API interface **1016** for configuring and managing various aspects of the broker. An AI Broker Database **1020** stores the results along with the meta-data information such as the request path. AI Broker Database **1020** creates an index of "derived requests" that may be used in future to select which set of "derived requests" an incoming request may fall into for further processing.

Referring now to FIG. **12** is an illustration of the combining h-LLMs in series, is described in more detail. User **1100** enters a prompt in user interface **1102**. The prompt **1104** is sent to an AI Input Broker **1106** which generates a derived prompt by adding more contextual information. The derived prompt is sent to multiple h-LLMs **1108** connected in series. The derived prompt goes to the first h-LLM in the sequence which generates results. The results of the first h-LLM are sent to the second h-LLM in the sequence for refinement/enhancement and then to the third h-LLM and so on. The AI Output Broker **1110** processes the results **1112** and sends the processed results to user **1200**.

Referring now to FIG. **13** is an illustration of combining h-LLMs in parallel, is described in more detail. User **1200** enters a prompt in user interface **1202**. The prompt **1204** is sent to an AI Input Broker **1206** which generates multiple derived prompts by adding more contextual information. The derived prompts are sent to multiple h-LLMs **1208** which process the prompt in parallel generating multiple results. The AI Output Broker **1210** processes the results and sends the processed results **1212** to the user **1200**.

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Referring now to FIG. 14 is an illustration of a hybrid approach of combining h-LLM in series and parallel, is described in more detail. User 1300 enters a prompt in user interface 1302. The prompt 1304 is sent to an AI Input Broker 1306 which generates multiple derived prompts by adding more contextual information. The derived prompts are sent to multiple h-LLMs 1308 which processes the prompts generating one or more results. The AI Output Broker 1310 processes the results and sends the processed results 1312 to the user 1300.

Referring now to FIG. 15 is an illustration of the lambda architecture for h-LLMs, is described in more detail. Lambda architecture is a way of processing massive quantities of data that provides access to batch-processing and stream-processing methods with a hybrid approach, often utilizing in-memory storage instead of disks for speedier processing. Such in-memory processing may be accomplished using a volatile memory device such as random-access memory (RAM) devices, static random-access memory (SRAM) devices, dynamics random-access memory (DRAM) devices, magnetoresistive random-access memory (MRAM) devices, and the like, or a non-volatile random-access memory (NVRAM) device. Such processing may be done partially or entirely in-memory.

This figure illustrates a lambda architecture for h-LLMs comprising batch layer 1402, real-time layer 1404 and a query layer 1406. New input data 1400 comes in continuously and is fed to the batch layer 1402 and real-time layer 1404 simultaneously. The batch layer 1402 maintains one or more h-LLMs which are updated/fine-tuned with the new data on a fixed schedule. Data is aggregated from the new input data 1400 over an aggregation duration that is tied to the fixed schedule. The real-time layer 1404 deals only with recent data which is not processed in the batch layer. The real-time layer 1404 maintains and updates smaller h-LLMs with incremental updates. The real-time layer 1404, also utilizes Map Reduce type analytics and computing and processing (See for example, tutorialspoint.com/map_reduce/map_reduce_introduction.htm) of tokens in the tokenization processes to improve speeds by which tokens are merged or otherwise aggregated in a distributed GPU computing environment, User 1412 sends a prompt 1408 through user interface 1410 to the query layer 1406. The query layer 1406 forwards the original prompt or creates one or more derived prompts which are sent to the batch and real-time layers. The query layer receives the results from the batch and real-time layers and performs tasks such as combining, ranking, filtering, assigning weights and priorities to the results and sends the best results to the user.

Referring now to FIG. 16 is an illustration of batch and real-time processing architecture for h-LLMs, is described in more detail. The input data stream 1500 is sent to batch layer 1506 and real-time layer 1526. The batch layer 1506 maintains a base h-LLM 1502 which is fine tuned 1504 in batch to generate fine-tuned h-LLM 1508. The real-time layer 1526 generates smaller h-LLMs with incremental updates 1514 in real-time increments 1512. The merger block 1516 combines and merges the h-LLMs from the batch layer and real-time layer to produce a combined h-LLM. The merged h-LLM is used with the query layer 1518 to respond to prompts 1520 sent by user 1524 through the user interface 1522.

Referring now to FIG. 17, an illustration of an in-memory processing architecture for h-LLMs, is described in more detail. The input data stream 1600 is sent to the data receiver 1602 which breaks the data into small batches 1604 which can be processed at least partially, and in some embodiments

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entirely, in-memory. The processing layer 1606 includes multiple h-LLMs which process the batches on input data and produce the batches of processed data 1608. Such batches may be produced after aggregating data from the input data stream 1600 over an aggregation duration.

Referring now to FIG. 18 is an illustration of the architecture of PDF label search tool with CatchUp GlassViewer, is described in more detail. User 1700 uploads a PDF document 1702 to the CatchUp document management system 1704. The text of the PDF document is extracted and indexed 1714 in the AEye backend system 1716. Such extraction and indexing may be performed using character recognition analysis, including optical character recognition analysis. The user opens the PDF document 1706 with the CatchUp GlassViewer application 1708 in a browser. User 1700 launches the label search tool 1710 within the CatchUp GlassViewer application 1708 and selects a label using the magnifier tool. The selected label is sent to the AEye backend system 1716 which retrieves and returns 1718 all occurrences of the label.

Referring now to FIG. 19 is an exemplary interface 1800 of the CatchUp platform showing the document management system, is described in more detail. Within this interface users can create new documents, upload existing documents, view and edit the documents.

Referring now to FIG. 20 is an exemplary interface 1900 of the CatchUp platform showing the PDF viewer (GlassViewer), is described in more detail. GlassViewer is a PDF viewer application with CatchUp that allows annotating and commenting PDF files. The annotations and comments are stored in a separate layer which is rendered above the PDF document.

Referring now to FIG. 21 is an exemplary interface 2000 of the CatchUp platform showing a magnifier tool 2002 within the GlassViewer for searching labels, is described in more detail. GlassViewer includes a PDF label searching tool called AEye Label Searcher that allows quickly searching for all occurrences of selected labels within the PDF. AEye Label Searcher uses a magnifier to select specific labels within a region of the PDF which are sent to the AEye backend for processing, and the results are then displayed, which include excerpts from the document where the labels are mentioned. In some embodiments, the AEye backend may lookup labels within multiple documents or return additional information generated from one or more h-LLM models as taught elsewhere in other embodiments of this invention. For example, a legal brief may be first generated using a local (in-house) database of briefs and then supplemented by h-LLMs that are trained on public-domain training sets of legal briefs, and the combination may be merged as needed.

Referring now to FIG. 22 is an exemplary interface of the CatchUp platform showing label search results within GlassViewer, is described in more detail. The labels selected using the magnifier within the AEye Label Searcher are sent to the AEye backend for processing and the results are then displayed as shown in this figure.

Referring now to FIG. 23, an illustration of the process used by organizations to decide the list of tasks for a project, is described in more detail. Given an objective 2200 such as applying for a business loan, the manager and team 2202 in an organization use their experience or current company process documents to come up with a plan in an ad-hoc manner. The plan includes a list of things to do 2204. The list of things to do may be informed by procedures and guidelines 2208 that may be provided by an administrative or

governing body, such as, for example, the Small Business administration, and may be retrieved from a website **2210** therefor.

Referring now to FIG. **24**, an illustration of the process of training LLMs to generate project, tasks and action items, is described in more detail. A Corpus of Company-Specific Processes and Procedures and Documents **2300** and Corpus of Public Processes and Procedures and Documents **2302** are used as input data for training of LLMs to generate Project, Tasks and Action Items **2304**. The content generated by the LLMs may be designated as supervised or labeled learning data **2308**. During the training process inputs such as requirements, constraints, assumptions, dependencies and risks **2310** are given. The trained LLMs are used to generate synthetic data which is then fed back to the models in feedback loop through a process called model refinement **2306**.

Referring now to FIG. **25**, an illustration of the process of creating a project within CatchUp using a prompt, is described in more detail. Within CatchUp, the user enters a prompt **2400** (such as create a project for implementing a pre-sales process in an IT Company). The prompt **2400** is sent to the AI Input Broker **2402** which generates one or more derived prompts **2410**. The derived prompts **2410** are sent to one or more h-LLMs **2404**, preferably several h-LLMs, which produce the results. The results **2412** are sent to the AI Output Broker **2406** which processes the results and performs tasks such as filtering, ranking, weighting, assigning priorities, and then sends the best results to the user. CatchUp **2408** creates a project and list of tasks and sub-tasks/action items within the project based on the response received from the AI Output Broker.

Referring now to FIG. **26**, an illustration of the process of Retrieval Augmented Generation (RAG), is described in more detail. RAG is a paradigm for augmenting LLM with custom data. A knowledge base **2500** is prepared from a corpus of documents (such as company-specific documents or public documents) in an indexing stage. The knowledge base **2500** can then be queried in the querying stage in which the relevant context **2502** is retrieved from the knowledge base to assist the LLM **2506** in generating a response **2508** to a query **2504**.

Referring now to FIG. **27**, an illustration of the process of indexing documents for RAG, is described in more detail. The data source **2600** comprises documents (such as company-specific documents or public documents). The data loader **2602** reads the data source files and parses them to create the parsed documents **2604**. The parsed documents **2604** are then indexed in an indexing stage **2606** in which the documents are converted into vector embeddings, and related metadata is inferred. The vector embeddings of documents are stored in a vector database which serves as the knowledge base **2608** for RAG.

Referring now to FIG. **28**, an illustration of the process of querying documents in RAG, is described in more detail. In the querying stage, the most relevant documents from the knowledge base **2700** are retrieved as the relevant context which is passed to the LLM along with the user's query. The relevant context allows the LLM to respond to the user query (including through use of the derived queries) even though the LLM may not be trained with the data from the knowledge base. The retrievers **2702** are used to retrieve the relevant context from the knowledge base **2700**. The retrievers **2702** may use different search and similarity matching strategies to search and retrieve the documents. Post-retrieval, the post-processor **2704** filters, transforms and ranks the relevant documents. The response synthesizer **2706**

generates a response from an LLM, using a user query. The query engine **2708** allows users to ask questions over their own data. The chat engine **2710** allows users to have chat conversations with their data. The agent **2712** is an automated decision maker that interacts with external tools.

The knowledge base contains a large set of documents (which can be private documents that are used to generate a "context" for the trained LLM". The Retriever may use efficient search algorithms, such as keyword-based or page-rank based searches for a cluster or subset, to identify the relevant context that is then used by the Retriever to generate the context that is they sent to the Query engine, or Chat Engine or LLM Agent. This is a two-step process that adds to how existing RAGs operate and removes the limitation that current LLMs start with an unsuitable context, and thus cannot do better. The same approach can also be used when training the LLM initially or fine-tuning it. Our approach thus combines traditional search algorithms of a set of documents to find the "best match" through traditional search mechanisms (e.g., page rank, or cluster search algorithms (both online, internet, and offline or local modes) to identify the suitable (e.g., based on a matching score of the subset of relevant documents or their portions/blocks/chunks) cluster, which may be ordered or reordered or transformed relevant to the prompt/query (or the derived queries and/or prompts) and associated context.

Referring now to FIG. **29**, an illustration of the RAG pipeline is described in more detail. RAG is a paradigm that allows combining LLMs with custom data. The RAG pipeline includes indexing and querying stages. In the indexing stage, documents and custom data **2800** (such as such as company-specific documents or public documents) are loaded and parsed. Next, the documents **2800** are split into smaller chunks using text splitters creating document splits **2804**. The splits **2804** are then stored in a vector database **2806**, such as ChromaDB, along with extracted metadata. Relationships between the document splits (or chunks, contexts, or blocks) may be stored in a graph database, such as Neo4j or GraphRAG (that was open-sourced by Microsoft in **2024**). The documents **2800** are stored in the form of vector embeddings in the vector database which allows fast searching, retrieval and comparison of documents using different similarity measures and/or through use of the metadata. This completes the indexing stage. In the querying stage, the user sends a query **2802**. The system retrieves splits from the vector database which are similar to the query. These retrieval splits **2818** and relevant associated metadata and processing results are then sent to the LLM as the context information along with the query as a prompt **2814**. The prompt **2814** comprises the user query **2802** and the retrieval splits **2818**. The LLM **2816** generates the answer **2812** based on the query **2802** and context information in the prompt **2814**.

Referring now to FIG. **30**, an illustration of an approach that uses multiple RAG pipelines with clusters of documents, is described in more detail. Existing LLMs and RAG pipelines have low accuracy and increased memory and performance penalties because of the way they develop context in normal high-level API use. A problem with LLMs includes the context being typically small (e.g. 4k tokens) as opposed to much larger context size, that could be 100k tokens or more. The context is not relevant and is created or selected through techniques that may not create the highest relevance score. To address these limitations an approach of using multiple retrieval augmented generation pipelines with clusters of documents is proposed. Based on a set of user queries **2900** the documents **2902** are searched **2904**

and clustered **2906**. The proposed approach thus combines traditional search of a set of documents to find the “best match” through traditional search mechanisms (e.g., page-rank), or cluster search algorithms (both online and offline modes) to identify the suitable (e.g., based on a matching Score of the subset of relevant documents) cluster and associated metadata relevant to the query and its context. For each document cluster **2906**, a separate RAG pipeline is created **2916**, **2918**, **2920** and produces its own result **2910**, **2912**, **2914**. The user queries are then routed to the most suitable RAG pipelines **2908** using a similarity/matching score and/or processing based on relationships between documents and/or their chunks and each other across the entire document set (utilizing, for example, a graph database as well for processing of relationships and/or metadata).

Referring now to FIG. **31**, an illustration of an approach that uses a prompt augmented with a search query, is described in more detail. Currently, in RAG pipelines, the prompt is used by both the Retriever and also the LLM. The language in the prompt may cause the retriever to generate low quality context that is then sent to the LLM, and results in poor quality results from the LLM. Instead, the prompt that we propose can be parsed or filtered or processed first to create a search query that is used to retrieve a subset of documents and/or their portions that have the best match with the search query derived from the prompt.

Prompt **3004** has a portion that is the prompt for the LLM **3006**, and a portion that is a search query **3002** for creating a subset of the documents to be used by the retriever for its matching function. The search query **3002** portion of the prompt is used to search a large corpus of documents **3000** and create small document clusters (such as corpus **3008** and **3010**). For each document cluster, a separate RAG pipeline may be used (such as **3020** and **3022**).

The prompt **3004** is processed to separate out a search query (before the documents are indexed into the vector database, and the full corpus of documents is broken (possible offline) into smaller clusters or chunks of documents. Based on the augmented prompt (that contains the prompt and search query), one or more clusters **3008**, **3010** are chosen to answer the query and the corresponding RAG pipelines **3020**, **3022** are then used to generate responses **3016**, **3018** to the prompts **3012**, **3014**. For example, consider a document corpus of a large number of PDF files related to real-estate contracts. The prompt is processed to separate out a search query and the full corpus of PDF files is broken into a rental contract subset/cluster and a purchase contract subset/cluster. Based on the query, the rental contract template subset/cluster is chosen and then the RAG pipeline processes the query in the context of rental contracts or their portions.

The processing of the prompt can be done in two ways—the prompt specifically and explicitly identifies “rental option” as a separate field or word, or the “rental option” may be derived from the prompt because the prompt uses “without a downpayment”. The context that results, e.g., “Rental Contracts”, is more accurate and responsive to the prompt that a combined context that is generated from all the PDFs in the original collection. This partitioning into subsets can be done offline, and the “search portion” of the prompt can be further processed by mapping or filtering or inference to the closest of the subsets of the documents in local or online databases, in case online creation of the subsets takes too long. Sometimes, online creation of the subsets of the collection of PDF files (or files or objects of other kinds, like images, videos, or text or songs) may be possible. In all cases accuracy is improved because the retriever is operating

on a more relevant subset of documents or their portions based on their content and/or associated metadata.

Referring now to FIG. **32**, an illustration of an approach that uses a search query to extract relevant context for the RAG pipelines, is described in more detail. A search query **3102** is used to extract documents **3108** from a large corpus **3100** and create smaller clusters of documents **3110** and **3112**. The search query **3102** is sent to an AI Input Broker **3104** which generates multiple derived queries which help in clustering documents into different clusters **3110** and **3112**. For each document cluster **3110**, **3112**, a separate RAG pipeline **3114**, **3116** is created. The prompt **3118** is sent **3120** to each of the RAG pipelines **3114**, **3116** and the responses from these pipelines are combined by the AI Output Broker **3122** which processes the results and performs tasks such as filtering, ranking, weighting, assigning priorities, and then sends the best results (in batch and/or real-time modes) **3124** to the user utilizing suitable APIs and supported by cloud-based container or virtualized systems, such as Kubernetes and Istio Service Mesh.

Referring now to FIG. **33**, an exemplary interface of the CatchUp platform for RAG, is described in more detail. The exemplary interface **3200** is designed for legal use cases and allows users to query public and private legal documents obtained from courts and other sources. Different query modes **3204** are supported for data sources such as public, private and company specific document clusters, online and offline. For querying documents, different query types **3206** are supported such as legal research, legal citation, legal case brief, legal writing and legal argument. User can optionally provide attributes **3202** such as legal issue, represented party, jurisdiction, case name, court name, year, statute name, area of law, for instance. The user query **3208** is sent to the CatchUp backend server which implements multiple RAG pipelines as shown in FIGS. **31** and **32**.

Referring now to FIG. **34**, an exemplary interface **3250** of the CatchUp platform showing response from Retrieval Augmented Generation (RAG) pipeline, is described in more detail. In response to a user query **3252** sent from the user interface **3250**, as shown in FIG. **33**, the response **3254** is returned as shown in FIG. **34**. The response **3254** includes the links to the source documents (or their portions or chunks, thereof) **3256** which were used as the relevant context with the RAG pipeline for generation of response.

Referring now to FIG. **35A**, an illustration of using a single LLM as known in the prior art is depicted. Currently a single large LLM **3302** is trained on all types of data and has large number of parameters (e.g. OpenAI GPT3.5 has 175 billion parameters and GPT-4 has over 1 trillion of parameters). The single large LLM **3302** responds individually to the prompt **3300** to generate the answer **3304**.

FIG. **35B** presents an illustration of using a network of LLMs. In an embodiment of the present invention, smaller LLMs (with 3 billion or 7 billion parameters, for example), each focused on a specific type of result (cost estimation, profit estimation, expense estimation or prediction) are provided in a network **3308**, and then network of LLMs **3308** is used to provide a composite result that is easier to prompt for, easier to optimize and easier to “explain” how it works by having smaller focused LLMs trained on specialized training sets. The smaller LLMs can be chosen from a library of LLMs and mixed and matched with other compatible LLMs using a communication protocol that exchange messages and exchange messages with a head-end or Manager LLM using an architecture that could be similar to Hadoop (where there is a master manager and bunch of worker nodes). Here is there is a master LLM **3312** and several

networked worker LLMs 3314. An adaptor module or a set of adapters 3316 may be configured to process these inter-LLM messages and inter-h-LLM messages and serve as a router for converting the format of a particular LLM's output to fit to the format of the API for another LLM and then route it on a cloud-container based virtualized network, similar to Istio Service Mesh that is used for routing services in typical cloud-based container implementations of computing microservices, that may be optionally based on GPUs. Therefore, if there are two choices available, one is to use an API Gateway that receives the APIs and converts and adapts the APIs for various LLMs and calls between LLMs, and another could be based on a service mesh that has agents, for example, that route the prompts and queries to the respective LLMs and process the results and redirect them to the appropriate destination. Other options are also available in batch or real-time modes.

The LLMs in a network can communicate with each other using client-server protocols such as HTTP/HTTPS, FTP, RPC, or peer-to-peer network protocols such as BitTorrent protocol, IPFS, WebRTC, for instance. Custom protocols may be developed for efficient communication of LLMs in a network. An example use case can be a network of Vision LLMs (one for each car on a road or in a platoon) which exchange information as to unusual traffic incidents that the autonomous car may not have been trained with to handle.

Referring now to FIG. 36, a comparison of the current and proposed approaches with the LASER system and SCORE-RAG system is described in more detail. FIG. 36A illustrates the prior art approach, FIG. 36B illustrates the proposed approach for processing long documents with LASER, and FIG. 36C illustrates the proposed approach for optimizing RAG with SCORE-RAG. In the prior art approach a User 3400 sends 3606 one or more long documents (as context) 3402 along with a prompt (containing the user's query) to an LLM 3406. The LLM processes the prompt and the context and generates a response 3408. The prior art approach is limited by the context length limit of the LLM, therefore, only a limited number of small documents or only a portion of a long document can be processed. Moreover, the LLM faces challenges when processing long documents, particularly in maintaining coherence and performing long-range reasoning due to the attention span problem which causes a noticeable drop in performance as the length of the input context increases, typically beyond 10,000 to 50,000 tokens.

As shown in FIG. 36B, an embodiment of the present invention, the LASER system 3414, uses an iterative attention-focusing technique that dynamically refines and condenses document context to improve model comprehension and coherence over extended inputs. The process begins with a User 3410 who has one or more Long Documents 3412 for processing. Instead of sending the document directly to an LLM 3418, long document(s) 3412 are first passed through a LASER system 3414. The LASER system 3414 employs the iterative attention-focusing technique to refine and distill the context of the long document and its chunks. After refinement, the refined documents (or refined generated documents and subsets/chunks/blocks) 3416 (containing processed and condensed information and also extracted metadata, optionally stored in a graph database) are then sent to the LLM 3418 for final analysis or generation tasks. One distinction between the approaches in FIGS. 36A and 36B is the introduction of the LASER system 3414 for attention span refinement. The LASER system 3414 acts as an intermediary processor that prepares the long document(s) 3412 for more effective handling by the LLM 3418. The LASER system 3414 addresses the limitations of the

prior art solution by introducing an intelligent preprocessing step which enables LLMs to effectively handle long documents that would otherwise exceed their attention capacity.

FIG. 36C illustrates another embodiment of the invention, a SCORE-RAG system 3430 which can be combined with a LASER system 3426 to create a solution for processing long documents and generating accurate responses that may perform better than the prior art solution. The process begins with a User 3422 who has one or more long documents 3424 for processing. The documents and/or their chunks are processed through the LASER system 3426 which produces refined documents 3428 which are then fed to the SCORE-RAG system 3430. The SCORE-RAG system 3430 uses context-optimized retrieval techniques to enhance the traditional RAG approach. This system enhances the traditional RAG approach by incorporating advanced techniques such as topic modeling, intelligent document chunking, citation analysis, and hybrid search methods. The SCORE-RAG system 3430 processes the refined documents 3428 to create optimized chunks of information (and associated metadata based relationship information between chunks themselves (within a document and across documents) and also between chunks and documents), which are then indexed and prepared for efficient retrieval and use in language generation by the LLM 3432. New documents or documentation may be assembled (including reordering) "on-the-fly" in batch mode or real-time modes based on this processing. At a high level, without restricting or limiting the invention, given a corpus of documents, the LASER system 3426 processes these documents, document to document, to look at portions of relevance and interest (not in response to a particular user query, usually) but based on categories or types of information that may be present in each document that may be of interest. A long document processed by the LASER system 3426 results in a refined document. The processing of long documents by the laser system 3426 is usually perform, document by document, offline. The refined document may also be a summary, or a subset, or portions (or chunks) of a document that are of most interest in general, by a category. The LASER system 3426 may itself utilize an LLM to help it identify portions or chunks of most relevance to a category of information in the process of creation of the refined documents. The refined documents 3428 may then be taken by the SCORE-RAG system 3430 which may, in response to a query, look at the refined documents 3428 and their tagged portions or chunks organized along with metadata and relationships to each other and to categories, for example, and carry out operations on ranking and scoring those chunks that are most relevant to a particular query or a set of queries, for example, a batch of queries or derived queries.

The LASER system 3426 usually performs offline preprocessing that is not directly related to user query or derived query. The LASER system 3426 processes all documents one by one and identifies portions (or chunks or contexts) of them that fall into one or more categories of interesting information, for example, receipts, bills, tax data, customer names, inventories, etc. The LASER system 3426 then redefines each document to call out these portions and adds meta-data to identify the categories. Additionally, the LASER system 3426 may summarize the portions/chunks/contexts. Multiple categories can exist for the same portion, for example, a single portion can be in the receipt and tax categories. The LASER system 3426 may then repeat this process for all long documents. It can use a LLM or other search means (e.g., cluster search) for this process. This entire process can be done once or relatively few times for

each document/Typically, the LASER system **3426** performs this process not responsive to a real-time query or prompt from the user. It can be done each time a long document is loaded into the system. The prompts from users can be used to create categories that are used in the LASER system **3426**.

The SCORE-RAG system **3430** may perform online real-time processing to create relevant context in response to a derived prompt or a user prompt. The SCORE-RAG system **3430** may then take the refined documents **3428**, which are categorized chunks from these documents and other meta-data, including relationships between chunks and documents, and then performs scoring, ranking, collection, and collation in response to a user query and/or its derived queries. The highly scored chunks may then be sent to the LLM **3432** for processing to create results for the user. The SCORE-RAG system **3430** can use graph and vector databases to create chunks that are most relevant to the user prompt of derived prompts and send the context to the LLM.

The combined approach using the LASER system **3426** and the SCORE-RAG system **3430** may provide improvements over prior solutions. By first refining the document through the LASER system **3426**, the subsequent RAG optimization process can work with a more focused and coherent input, potentially improving its effectiveness. The refined documents **3428** (or refined generated documents and subsets/portions/chunks/blocks) allows for more accurate topic modeling and chunking in the SCORE-RAG system **3430**, leading to better retrieval results. The iterative refinement process helps maintain important contextual information, which can then be effectively utilized in the RAG system's retrieval and generation stages. Both the LASER system **3426** and the SCORE-RAG system **3430** are designed to handle large volumes of text, making this combined approach suitable for processing extensive document collections or extremely long individual documents. This integrated system can be applied to a wide range of document types and domains, such as scientific literature, legal documents, and technical manuals, for instance.

Referring now to FIG. 37, an illustration of the high-level architecture of a LASER system **3500** according to an embodiment of the invention is described in more detail. The LASER system **3500** architecture comprises five primary modules, including an input module **3502**, a model module **3504**, an iteration controller **3506**, a knowledge module **3508**, and an output handler **3510**. Each of these modules contains sub-components that perform specific functions within the overall system. The input module **3502** serves as the entry point for document processing. It comprises a long document parser, which is operable to ingest long documents and breaking them down into manageable blocks, and an iterative context generator, which is operable to create and refine batches of blocks from the long document processor for processing in subsequent stages. The model module **3504** forms the core processing unit of the system. It comprises, and in some embodiments consists of, an attention model which is an LLM that is operable to process and comprehend the document content, and a ranking Unit that is operable to evaluate and score the outputs produced by the attention model, identifying the most relevant and coherent content. The iteration controller **3506** is operable to control the flow of the iterative process. It comprises a loop control, which is operable to determine when to continue or terminate the iteration process based on predefined criteria, and an output aggregator, which is operable to collect and consolidate the results from multiple iterations. The knowledge module **3508** is operable to enhance the ability of the LASER system **3500** to distill and organize information. The

knowledge module **3508** comprises an extractive summarizer, which is operable to identify and extract key sentences or passages from the processed content, a document clustering component, which is operable to group related content/blocks/chunks based on semantic similarity or relevance to a topic, and a virtual document creator which assembles chunks (in any order or sequence or through creation of summaries) together to create an improved document on a topic relative to the long document(s). The output handler **3510** is operable to prepare the processed information for presentation. The output handler **3510** comprises a response filter, which is operable to perform a final validation of the output to ensure quality and relevance, and a result presentation component, which is operable to format the output for user consumption, potentially including visualizations or explanations. This architecture enables the LASER system **3500** to progressively refine and focus attention on the most relevant parts of long documents through multiple iterations.

Referring now to FIG. 38, a flow chart of the LASER system depicting a process by which long documents are processed and refined to enhance the attention span of LLMs according to an embodiment of the invention is described in more detail. The process begins with a long document **3600**, which serves as the input to the LASER system. It is contemplated that multiple long documents may be provided. At step **3602**, the document is split into smaller, manageable blocks/chunks. The block/chunk size is configurable, typically ranging between 10,000 to 50,000 tokens. This step facilitates breaking down the long document into sizes that are more aligned with the typical attention span of current LLMs. Following the splitting operation, the blocks/chunks undergo a shuffling process at step **3604** to mitigate any potential position bias that might be present in the original document structure. By randomizing the order of the blocks/chunks, the system ensures a more uniform treatment of the content during subsequent processing stages. Alternatively, reordering of blocks/chunks may be done such that most similar blocks/chunks are either at the beginning or at the end, to avoid attention span problems with LLMs. The shuffled/reordered blocks/chunks are then organized into initial batches at step **3606**. These batches serve as the first set of inputs for the LLM processing stage. The batching strategy allows for efficient processing while maintaining a balance between context size and manageability. An iterative process **3608** begins at step **3610**, where the batches are processed through an LLM. The processing step **3610** involves applying the capabilities of the LLM to comprehend and reason over the content of each batch. The outputs generated by the LLM are then subjected to a ranking process at step **3612**. This step evaluates the quality of the LLM's outputs, typically based on factors such as coherence, relevance, and alignment with the original document's intent. At step **3614**, the system determines whether the iteration process should continue or conclude. This decision is based on predefined criteria, which may include factors such as the number of iterations completed, the degree of content refinement achieved, or specific quality thresholds being met. If the decision is to continue the iteration, the process moves to step **3618**, where the top K of the ranked blocks/chunks are selected, where K is a configurable parameter, typically between 10 to 50%. This selection process ensures that only the most relevant and coherent content is carried forward to the next iteration. The selected blocks/chunks then undergo a clustering operation at step **3618**. This step groups related content together, further refining the organization of the information based on

semantic similarities. At step **3622**, new batches are formed from these clustered blocks, creating a more focused and refined set of inputs for the next iteration. These new batches are then fed back into the LLM processing stage **3610**, beginning the next iteration cycle. The iterative process **3608** continues until the termination criteria are met, at which point the process concludes, outputting the final condensed version of the document **3616**. This output represents a distilled version of the original long document, containing the most relevant and coherent information as determined by the iterative attention focusing process of the LASER system.

Through this iterative approach, the system progressively refines and focuses the content of long documents, enabling LLMs to more effectively process and reason over extended contexts. This method addresses attention span limitations of current LLMs and provides a novel solution for handling long-form content across various domains. The LASER system is operable to iteratively refine the context to focus on the most relevant information to help the LLM maintain coherence and attention.

Aspects of a LASER system according to an embodiment of the invention include splitting long inputs into multiple blocks/chunks (e.g. 10k-50k tokens each) and processing the blocks/chunks separately, ranking and/or scoring the outputs from each block/chunk to determine most relevant content, constructing new condensed blocks/chunks from the highest scoring outputs for next round and/or delete irrelevant information, iteratively processing the blocks/chunks through an LLM, selecting the best outputs, and condensing or summarizing the best outputs, and gradually concentrating the context into fewer but more focused subsets of documents or chunks or summaries.

The LASER system allows an LLM to digest manageable sized contexts, select useful results/signals, and carry those forward in a refined set of inputs. The LASER system works similar to a funnel by distilling the context. The ranking/selection steps of the LASER process improve the ability of an LLM to identify and focus on the pertinent content across rounds of processing. The LASER system guides the LLM's limited attention by incrementally removing irrelevant information and emphasizing the best results/signals from prior rounds. LASER system can have different variations, such as having multiple rounds of a large number of small blocks before condensing, however, the incremental refinement of context and attention through derived inputs remains the core idea.

Referring now to FIG. **39**, an illustration of an adaptation of the MapReduce model for the LASER system, is described in more detail. The figure presents a flow chart depicting how the concepts of mapping, translating, shuffling, and reducing are applied to the iterative attention focusing approach of the LASER system, enabling efficient processing of long documents. The process begins with a long document **3700**, which is the input to the system. The long document **3700** is split into blocks/chunks at step **3702**. Step **3702** prepares the document for distributed processing. The system then enters the map phase at step **3704**, which is a component of the MapReduce paradigm. In the map phase, each block/chunk is independently processed by the LLM. This is represented by the parallel processes **3706**, **3708** and **3710**, each handling a separate block/chunk. The number of these parallel processes can be scaled based on the available computational resources, allowing for efficient handling of long documents. Following the map phase, the system moves to the shuffle & sort stage at step **3712**. This stage facilitates reorganizing the processed information from

the individual blocks/chunks. It allows for the redistribution of related content across the processed blocks, preparing for the subsequent reduce phase. The reduce phase begins at step **3714**, where the shuffled and sorted data is consolidated. The reduce phase is represented by processes **3716**, **3718** and **3720**, which perform ranking and clustering operations on batches of the processed and reorganized blocks/chunks. The number of reduce processes can be adjusted based on the desired granularity of the reduce operation. The results from the reduce phase are then combined at step **3722**, creating a consolidated view of the processed document. At this point, the system evaluates whether the iteration is complete at step **3724** based on predefined criteria such as the number of iterations, the degree of refinement achieved, or specific quality thresholds. If the iteration is not complete, the system forms new blocks/chunks at step **3728** based on the reduced and refined content. These new blocks/chunks are then fed back into the map phase **3704**, beginning another iteration cycle. This iterative loop allows for progressive refinement of the document's content, gradually focusing on the most relevant and coherent information. When the iteration criteria are met, the process concludes, outputting the final refined output at step **3726**. This output represents a distilled version of the original long document, containing the most relevant and coherent information.

This adapted MapReduce model may be advantageous for the proposed attention span refinement system. The parallel nature of the map phase allows for efficient processing of extremely long documents by distributing the computational load. The number of map-and-reduce processes can be adjusted based on available resources and specific requirements of the task. The cyclical nature of the process, facilitated by the iteration check and feedback loop, allows for progressive improvement of the document's focus and coherence. This model enables the system to leverage distributed computing architectures, potentially improving processing speed for very large documents. Processing of documents can be done in a fault tolerant manner, allowing for robust processing of long documents even in the face of potential hardware or software failures. By adapting the MapReduce model to the attention span refinement process, the LASER system provides an efficient and scalable approach to handle long documents, and address the limitations of current LLMs.

Referring now to FIG. **40**, an illustration of the architecture of a SCORE-RAG system according to an embodiment of the invention is described in detail. The SCORE-RAG system **3800** comprises several modules, each designed to perform specific functions in the overall process. A document processor **3802** serves as the entry point for the system. It includes an input handler operable to receive documents in various formats, a text extractor operable to parse and prepare the text comprised by the received documents for further processing, for example, by creating chunks from the text, and a parent document tracker operable to maintain relationships between original documents and their chunks. A topic modeling engine **3804** is operable to analyze the received document to identify main topics and themes. It comprises, and in some embodiments consists of, a topic analyzer and a theme identifier, working in tandem to create a semantic map of the document. A document chunking module **3806** is operable to break down the received document into semantically meaningful segments. It includes a semantic segmenter for intelligent text division, a topic tagger operable to associate relevant topics with each chunk, and a multi-format generator operable to create multiple representations for each chunk (e.g., summaries, hypotheti-

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cal questions related to the chunk). A citation analyzer **3808** comprises a reference extractor and an impact assessor. The citation analyzer is operable to identify and evaluate citations within the received document, contributing to the overall relevance scoring of different text segments. A chunk selection and ranking module **3810** employs a relevance scorer in being operable to assess the importance of each chunk, a top chunk selector in being operable to identify the most pertinent segments for indexing and retrieval, and a time-weighted scorer which is operable to incorporate the timestamps into chunk scoring (where more recent chunks may be ranked higher). A metadata enrichment module **3812** is operable to enhance the selected chunks with meta-data. This module uses a tag assigner to append relevant metadata, a context enhancer to add additional contextual information, and a timestamp tracker to record creation and access times for time-weighted retrieval. An indexing engine **3814** comprises a vector database, a graph database and a full-text search engine. This module is operable to prepare the processed chunks for efficient storage and retrieval. Multiple vector representations of the same document or chunk are stored (e.g. summarized versions, tagged versions, hypothetical questions for document/chunk, etc). A query processor **3816** is operable to handle user inputs. This module includes a query analyzer for understanding query structure, an intent classifier for determining the user's information needs, and a derived-query/prompt generator which is operable to create multiple prospective queries/prompts from a user input. A retrieval orchestrator **3818** is operable to coordinate the retrieval process. This module includes a cache checker for rapid information access and a search dispatcher for initiating more complex searches when needed. The search dispatcher implements different search and retrieval techniques such as vector similarity search, maximal marginal relevance (MMR) search, keyword-based search, cluster search (which groups similar documents/chunks into clusters based on their similarity and the query is match with the most relevant clusters) and ensemble search (which combines results from multiple search methods, for example, a combination of vector similarity and keyword based search). As disclosed in U.S. Provisional Patent Application Ser. No. 63/463,913, which is incorporated by reference herein above, a label or a search query results in a search to a set of documents to obtain the most relevant set.

An augmentation engine **3820** may be operable to employ a context integrator to combine retrieved information with the original query and a query enhancer to enhance the query based on this additional context. A generation module **3822** may be configured to interface with an LLM. This module includes an LLM interface for query processing and a response generator for producing coherent and relevant responses. An evaluation and fine-tuning module **3824** includes a performance analyzer which is operable to monitor system-wide metrics and evaluate the response quality, and a model fine-tuner which is operable to adjust the system parameters for optimization.

Referring now to FIG. 41, a flow chart of a process performed by a SCORE-RAG system according to an embodiment of the invention is described in more detail. The process begins with a document **3900**, which may be a refined output from the LASER system or any other textual input requiring processing. The document **3900** undergoes topic modeling at step **3902** to identify key topics and categories within the text. The document is segmented into semantically meaningful chunks at step **3904**, with each chunk tagged according to the identified topics. Citation

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Analysis is performed at step **3906** which involves analyzing citations or references within the document to identify the most influential or frequently cited sources. Based on the topic relevance and citation analysis, the most important chunks of the document are selected for further processing at step **3908**. The selected chunks are enriched with meta-data at step **3910**, including topic tags, citation information, and other relevant attributes. At step **3912** the processed and tagged chunks are indexed in a vector/graph database and/or a full-text search engine (such as, for example, Elasticsearch or Solr), enabling efficient retrieval. Indexing can be done in a combination of databases (vector/graph) and full-text search engines to enable efficient hybrid search at a later stage. At step **3914**, the system waits for a query/prompt input. When User **3916** provides the query it triggers the retrieval process. The retrieval stage begins at step **3918**. Upon receiving a query, the system initiates the retrieval process to find relevant information. At step **3920**, a cache check is performed. The system first checks if the required information is available in the cache/in-memory. If there's a cache hit, i.e. the required information is available in-memory, the information is fetched directly from the cache at step **3922**. In case of a cache miss, i.e. the required information is not available in-memory, the system performs a hybrid search at step **3924** to retrieve relevant information. At augmentation stage is performed at step **3926** where the retrieved information, whether from cache or search, is used to augment the original query. A generation stage is performed at step **3928** where the system generates a response using the augmented query. The generated response undergoes an evaluation process at step **3930** to assess its quality and relevance. The system determines if the generated response meets the required performance criteria at step **3932**. If satisfactory, a final response is produced at step **3934**. If unsatisfactory, the system initiates a fine-tuning process at step **3936**. In the fine-tuning process, the system can tweak the models and parameters used in the topic modeling, document chunking and tagging, citation analysis, chunk selection, meta-data tagging, indexing and hybrid search stages, to improve the overall performance. The system may then iterate steps **3902** through **3912** and **3918** through **3932** until the generated response is determined to b

The SCORE-RAG system of FIG. 41 may be advantageous over previous solutions. Through topic modeling and intelligent chunking, the system can better understand and retrieve relevant information. The use of caching and hybrid search techniques may enhance retrieval speed and accuracy. The fine-tuning process may allow the system to continuously improve its performance based on evaluation results.

In the context of the SCORE-RAG system, it should be noted that the RAG optimizations can occur before or after a document is stored/indexed in a Vector database.

In the context of the present invention, it should be noted that the term "Document" is used in a broad and inclusive manner. A person skilled in the art should understand that "Document" may refer to, but is not limited to: a traditional text-based document in its entirety; a portion or chunk of a larger document along with extracted metadata; a block of text, regardless of its source; a summary or abstract of a document; a combination of multiple distinct documents; a combination of summaries or chunks from multiple documents; a context or set of contextual information, including metadata; any digital content that can be processed as text, including web pages, emails, or social media posts; structured or semi-structured data that can be converted into a

textual format; and/or a collection of related information, regardless of its original format or source.

Referring now to FIG. 42, an illustration of using the SCORE-RAG system for multi-modal RAG is described in more detail. The SCORE-RAG system 4002 of the present embodiment is designed with advanced multimodal capabilities, enabling it to process and generate a wide array of content types, including text, audio, video, software code, and images. This system handles multi-modal inputs 4000, including but not limited to text, image, video, audio, and code, making it versatile for diverse applications. To achieve improved performance across these varied modalities, the SCORE-RAG system 4002 employs multiple embedding models 4004, 4006, 4008, 4010, 4012 and specialized vector databases 4014, 4016, 4018, 4020, 4022 each fine-tuned for a specific content type such as text, image, video, audio, or code. This specialized approach ensures that the unique characteristics and nuances of each modality are accurately captured and indexed. For the generation phase, the SCORE-RAG system 4002 utilizes an ensemble of LLMs or h-LLMs 4024, each specialized for different tasks. These may include models optimized for question-answering, code generation, image interpretation, audio transcription, and video analysis, among others. This multi-faceted approach allows CORE-RAG to process a wide range of input types and also generate appropriate and context-aware multi-modal outputs. A User 4026 sends a query 4028 (comprising multi-modal inputs) to the SCORE-RAG system 4002 and received a multi-modal response 4030. The multimodal capability of the SCORE-RAG system 4002 enhances the system's utility across diverse fields such as education, research, software development, multimedia content creation, and data analysis, for instance.

Referring now to FIG. 43, an illustration of a process of using a LASER system and a SCORE-RAG system for multi-modal RAG according to an embodiment of the invention is described in more detail. The process starts with a multi-modal input (for example, text, image, audio, video, or code) being ingested by the LASER system 4102 which is then refined/processed/distilled/filtered. The processed or refined data/documents (text, image, video, audio, code) are then fed to the SCORE-RAG system 4104. The SCORE-RAG system 4104 can integrate information from various sources and modalities, providing comprehensive and accurate responses that can combine textual explanations, code snippets, audio interpretations, and visual analyses as needed. For generation tasks, the SCORE-RAG system 4104 utilizes an ensemble of LLMs or h-LLMs 4106, each specialized for different tasks. A User 4108 sends a query 4110 (comprising multi-modal inputs) to the SCORE-RAG system 4104 and receives a multi-modal response 4112. The output from the SCORE-RAG system 4104 can also be fed back to the LASER system 4102 for iterative processing/refinement.

Referring now to FIG. 44, an illustration of the different meanings of "Document" in the context of the present invention, is described in more detail. A person skilled in the art should understand that "Chunk" or "Block" may refer to, but is not limited to: traditional forms 4232, such as entire text-based document 4234, a portion/chunk of larger document 4238, or a block of text from any source 4236; data forms 4202, such as structured/semi-structured data convertible to text 4204, or related information collection 4206; digital forms 4208, such as social media posts 4218, web pages 4216, emails 4210, images 4212, and audio/video 4214; contextual forms 4220, such as context or contextual information set 4222; and derived forms 4224, such as a

summary/abstract of a document 4226, a combination of summaries/chunks 4228, or a combination of multiple documents 4230.

The meaning of "document" in any specific instance is dependent upon the context within which it is used, and the meaning may be expressly modified within the description of particular embodiments. This broad definition is intended to encompass the various ways in which textual or informational content may be presented, processed, or manipulated within the scope of the present invention.

Referring now to FIG. 45, an illustration of the different meanings of "Chunk/Block" in the context of the present invention, is described in more detail. In the context of the present invention, it should be noted that the terms "chunk" and "block" are used in a broad and inclusive manner. A person skilled in the art should understand that "chunk" or "block" may refer to, but is not limited to: document-based 4302, such as a portion or segment of a larger document or text 4304, a fragment of a document that maintains some level of context or meaning 4306, a semantically coherent section of text 4308, regardless of its size, a section of a document defined by structural elements 4310 (e.g., headers, chapters, sections); content-based 4318, such as a unit of text 4324 defined by a specific number of tokens, words, sentences, or paragraphs, a piece of information extracted from a larger context 4320; processing-based 4334, such as a unit of data processed or analyzed in a single operation, a unit of information in any digital format that can be processed as a discrete entity 4340, a unit of text used for processing, indexing, or retrieval purposes 4338; or organizational 4326, such as a logical division of content that may span across multiple physical documents 4328, a section of text or data defined by temporal or sequential ordering 4332, a portion of a document or dataset selected based on specific criteria or algorithms 4330.

The terms "chunk" and "block" may be used interchangeably or with distinct meanings depending on the specific context within the description of the invention. The precise definition, size, or characteristics of a "chunk" or "block" may vary based on the particular embodiment, implementation, or application of the invention being described.

Referring now to FIG. 46, an illustration of the different meanings of "meta-data" in the context of the present invention, is described in more detail. A person skilled in the art should understand that "meta-data" may refer to, but is not limited to: identification 4432, such as filename 4436, unique ID 4434, version number 4440, and hash value 4438; authorship 4422, such as author name 4424, creator 4428, contributors 4426, and rorganization 4430; temporal 4402, such as creation date 4404, last modified date 4408, publication date 4410, and expiration date 4404; source 4412, such as origin URL 4416, database source 4420, file path 4414, and reference information 4418; content 4452, such as title 4458, abstract 4454, keywords 4460, language 4456, and genre/category 4462; technical 4442, such as file type 4450, file size 4444, character encoding 4446, and word/page count 4448; rights 4484, such as copyright information 4490, license 4488, access permissions 4492, and usage restrictions 4486; relational 4474, such as parent document 4482, related documents 4478, position within document 4480, and hierarchy information 4476; contextual 4464, such as project name 4468, department 4472, subject area 4466, and intended audience 4470; and processing 4494, such as processing status 3800, indexing information 3802, classification tags 4498, and confidence score 4496.

Referring now to FIG. 47, an illustration of the different meanings of "document processing" or "chunk/block pro-

cessing” in the context of the present invention, is described in more detail. In the context of the present invention, it should be noted that “document processing” and “chunk processing” are used in a broad and inclusive manner. A person skilled in the art should understand that “document processing” or “chunk/block processing” may refer to, but is not limited to: document manipulation **4556**, such as documents being split into chunks **4563**, and documents can be summarized into one or more chunks **4558**; chunking process **4500**, which can vary based on specific needs **4502**, for example a ten-page PDF can be broken into 5 three-page chunks (with overlapping text) **4506**, the same PDF could be divided into 5 two-page chunks **4510**, the same PDF could be trimmed and split into just three 2-page chunks **4508**, and/or, alternatively, the same PDF could be condensed by summarization into two 3-page chunks **4504**; adding meta-data **4512**, where rich metadata is associated with each chunk, such as descriptive metadata about the chunk itself **4516**, relationship metadata linking the chunk to its parent document **4518**, and relationship metadata connecting the chunk to other chunks and their parent documents **4520**; grouping **4522**, where chunks and documents can be grouped based on similarities **4524** (affine chunk groups **4528** and affine document groups **4526** representing similar content) or based on differences **4530** (anti-affine chunk groups **4534** and anti-affine document groups **4532** representing different viewpoints or contrasting content; complex relationships exist within the system **4536**, such as documents linking to other documents **4538**, documents linking to their constituent chunks **4548**, chunks linking to other chunks **4550** within the same document **4552** and across different documents **4554**, and relationship types **4540** which can be inherited **4542**, derived on the fly **4546**, or captured in a graph structure **4544**.

The precise definitions, sizes, characteristics, and relationships of “documents”, “chunks”, “blocks”, “meta-data” and “document processing” may vary based on the particular embodiment, implementation, or application of the invention being described.

Referring now to FIG. 48, an illustration of an approach of generating derived prompts and using them with a RAG system or a SCORE-RAG system **4606** to generate the best results, is described in more detail. User **4600** provides a prompt **4604**. The prompt is sent to the AI Input Broker **4602** which generates multiple derived prompts. Existing RAG systems often rely on single, user-provided prompts for information retrieval. This approach can be limited by the specificity of the prompt and may not capture the full spectrum of relevant information. The present invention addresses this limitation by automatically generating a set of derived prompts that explore various aspects, phrasings, and contextual interpretations of the original query. The prompt derivation engine employs several techniques to generate derived prompts:

1. Paraphrasing: The original prompt is rephrased while maintaining its core meaning. This captures different ways of expressing the same query.
2. Query Expansion: Relevant terms or synonyms are added to the original query to broaden its scope and capture related concepts.
3. Aspect-based Prompts: The original query is broken down into different aspects or sub-topics, with separate prompts created for each.
4. Question Transformation: Statements are converted into questions and vice versa. The query is also transformed into different question types (e.g., who, what, when, where, why, how).

5. Persona-based Prompts: The query is rephrased from different persona perspectives (e.g., expert, novice, skeptic).
6. Contextual Variations: Context is added or modified to the original prompt to explore different angles.
7. Abstraction and Specification: More general and more specific versions of the original query are created.

The knowledge documents **4622** are fed to a RAG or SCORE-RAG system **4606** which creates embeddings of the documents and indexes in the vector database **4624**. If a SCORE-RAG system is used, it leverages context-optimized retrieval techniques to enhance the traditional RAG approach. This system enhances the traditional RAG approach by incorporating advanced techniques such as topic modeling, intelligent document chunking, citation analysis, and hybrid search methods. Upon receiving the derived prompts **4608**, the RAG or SCORE-RAG system **4606**, proceeds with the retrieval and augmentation tasks creating refined contexts **4616**, **4618**, **4620**. These refined contexts are then transmitted to one or more LLMs **4610**, **4612**, **4614** for the generation task.

Referring now to FIG. 49, an illustration of an approach of generating derived prompts and using a LASER system **7408** and a RAG or SCORE-RAG system **4714** to generate the improved results, is described in more detail. User **4700** provides a prompt **4704**. The prompt is sent to an AI Input Broker **4702** which generates multiple derived prompts. The knowledge documents **4706** are passed through the LASER system **4708**. The LASER system **4708** employs the iterative attention focusing technique to refine and distill the context of the long document. After refinement, the refined documents **4710** fed to the RAG or SCORE-RAG system **4714** which creates embeddings of the documents and indexes in a vector database **4728**. Upon receiving the derived prompts **4712**, the RAG or SCORE-RAG system **4714**, proceeds with the retrieval and augmentation tasks creating refined contexts **4722**, **4724**, **4726**. These refined contexts are then transmitting to one or more LLMs **4716**, **4718**, **4720** for the generation task. Distinctions between the approaches in FIGS. 48 and 49 lie in the introduction of the LASER system **4710** for attention span refinement. The LASER system **4710** acts as an intermediary processor that prepares the long documents (knowledge documents **4706**) for more effective handling by the LLM. The LASER system **4710** addresses the limitations of the prior art by introducing an intelligent preprocessing step which enables LLMs to effectively handle long documents that would otherwise exceed their attention capacity.

Referring now to FIG. 50, an illustration of an exemplary set of APIs of a LASER system according to an embodiment of the invention are described in more detail. The illustration depicts APIs categorized based on modules: Document Intake **4802**; Document Splitting **4804**; Batch Processing **4806**; Model Processing **4808**; Ranking **4810**; Clustering **4812**; Iteration Control **4814**; and Result Aggregation **4816**. The Document Intake module **4802** includes APIs for document submission and document retrieval. These endpoints facilitate the ingestion and initial metadata retrieval of documents within the system. The Document Splitting module **4804** includes APIs for splitting a document and getting document blocks/chunks. The Batch Processing module **4806** includes APIs for creating a batch for processing and getting batch information. The Model Processing module **4808** includes APIs for processing a batch using the model and getting the batch output. The Ranking module **4810** includes APIs for ranking batch outputs and getting batch rankings. The Clustering module **4812** includes APIs for

clustering blocks/chunks and getting document clusters. The Iteration Control module **4814** includes APIs for initiating subsequent iterations and status retrieval. These endpoints enable the system to manage and monitor the iterative refinement process. The Result Aggregation module **4816** has APIs for retrieving the refined documents, representing the final output of the long document attention span refinement process.

Referring now to FIG. **51**, an illustration of an exemplary set of APIs of the SCORE-RAG system are described in more detail. The illustration depicts the APIs categorized based on modules: Document Processing **4902**; Metadata Tagging **4904**; Indexing **4906**; Query Processing **4908**; Generation **4910**; Evaluation **4912**; and System Management **4914**. The Document Processing module **4902** includes APIs for document ingestion, analysis, and pre-processing. The Metadata Tagging module **4904** includes APIs for enriching document chunks with metadata. The Indexing module **4906** includes APIs for managing the indexing of processed document chunks. The Query Processing module **4908** includes APIs for handling user queries and search functionalities. The Generation **4910** module includes APIs for response generation. The Evaluation **4912** module includes APIs for assessing the quality of generated responses. The System Management module **4914** includes APIs for system-level operations and monitoring.

The Application Programming Interfaces (APIs) described in FIG. **50** and FIG. **51** herein are not limited to any particular implementation technology or architecture. It should be understood that these APIs can be implemented using various technological approaches, including but not limited to microservices architectures, serverless computing platforms (e.g., AWS Lambda, Azure Functions, Google Cloud Functions), Function-as-a-Service (FaaS) models, containerization technologies (e.g., Docker, Kubernetes), remote procedure call frameworks such as gRPC (gRPC Remote Procedure Call), RESTful web services, GraphQL endpoints, or any combination thereof. The choice of implementation technology may depend on factors such as scalability requirements, performance needs, deployment environment constraints, and integration with existing systems. Furthermore, the APIs may be implemented using different programming languages, frameworks, and runtime environments, as appropriate for the specific use case and technological ecosystem. The flexibility in implementation allows for optimal adaptation of the system to various operational contexts while maintaining the core functionality and architectural principles described in this specification. The textbook *A First Course in Cloud-Based Microservices* (Arshdeep Bahga and Vijay Madisetti, VPT Press, 2024) is incorporated herein by reference in its entirety.

Throughout the application, reference may be made to various computer hardware, including servers, GPUs, storage, cloud storage, and the like. It is contemplated and included within the scope of the invention that the LASER and SCORE-RAG systems and their various components may be software executed on computer devices, including servers, personal computers, smartphone devices, and the like, each comprising a processor configured to execute commands received from software (such as microprocessors, field-programmable gate arrays, integrated circuits, and the like), a non-transitory computer-readable storage medium positioned in electrical communication with the processor and operable to store software and other digital information thereupon in one or both of transitory and non-transitory status (such as hard disk drives, solid state drives, flash drives, compact flash drives, SD drives,

memory, and the like), and a network communication device operable to communicate across computer networks as are known in the art, including, but not limited to, wide area networks such as the Internet and mobile data networks, local area networks such as Ethernet and Wi-Fi networks, and personal area networks such as Bluetooth networks. Accordingly, it is contemplated and included within the scope of the invention that the computer hardware performing the above-described LASER and SCORE-RAG systems includes hardware necessary for such performance as is known in the art.

Referring now to FIG. **52** is an illustration of prefill and decode phases in an LLM inference process is described in more detail. LLM inference is the process of using a trained LLM to generate text or perform other language-related tasks. It involves two main phases; prefill **5000** and decode **5020**. The prefill phase **5000** occurs at the beginning of the inference process when the model is presented with an initial prompt **5002** or context. The prefill phase **5000** processes all input tokens **5002** in parallel, making it computationally intensive but requiring only one execution per inference session. The input **5002** undergoes tokenization **5004** to convert it into a format suitable for processing. These tokens are transformed into embeddings through an embedding lookup step **5006**. The embedded tokens pass through a series of neural network layers at step **5008**, where each layer computes query (Q), key (K), and value (V) vectors **5010**, resulting in pluralities of Q vectors, K vectors, and V vectors. These pluralities of vectors are used in a self-attention mechanism at step **5012**, followed by processing through a feed-forward network at step **5014**. After layer normalization at step **5016**, where normalized K and V vectors are produced, the normalized key and value vectors are stored in a KV cache at step **5018** for later use.

The decode phase **5020** follows the prefill phase **5000** and is responsible for generating new tokens one at a time. The decode phase **5020** is characterized by autoregressive generation, where each new token depends on all previously generated tokens. The decode phase begins by generating a new token at step **5040** based on the context provided by the prefill phase **5000**. This new token undergoes embedding lookup at step **5022** and is processed through the neural network layers at step **5024**, similar to the prefill phase **5000**. The neural network computes query (Q), key (K), and value (V) vectors **5026** for the new token at each layer, resulting in pluralities of Q, K, and V vectors. An attention computation at step **5029** is performed using the newly computed Q vector and the K and V vectors from both the KV cache **5028** (populated during the prefill phase **5000** and previous decode steps) and the newly computed K and V for the current token. The resulting output passes through a feed-forward network **5030** and layer normalization **5032**, producing normalized K and V vectors. The model then generates token probabilities **5034** and selects the next token **5036** based on these probabilities. The newly generated token's normalized K and V vectors are added to the KV cache **5038**, which grows with each new token. The decode phase **5020** iterates, using the newly generated token **5040** as input for the next cycle, until the desired output length is reached or a stop condition is met. The final output of this process is the generated text **5042**, which represents the model's response.

Referring now to FIG. **52** and also referring back to FIG. **15**, the lambda architecture for h-LLMs can be further improved by incorporating prefill and decode strategies. In the batch layer **1402**, the base h-LLM can be improved to handle prefill operations more efficiently. By implementing

chunked-prefills, where a prefill request is split into near-equal sized chunks, the batch layer can process large prompts more effectively. This approach allows for improved utilization of GPU resources and can improve throughput for batch processing of historical data. The real-time layer **1404** is improved by using dynamic or hybrid scheduling techniques. By allowing new requests to join a running batch without pausing ongoing decodes, the real-time layer can maintain lower latency while achieving higher throughput. This is beneficial for handling streaming data and providing quick responses to user queries. These improvements to the batch and real-time layers help in balancing the throughput-latency tradeoff by interleaving prefill and decode operations.

Referring now to FIG. **52** and also referring back to FIG. **17**, the in-memory processing architecture can be further improved by leveraging High Bandwidth Memory (HBM) technologies. HBM offers higher bandwidth and lower power consumption compared to traditional DRAM, making it ideal for LLM inference workloads. In the processing layer **1606**, multiple h-LLMs can be loaded into HBM, allowing for faster access to model parameters. This is particularly beneficial for the decode phase, which requires frequent, small memory accesses. By storing frequently accessed data, such as attention keys and values (KV Cache), in HBM, the system can reduce memory access latency and improve overall inference speed. Furthermore, the batches of input data **1604** can be more efficiently managed in HBM, allowing for rapid prefill operations. The high bandwidth of HBM enables the system to quickly load and process large input prompts.

Referring now to FIG. **52** and also referring back to FIG. **15** and FIG. **17**, additional memory-related optimizations can be implemented to further enhance the efficiency of LLM inference within the lambda architecture and in-memory processing framework. These enhancements focus on three areas: KV Cache Optimizations, Tiered Memory Systems, and Progressive Loading.

KV Cache Optimizations are crucial for improving memory efficiency during the decode phase of LLM inference. One approach involves implementing KV Cache Pruning, where less important entries are removed from the cache based on factors such as attention scores or token positions. This pruning can be adaptive, with thresholds that adjust based on available memory and sequence length. Additionally, periodic cache cleanup can be performed to remove entries that haven't been accessed recently. Another strategy is Adaptive KV Cache Sizing, where the size of the KV cache is dynamically adjusted based on the current sequence length, available system memory, and model complexity. This can be implemented using a sliding window approach, keeping only a fixed number of recent tokens in the cache.

Tiered Memory Systems leverage different types of memory to balance performance, capacity, and cost, which is particularly relevant for large-scale LLM inference. One approach involves the integration of HBM, DRAM and Non-Volatile Memory (NVM). DRAM for frequently accessed data like the active parts of the model and the KV cache, while leveraging NVM for storing less frequently accessed model parameters. Intelligent data movement algorithms can be implemented to predict which parts of the model will be needed next and preemptively move them to DRAM. A more complex tiered caching system can be developed where different levels of cache use different memory technologies. For example, L1 Cache could use on-chip SRAM for immediate access, L2 Cache could employ High Bandwidth Memory (HBM) for high-speed,

larger capacity, L3 Cache could utilize DRAM for a balance of speed and capacity, and L4 Cache could leverage NVM for large capacity. Predictive prefetching algorithms can be implemented to move data between tiers based on usage patterns and model architecture. For multi-GPU or distributed systems, heterogeneous memory management strategies can be developed to efficiently use different memory types across devices, using faster memory (e.g., HBM) on primary computation devices while offloading less critical data to devices with larger but slower memory.

Progressive Loading techniques aim to optimize memory usage by loading only the necessary parts of the model as needed, rather than loading the entire model upfront. Layer-wise Progressive Loading can be implemented, where model layers are loaded into memory sequentially as they are needed for computation. For transformer models, this could mean loading encoder layers progressively for the input processing, then loading decoder layers as needed for generation. Efficient layer swapping mechanisms can be developed to manage memory when the model size exceeds available memory. Adaptive Model Pruning techniques can also be implemented, where the model size is adapted based on input complexity. For simpler inputs, a smaller, pruned version of the model can be loaded, progressively loading more complex model components as needed for challenging inputs.

These memory optimization techniques, when integrated with the lambda architecture described in FIG. **15** and the in-memory processing architecture outlined in FIG. **17**, can enhance the efficiency and performance of LLM inference. By implementing these strategies, the system can better manage memory resources, reduce latency, and improve overall throughput, particularly for large-scale language models and high-volume inference tasks.

Referring now to FIG. **53**, an illustration of an approach for creating, managing, and refining superchunks and using them a SCORE-RAG system according to an embodiment of the invention is described in more detail. While describing the superchunk approach, reference back to the SCORE-RAG architecture illustrated in FIG. **40** will also be made. The superchunk approach, according to an embodiment of the invention, represents an improved method for enhancing the indexing and retrieval of documents and their chunks and utilization of information in the context of RAG systems. The process begins with multiple source documents **5100**, **5102**, **5104**, each of which undergoes a chunking process to create chunks **5108**, **5110**, **5112**, **5114**, **5116**, **5118**, **5120**, **5122**, and **5124**. This chunking process divides each document into smaller, semantically coherent units referred to as chunks in certain embodiments, and in other embodiments they may correspond to paragraphs in a document (and/or images and audio content) processed in a sequential or predetermined order. Each chunk is then associated with metadata, which may include, but is not limited to, information about the parent document, creation date, category, topics, and citations or links to other documents and/or their chunks. The chunking can be performed, for example, by Document Chunking module **3806** (from FIG. **40**).

The chunks may then be analyzed to evaluate the importance and relevance of each chunk. This analysis takes into account factors such as the chunk's content, its metadata, and its relationships with other chunks and documents. The Chunk Selection and Ranking module **3810** (from FIG. **40**) assigns a score or rank to each chunk based on these factors.

A superchunk creator module may be additionally comprised by the SCORE-RAG system illustrated in FIG. **40**.

Other embodiments may comprise a superchunk creator module as part of a LASER system/module as described above. The superchunk module may be operable to select the highest-ranked chunks from across multiple documents to form a superchunk **5106**. A superchunk may be defined as a collection of the most important and relevant chunks related to a specific topic or category, regardless of their source documents. Each superchunk may be assigned a unique identifier (UUID), which is used to manage and retrieve the superchunk. The superchunk may be created in response to a user query or a type of user query that is expected, or in response to the domain-specific nature of the Generative AI application, e.g., financial analysis or code generation, for example.

The Indexing Engine **3814** (from FIG. **40**) may then process the created superchunks for efficient storage and retrieval. This indexing takes into account the composite nature of superchunks, allowing for rapid access to highly relevant information across multiple source documents. The memory access architecture (e.g., high bandwidth memories and their widths) may also be optimized for storage and retrieval of superchunks.

When a user query is received, for example, the query processor **3816** (from FIG. **40**) analyzes the query to identify relevant topics or categories. The retrieval orchestrator **3818** (from FIG. **40**) then prioritizes the retrieval of superchunks that match these identified topics. This approach ensures that the most pertinent information from across multiple documents is readily available for the RAG process.

The generation module **3822** (from FIG. **40**) utilizes the retrieved superchunks to generate a response to the user query. By working with superchunks, the Generation module **3822** has access to a concentrated set of relevant and/or highly relevant information, thus improving the quality and relevance of the generated response for the particular domain.

The Evaluation & Fine-tuning module **3824** (from FIG. **40**) continuously assesses the performance of superchunks in query responses and refines their composition based on usage patterns and effectiveness metrics. This ongoing optimization ensures that the system adapts to changing information landscapes and user needs.

The superchunk approach offers several advantages over traditional RAG systems:

1. Enhanced Relevance: By combining the most important chunks from multiple documents, superchunks provide a more comprehensive and focused information set for the RAG process.
2. Improved Efficiency: The use of superchunks reduces the volume of data that needs to be processed during query time, potentially leading to faster response generation. Superchunks can be considered as refined documents in this context. Instead of using with the original set of documents for RAG, a more efficient approach uses the superchunks.
3. Cross-Document Insights: Superchunks enable the system to draw connections and insights from across multiple documents, thus uncovering relationships that might be missed in a document-centric approach.
4. Adaptability: The continuous evaluation and optimization of superchunks allow the system to evolve and improve its performance over time.
5. Scalability: As the volume of source documents grows, the superchunk approach becomes increasingly valuable, providing a method to distill large document collections into manageable, highly relevant information units.

Referring now to FIG. **54**, an illustration of the different non-limiting embodiments of “superchunk” in the context of the present invention, is described in more detail. A person skilled in the art should understand that “superchunk” may refer to, but is not limited to: a collection of pointers **5228** to individual chunks **5230** from one or more documents **5232**; a fully populated refined document **5206** composed of selected chunks **5208**; a combination **5222** of pointers to chunks **5224** along with their respective summaries **5226**; a set of chunks **5272** selected based on specific criteria such as relevance **5276**, importance **5280**, or recency **5278**; a dynamically assembled collection **5202** of information tailored to a particular query or context **5204**; a hierarchical structure **5266** of chunks **5270** and sub-chunks **5268**; a semantically related group of chunks **5216**, regardless of their source documents; a compressed or encoded representation **5252** of multiple chunks **5254**; a versioned collection of chunks **5256** that evolves over time; a multi-modal collection **5264** incorporating text **5262**, metadata **5282**, and references to non-textual content **5260**; a unit of information **5210** optimized for specific memory access patterns **5212** or caching strategies **5214**; a privacy-enhanced collection **5234** of information with sensitive data removed or masked **5236**; a dynamically adjusted set of information **5246** based on privacy **5248** and security requirements **5250**; a content unit **5242** that can be associated with targeted advertisements **5244**; and a categorized information set **5238** that facilitates content-relevant marketing **5240**.

Referring now to FIG. **55**, an illustration of the characteristics of “superchunks” in the context of the present invention is described in more detail. Superchunks may include composition characteristics **5302**. The composition of superchunks **5300** within the system can vary based on different factors and implementation choices, including, but not limited to:

- a) Accuracy/Relevance Based **5312**: Superchunks may be assembled based on accuracy or relevance metrics, with the potential for more accurate or comprehensive superchunks to be made available to users with higher service tiers or permissions.
- b) Internal Structure Variation **5314**: The internal structure of a superchunk may vary from a simple list of references to a complex data structure incorporating multiple layers of information and metadata.
- c) Pre-computed or Dynamic **5316**: Superchunks may be pre-computed and stored, or they may be dynamically generated in response to specific queries or user interactions.
- d) Machine Learning Refined **5318**: The process of creating and maintaining superchunks may involve machine learning algorithms that continuously refine the selection and organization of chunks based on usage patterns and feedback.
- e) Version Controlled **5320**: Superchunks may incorporate version control mechanisms to track changes over time and allow for rollback or comparison between different states.
- f) Adjustable Granularity **5322**: The granularity of information within a superchunk may be adjustable, allowing for different levels of detail to be presented based on user needs or system capabilities.
- g) Relationship Metadata **5324**: Superchunks may include metadata about the relationships between contained chunks for capturing complex networks of information.
- f). The memory and caches hierarchy and storage architecture interconnecting the GPUs to the high-high bandwidth memories.

Superchunks may include utilization characteristics **5304**. The utilization of superchunks within the system can vary based on different factors and implementation choices, including, but not limited to:

- a) Domain-Specific Strategies **5326**: The system may employ different strategies for superchunk creation and utilization based on factors such as the domain of information.
- b) User Preference Based **5328**: The system may employ different strategies for superchunk creation and utilization based on factors such as user preferences.
- c) Resource-Aware **5330**: The system may employ different strategies for superchunk creation and utilization based on factors such as computational resources available.
- d) High-Speed Memory Cached **5332**: Superchunks may be preferentially stored in high-speed memory or caches to optimize access times and system performance. The system may employ intelligent caching strategies that prioritize frequently accessed or highly relevant superchunks for rapid retrieval.

Superchunks may further comprise privacy and security characteristics **5306**. The creation and maintenance of superchunks may involve privacy protection and security mechanisms, including, but not limited to:

- a) PII Detection & Removal **5334**: Automatic detection and removal or masking of Personally Identifiable Information (PII) before the superchunk is used in downstream processes such as feeding to LLMs or in a RAG pipeline.
- b) Unlearning Techniques **5336**: Implementation of “unlearning” or “targeted catastrophic forgetting” techniques that can dynamically adjust the content of superchunks based on privacy requirements or user queries.
- c) Differential Privacy **5338**: Application of differential privacy techniques to add noise to sensitive data while maintaining overall statistical properties of the information.
- d) Content Safety Screening **5340**: Content safety screening to prevent the inclusion of explicit, violent, or otherwise inappropriate material.
- e) Security Checks **5342**: Security checks to detect and remove potential malware, phishing attempts, or other security threats.
- f) Fact-Checking **5344**: Fact-checking to flag or filter out misinformation or unverified claims.
- g) Bias Detection **5346**: Bias detection and mitigation to ensure a balanced representation of information.

Superchunks may further comprise monetization characteristics **5308**. Superchunks may be associated with advertising content based on their composition or categorization, including (but not limited to):

- a) Content-Based Ads **5348**: Inserting relevant advertisements into superchunks based on the topical content. For example, a superchunk about natural farming practices might include or be associated with advertisements from organic fertilizer companies.
- b) Category-Linked Ads **5350**: Linking advertisements to the category or classification of the superchunk. For instance, a superchunk related to rent control laws in a specific city might be associated with advertisements for local legal services.
- c) Ad Bidding System **5352**: Implementing a bidding or auction system for advertisers to target specific types or categories of superchunks.

This advertisement feature can generate advertisements that are relevant to the user query, user interests, user intentions, or user past history of interactions. As part of the derived queries, certain queries may be made by the AI brokers to the user to identify their specific goals and intentions (for example, the AI brokers may ask the user if they are interested in buying a new car in the next six months, given that the user query appears to research and compare various brands of automobiles). The two-way interaction between the AI brokers and the user is seen as another novelty of certain embodiments of the present invention, compared to the one way interaction users currently have with generative AI LLMs.

Superchunks may further comprise processing characteristics **5310**. The processing of superchunks may include multiple stages of enhancement and screening, including, but not limited to:

- a) Privacy Enhancements **5354**: Implementation of privacy enhancements, including, detection and removal of PII, data anonymization or pseudonymization techniques.
- b) Ad Integration **5356**: Ad generation and integration based on content analysis and categorization.
- c) Safety Guardrails **5358**: Application of safety and security guardrails to filter or flag potentially harmful or inappropriate content.
- d) Tiered Access **5360**: The system may employ different strategies for superchunk creation, maintenance, and utilization based on factors such as user authentication level, subscription tier, or specific privacy and security requirements of the use case.

These variations and implementations of superchunks are not mutually exclusive, and the present invention incorporates systems that may include multiple approaches or allow for dynamic switching between different superchunk paradigms based on context or requirements.

Referring now to FIG. 56, an illustration of an architecture of a Hybrid-RAG system is described in more detail. A Hybrid Retrieval-Augmented Generation (Hybrid-RAG) system is designed to process and generate multi-modal data, including but not limited to text, documents, images, audio, video, and code. The system leverages a combination of various database types and LLMs to overcome the limitations of traditional Vector-RAG or Graph-RAG systems, providing enhanced performance and versatility across diverse data types and query scenarios. The Hybrid-RAG system comprises multiple components designed to efficiently process, store, retrieve, and generate multi-modal data.

To achieve optimal performance for multi-modal data, Hybrid-RAG employs multiple embedding models and specialized databases, each fine-tuned for a specific content type such as text, audio, images, video, or code. This specialized approach ensures that the unique characteristics and nuances of each content modality are accurately captured and indexed. The system begins with the indexing of multi-modal data **5400**, which may include text, documents, audio, video, code, and other data types. The indexing process **5402** involves several steps:

1. Data Ingestion: Raw multi-modal data is ingested into the system.
2. Preprocessing: This step includes chunking, filtering, and cleaning of the ingested data.
3. Embedding Generation: Specialized embedding models generate vector representations for each data type.

The processed and embedded data is stored in a variety of database types **5404**, including: a Vector Databases **5406**

(e.g., Pinecone, Milvus) for efficient similarity search; Graph Databases **5408** (e.g., Neo4j, TigerGraph) for relationship-based queries; Document Databases **5410** (e.g., MongoDB, Couchbase) for unstructured data; Relational Databases **5412** (e.g., PostgreSQL, MySQL) for structured data; Non-Relational Databases **5414** (e.g., DynamoDB) for unstructured or semi-structured data; Time-Series Databases **5416** (e.g., InfluxDB, TimescaleDB) for temporal data; In-Memory Databases **5418** (e.g., Redis, Memcached) for high-speed data access; Spatial/GIS Databases **5420** (e.g., PostGIS) for location-based data; Object-Oriented Databases **5422** (e.g., ObjectDB) for complex object storage; Column-Oriented Databases **5424** (e.g., Apache Cassandra) for wide-column storage; Full-Text Search Engines **5426** (e.g., Elasticsearch, Solr) for keyword-based retrieval; and Other specialized database types **5428** (e.g. NewSQL, multi-modal databases, RDF stores, XML databases, etc).

When a user **5444** submits a query **5446**, the system pre-processes the query. Query Preprocessing **5448** involves filtering, embedding generation, and the creation of derived queries. Based on the preprocessed query, the system determines which database(s) are most suitable for retrieval in a query routing process. The system then queries the selected databases to retrieve relevant context **5430**. The retrieved context **5430** undergoes processing **5432** including filtering, cleaning, and ranking to generate the refined context **5434**. One or more appropriate LLMs or h-LLMs **5436** are then selected based on the query type and refined context **5434**. The responses **5438** generated by the LLMs or h-LLMs **5436** undergo filtering, cleaning, and ranking at a post-processing step **5440**. The final processed response **5442** is then delivered to the user. The previously used contexts may also be stored in-memory (for example, a cache) for faster and more accurate processing times.

For the generation phase, the Hybrid-RAG utilizes an ensemble of LLMs or h-LLMs **5436**, each specialized for different tasks. These may include models optimized for question-answering, code generation, image interpretation, audio transcription, and video analysis, among others. This multi-faceted approach allows Hybrid-RAG to not only process a wide range of input types but also to generate appropriate and context-aware multi-modal outputs.

The Hybrid-RAG system addresses limitations of traditional RAG systems by utilizing the most appropriate database(s) for each data type and query scenario. Hybrid-RAG enables multi-modal data processing and generation, thus providing more comprehensive and accurate responses through the integration of multiple data sources and LLMs.

Referring now to FIG. **57**, an illustration of an architecture of a NoRAG system according to an embodiment of the invention is described in more detail. LLMs face limitations in accessing current information, maintaining factual accuracy, and providing transparent, attributable responses. RAG systems address these limitations of LLMs. RAG is useful for tasks requiring current or specialized knowledge, as it allows language models to draw upon external, updatable sources of information. However, RAG often introduces complexities in implementation and maintenance. Users may, depending on their design options, have to address complexities of chunking, embedding, indexing documents, maintaining vector databases, for instance. NoRAG system provides an approach to enhance LLMs without the need for RAG systems, hence the name NoRAG. By integrating functionalities directly in a plug-in manner into the LLM architecture, in some embodiments as a license-able plugin

to LLMs, NoRAG offers improved performance, reduced complexity, and enhanced user experience compared to traditional RAG systems.

The NoRAG system begins with ingesting multi-modal data **5500**, which may include text, documents, images, audio, video, code, and other data types. The NoRAG system **5502** comprises several modules, each designed to perform specific functions in the overall process of enhancing LLM capabilities.

The NoRAG system comprises a Document/Input Processor **5504** module. The Input Processor module **5504** is responsible for processing input documents and data sources. It handles various file formats, extracts relevant information, and prepares the data for integration into the NoRAG system.

The NoRAG system further comprises a Query Processor module **5506**: The Query Processor module **5506** handles user queries, performing sophisticated analysis to improve them for the LLM **5536**. It breaks down complex queries into manageable parts and generates derived queries when necessary.

The NoRAG system further comprises a Response Processor module **5508**. The Response Processor module **5508** performs post-processing on the LLM's **5536** output **5534** before sending it to the user. This module refines the generated content, ensures coherence, and applies any necessary formatting or style adjustments to enhance the quality and relevance of the final response.

The NoRAG system further comprises Dynamic Knowledge Integrator component **5510**. The Dynamic Knowledge Integrator component **5510** interfaces directly with the LLM **5536**, providing relevant information during the generation process. It acts as a bridge between the LLM's **5536** inherent knowledge and the additional information processed by the NoRAG system, improving integration of external knowledge into the LLM's **5536** responses **5534**.

The NoRAG system further comprises a Domain Specific Agents module **5512**: The Domain Specific Agents module **5512** comprises several domain specific agents which retrieve appropriate specialized knowledge on the query context (e.g. web search agent, stock market agent, weather data agent, IoT data, etc). It enables the NoRAG system to adapt its responses to specific domains, improving accuracy and relevance in specialized fields.

The NoRAG system further comprises an Internal Indexing module **5514**. The Internal Indexing module **5514** utilizes a combination of diverse database types, including, but not limited, to vector databases, graph databases, document databases, time-series databases, full-text search engines, in-memory databases, object databases, spatial databases, SQL databases, NoSQL databases, and column databases. This approach ensures efficient indexing and retrieval of information, improving the NoRAG system's performance across various data types and query patterns.

The NoRAG system further comprises Specialized Domain Adapters modules **5516**: These plug-in modules **5516** contain specialized knowledge for specific domains. They can be dynamically loaded and unloaded based on the query context, allowing the NoRAG system to provide expert-level responses in various fields without overburdening the core LLM.

The NoRAG system further comprises a Self-Verification system **5518**. The Self-Verification system **5518** checks facts and reduces hallucinations in the LLM's **5536** outputs **5534**. It employs internal consistency checks and compares generated content against the system's knowledge base to ensure accuracy and reliability in the responses.

The NoRAG system further comprises a Source Attribution module **5520**: The Source Attribution module **5520** tracks and cites internal knowledge sources used in generating responses. It enhances the transparency and credibility of the NoRAG system's outputs by providing citations for the information used.

The NoRAG system further comprises a Personalization Engine **5522**. The Personalization Engine **5522** adapts responses **5542** based on user preferences and interaction history. It maintains user profiles and adjusts the system's outputs to match individual user needs, enhancing the relevance and usefulness of the responses. This module may optionally inject advertisements in responses based on the user's subscription tier or preferences or queries sent to the user **5538** by the LLM **5536** to identify the user's attitudes, intentions, and predict behavior and future actions.

The NoRAG system further comprises a Bias Detection & Mitigation module **5524**. The Bias Detection & Mitigation module **5524** identifies potential biases in the NoRAG system's responses and works to balance them. It employs advanced algorithms to recognize various types of bias and adjusts the output to provide more neutral and fair responses.

The NoRAG system further comprises a Prompt, Derived Prompts, and Context Caching module **5526**: This module **5526** caches user queries, derived prompts, and the relevant context used (including previously used contexts) that may be used to generate responses. By storing this contextual information for in-memory processing, the NoRAG system can improve response times for similar queries and maintain consistency in its outputs over time.

The NoRAG system further comprises a Continuous Learning Orchestrator **5528**: The Continuous Learning Orchestrator **5528** manages the ongoing learning process of the model. It identifies knowledge gaps, prioritizes learning objectives, and coordinates the integration of new information across all modules, ensuring that the NoRAG system remains up-to-date and continues to improve over time.

The NoRAG system further comprises a Security and Privacy Guardian module **5530**: The Security and Privacy Guardian module **5530** ensures data privacy and security in knowledge storage and retrieval. Privacy and security guardrails are implemented to filter sensitive data in the query and responses (such as personally identifiable information (PII)).

When a user **5538** submits a query **5540**, the NoRAG system processes the query and generates a relevant context **5532** which is passed to one or more LLMs or h-LLMs **5536**. NoRAG utilizes an ensemble of LLMs or h-LLMs **5536**, each specialized for different tasks. These may include models optimized for question-answering, code generation, image interpretation, audio transcription, and video analysis, among others. The processed response **5542** is then returned to the user.

The NoRAG plug-in and/or integrated LLM system may provide several advantages over traditional RAG approaches:

1. **Reduced Complexity**: By integrating functionalities directly into the LLM architecture, the NoRAG system eliminates the need for external retrieval systems, simplifying implementation and maintenance. The NoRAG system works like a plugin system enhancing the capabilities of an LLM.
2. **Improved Performance**: The tight integration of agents, domain adapters, knowledge and processing modules allows for faster response times and more coherent outputs.
3. **Enhanced Customization**: The modular architecture of the NoRAG system allows for easy addition or modification

of specialized knowledge domains without requiring changes to the core LLM.

4. **Improved Privacy and Security**: By internalizing data storage and retrieval, the NoRAG system provides improved control over sensitive information and reduces potential vulnerabilities associated with external data sources.
5. **Seamless Updates**: The Continuous Learning Orchestrator module **5528** enables the NoRAG system to incorporate new information more efficiently than traditional RAG systems, which often require separate update processes for external knowledge bases.
6. **Use in Network of LLM Agents**: The NoRAG plug-in module can be used as a series or parallel network when connected to LLMs that operate as a network of LLM agents performing specialized tasks in a coordinated sequence (managed by AI brokers or LLMs, for example). Each specialized LLM agent may use a different NoRAG plugin, and NoRAG plugins may be mapped to different LLMs, depending on the type of task being done. A library of NoRAG modules may be developed in a generic manner and also with a target LLM family as an objective, and NoRAG modules for billing, advertisement generation, fault-tolerance, and security may also be added on in a plug-in manner.

Some of the illustrative aspects of the present invention may be advantageous in solving the problems herein described and other problems not discussed which are discoverable by a skilled artisan.

While the above description contains much specificity, these should not be construed as limitations on the scope of any embodiment, but as exemplifications of the presented embodiments thereof. Many other ramifications and variations are possible within the teachings of the various embodiments. While the invention has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best or only mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited. Moreover, the use of the terms first, second, etc. do not denote any order or importance, but rather the terms first, second, etc. are used to distinguish one element from another. Furthermore, the use of the terms a, an, etc. do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item.

Thus the scope of the invention should be determined by the appended claims and their legal equivalents, and not by the examples given.

The claims in the instant application are different than those of the parent application or other related applications. Applicant therefore rescinds any disclaimer of claim scope made in the parent application or any predecessor application in relation to the instant application. Any such previous disclaimer and the cited references that it was made to avoid,

may need to be revisited. Further, any disclaimer made in the instant application should not be read into or against the parent application.

What is claimed is:

1. A method of improving performance of large language models (LLMs) comprising:
 - receiving one or more context files via an application programming interface (API) at an input broker from a user interface;
 - generating one or more refined context files from the one or more context files using one or more refining LLMs;
 - sending the one or more refined context files to one or more h-LLMs via a cloud service API, the one or more h-LLMs being hosted in a cloud container environment;
 - receiving a user prompt via the API at the input broker from the user interface;
 - generating a plurality of derived prompts from the user prompt at the input broker;
 - transmitting the plurality of derived prompts to the one or more h-LLMs via the cloud service API;
 - receiving a plurality of h-LLM results at an output broker, the plurality of h-LLM results being generated responsive to both the one or more refined context files and the plurality of derived prompts;
 - processing the plurality of h-LLM results at the output broker to generate a responsive result; and
 - transmitting the responsive result to the user interface via the API.
2. The method of claim 1 wherein generating the one or more refined context files comprises:
 - splitting content of the one or more context files into a plurality of token blocks;
 - reordering the plurality of token blocks;
 - grouping token block subsets of the plurality of token blocks into a plurality of token block batches; and
 - iteratively processing the plurality of token block batches to generate one or more refined context files.
3. The method of claim 2 wherein iteratively processing the plurality token block batches comprises:
 - processing each token block batch of the plurality of token block batches through the one or more refining LLMs;
 - ranking outputs of the one or more refining LLMs for each token block of the plurality of token blocks;
 - determining whether an iteration completion criterion has been met;
 - upon determining the iteration completion criterion has been met, generate the one or more refined context files from the plurality of token block batches; and
 - upon determining the iteration completion criterion has not been met:
 - selecting a subset of high-ranking token blocks from which a predetermined number of the highest ranked outputs were generated;
 - performing a clustering operation on the subset of high-ranking token blocks; and
 - generating a plurality of refined token block batches responsive to the clustering operation;
 - wherein the plurality of refined token block batches is iteratively processed in the same manner as the plurality of token block batches until the iteration completion criterion is met.
4. The method of claim 1 wherein generating one or more refined context files comprises identifying one or more categories of information comprised by the one or more context files using topic modeling techniques.

5. The method of claim 4 wherein the plurality of derived prompts is generated responsive to the one or more categories of information identified in the one or more context files.

6. The method of claim 4, wherein at least one of the input broker and the output broker is configured to generate one or more advertisements to be provided as part of the responsive result responsive to the one or more identified categories of information.

7. The method of claim 1 wherein the one or more refined context files are generated offline and stored prior to receiving the user prompt.

8. The method of claim 1 wherein the one or more refined context files are retrieved from a cached memory storage.

9. The method of claim 1 further comprising performing pre-prompt processing on the one or more refined context files using one or more RAG enhancement engines or RAG enhancement modules, wherein generating the plurality of h-LLM results comprises:

- parsing the user prompt;
- identifying one or more relevant information items to be retrieved;
- checking an information cache for the one or more relevant information items;
- retrieving the one or more relevant information items from the cache responsive to locating the one or more relevant information items in the cache;
- conducting a hybrid search to locate the one or more relevant information items responsive to not locating the one or more relevant information items in the cache;
- generating an augmented user prompt by combining the user prompt with the one or more relevant information items; and
- providing the augmented user prompt to the one or more h-LLMs to generate the plurality of h-LLM results.

10. The method of claim 9 wherein performing pre-prompt processing on the one or more refined context files comprises:

- identifying one or more topics comprised by the one or more refined context files using a topic modeling engine;
- segmenting the one or more refined context files into one or more semantic chunks using a file chunking module;
- identifying cited sources within the one or more refined context files using a citation analyzer module;
- selecting one or more selected chunks responsive to the one or more topics and the cited sources using a chunk selection and ranking module;
- generating enriched chunks by adding metadata to the one or more selected chunks using a metadata enrichment module; and
- indexing the enriched chunks using an indexing engine.

11. The method of claim 10 wherein the enriched chunks are saved in a graph database.

12. The method of claim 1 wherein generating the plurality of h-LLM results comprises:

- performing a prefill process that simultaneously generates a key vector and a value vector from each derived prompt of the plurality of derived prompts in parallel; and
- performing a decode process that serially generates each h-LLM result of the plurality of h-LLM results from the key vector and the value vector generated for each derived prompt.

13. The method of claim 12 wherein the prefill process is performed by at least one h-LLM of the one or more h-LLMs and comprises:

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receiving the plurality of derived prompts;
 generating a plurality of tokens by tokenizing each
 derived prompt of the plurality of derived prompts;
 performing an embedding lookup for the plurality of
 tokens;
 processing the plurality of tokens through a series of
 neural network layers comprised by the at least one
 h-LLM, comprising:
 computing a query vector, a key vector, and a value
 vector for each token of the plurality of tokens at
 each neural network layer of the series of neural
 network layers, resulting in a plurality of query
 vectors, a plurality of key vectors, and a plurality of
 value vectors;
 performing a self-attention mechanism using the query
 vector, the key vector, and the value vector for each
 token; and
 processing each of the query vector, the key vector, and
 the value vector for each token through a feed-
 forward network comprised by the at least one
 h-LLM;
 generating a normalized key vector and a normalized
 value vector by performing a layer normalization pro-
 cess on the plurality of key vectors and the plurality of
 value vectors; and
 storing the normalized key vector and the normalized
 value vector in a key-value cache.
14. The method of claim **13** wherein the decode process
 is performed by at least one h-LLM of the one or more
 h-LLMs and comprises:
 receiving a first derived prompt of the plurality of derived
 prompts, the prefill process having been completed for
 the first derived prompt;
 generating a first decode token from the first derived
 prompt;
 performing an embedding lookup on the first decode
 token;
 processing the first decode token through the series of
 neural network layers, wherein processing comprises:
 computing a new query vector, a new key vector, a new
 value vector for the first decode token at each neural
 network layer of the series of neural network layers,
 resulting in a plurality of new query vectors, a
 plurality of new key vectors, and a plurality of new
 value vectors;
 retrieving a retrieved key vector and a retrieved value
 vector from the key-value cache, the retrieved key
 vector and the retrieved value vector having been
 computed for the first derived prompt in the prefill
 process;
 performing a self-attention mechanism using a new
 query vector of the plurality of new query vectors, a
 new key vector of the plurality of new key vectors,
 a new value vector of the plurality of new value
 vectors, the retrieved key vector, and the retrieved
 value vector; and
 processing each of the new query vector of the plurality
 of new query vectors, the new key vector of the
 plurality of new key vectors, and the new value
 vector of the plurality of new value vectors through
 the feed-forward network;
 generating a normalized new key vector and a normalized
 new value vector by performing a layer normalization
 on the plurality of new key vectors and the plurality of
 new value vectors;
 generating first token probabilities from the normalized
 new key vector and the normalized new value vector;

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generating a first output token responsive to the first token
 probabilities;
 adding the first output token to an h-LLM response of the
 plurality of h-LLM results; and
 adding each of the normalized new key vector and the
 normalized new value vector to the key-value cache.
15. The method of claim **14** wherein the decode process
 further comprises performing a second iteration of the steps
 of claim **14** with the first decode token being generated from
 the first output token.
16. The method of claim **15** further comprising perform-
 ing the decode process until at least one of:
 the h-LLM response reaches a desired output length; or
 a predefined stop condition is met.
17. The method of claim **1** wherein the one or more
 refined context files comprises a plurality of refined context
 files, the method further comprising performing pre-query
 processing on the plurality of refined context files by:
 identifying one or more topics comprised by the plurality
 of refined context files using a topic modeling engine;
 segmenting the plurality of refined context files into one
 or more semantic chunks using a file chunking module;
 generating a plurality of enriched chunks by adding
 metadata to the one or more semantic chunks using a
 metadata enrichment module;
 analyzing the plurality of enriched chunks to determine at
 least one of an importance score and a relevance score
 for each enriched chunk of the plurality of enriched
 chunks using a chunk selection and ranking module;
 forming at least one superchunk by selecting a subset of
 enriched chunks from the plurality of enriched chunks
 based upon the subset of enriched chunks having the
 highest importance score or relevance score and asso-
 ciating the subset of the enriched chunks with each
 other; and
 indexing the at least one superchunk using an indexing
 engine in a superchunk index;
 wherein generating the plurality of h-LLM results com-
 prises:
 identifying at least one of an identified topic or an
 identified category by analyzing the user prompt
 using a query processing module;
 retrieving one or more retrieved superchunks from the
 superchunk index responsive to the identified topic
 or identified category; and
 generating an h-LLM result responsive to the one or
 more retrieved superchunks and the user prompt; and
 wherein each superchunk is formed from enriched chunks
 generated from semantic chunks originating from at
 least two different refined context files of the plurality
 of refined context files.
18. The method of claim **1** further comprising:
 performing pre-prompt processing on the one or more
 refined context files using an input processing module
 comprising:
 receiving multimodal data from a multimodal data source;
 chunking the multimodal data to produce chunked mul-
 timodal data;
 filtering the chunked multimodal data to produce filtered
 multimodal data;
 generating embeddings for the filtered multimodal data in
 the form of vector representations responsive to a data
 modality of the data comprised by the filtered multi-
 modal data; and
 storing the filtered multimodal data and the embeddings in
 a plurality of databases;

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generating a plurality of prompt embeddings using a query processing module from the user prompt;
 selecting one or more selected databases from the plurality of databases from which to retrieve context files responsive to the plurality of prompt embeddings using the query processing module;
 querying the one or more selected databases with the plurality of prompt embeddings using the query processing module;
 receiving a plurality of retrieved context files from the one or more selected databases at a response processing module; and
 processing the plurality of retrieved context files to produce the one or more refined context files using the response processing module;
 wherein transmitting the plurality of derived prompts to the one or more h-LLMs comprises transmitting each derived prompt of the plurality of derived prompts to an h-LLM specialized for a different task.

19. The method of claim 18 wherein the input broker, the output broker, the input processing module, the query processing module, and the response processing module are comprised by a single system configured to be integrated with various LLMs.

20. A system for improving performance of large language models (LLMs) comprising:

- a processor configured to:
 - generate one or more refined context files from one or more context files using one or more refining LLMs;
 - generate a plurality of derived prompts from a user prompt at an input broker; and
 - process a plurality of h-LLM results at an output broker to generate a responsive result; and
- a communication device configured to
 - receive the one or more context files via an application programming interface (API) at the input broker from a user interface;
 - send the one or more refined context files to one or more h-LLMs via a cloud service API, the one or more h-LLMs being hosted in a cloud container environment;

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receive a user prompt via the API at the input broker from the user interface;

transmit the plurality of derived prompts to the one or more h-LLMs via the cloud service API;

receive the plurality of h-LLM results at the output broker, the plurality of h-LLM results being generated responsive to both the one or more refined context files and the plurality of derived prompts; and

transmit the responsive result to the user interface via the API; and

a non-transitory computer-readable medium comprising software that is executable by the processor.

21. A system for improving performance of large language models (LLMs) comprising:

means for receiving one or more context files via an application programming interface (API) at an input broker from a user interface;

means for generating one or more refined context files from the one or more context files using one or more refining LLMs;

means for sending the one or more refined context files to one or more h-LLMs via a cloud service API, the one or more h-LLMs being hosted in a cloud container environment;

means for receiving a user prompt via the API at the input broker from the user interface;

means for generating a plurality of derived prompts from the user prompt at the input broker;

means for transmitting the plurality of derived prompts to the one or more h-LLMs via the cloud service API;

means for receiving a plurality of h-LLM results at an output broker, the plurality of h-LLM results being generated responsive to both the one or more refined context files and the plurality of derived prompts; and

means for processing the plurality of h-LLM results at the output broker to generate a responsive result; and

means for transmitting the responsive result to the user interface via the API.

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